

LYCÉE GUILLAUME KROLL

BTS Cloud Computing

MODULE VIRCL · TEAM 5

SERVER VIRTUALISATION

Comparing Proxmox VE and VMware vSphere



SPECIALISATION · SNAPSHOTS & TEMPLATES

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INSTALLATION OF THE HYPERVISOR

VMWARE INSTALLATION



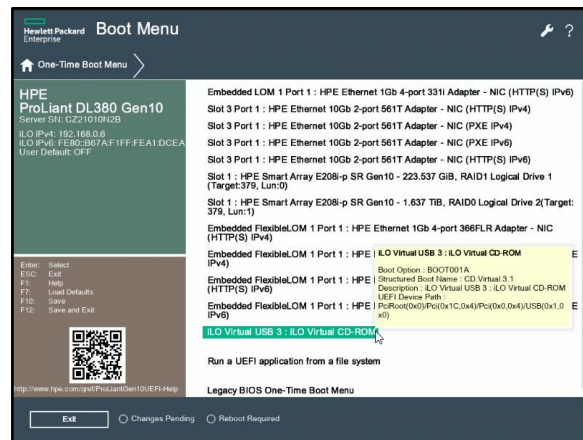
OS INSTALLATION

We're installing VMWare on HP06 first, so we connect to the ILO for the server to configure it remotely.

Starting the server, we enter the console from the ILO and we remotely install the VMWare iso onto the server then select the virtual disk in the boot menu.

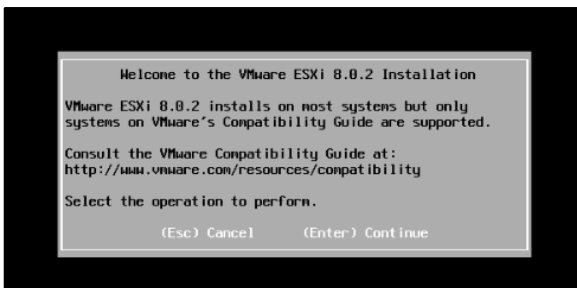


VMWare - Screenshot 1: Loading ESXi ISO

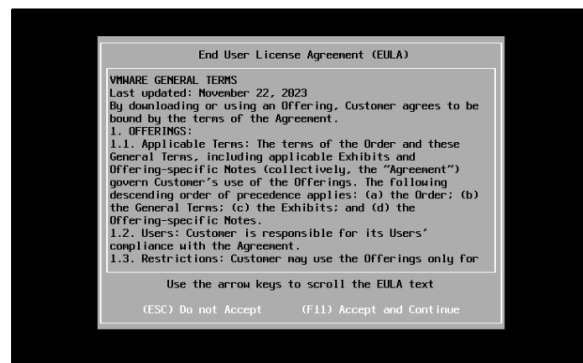


VMWare - Screenshot 2: Selecting Boot Media

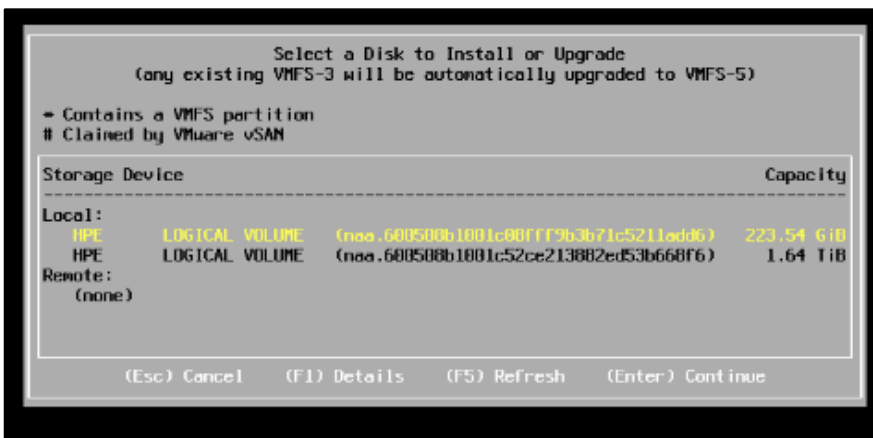
In the installation of VMWare, we continue past the welcome and EULA screen to the installation disk, here we select our Array of SSDs, which here is our lower capacity device.



VMWare - Screenshot 3: Installation welcome screen

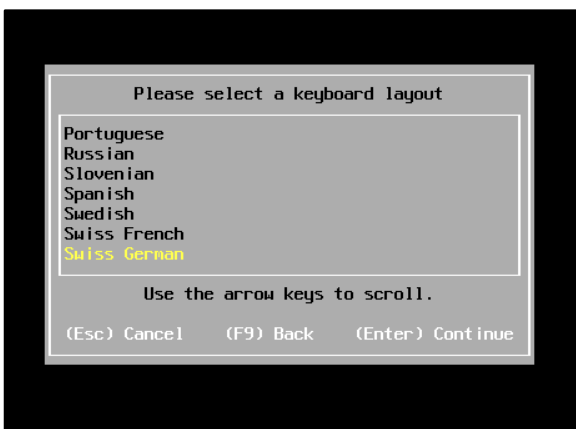


VMWare - Screenshot 4: ESXi EULA screen

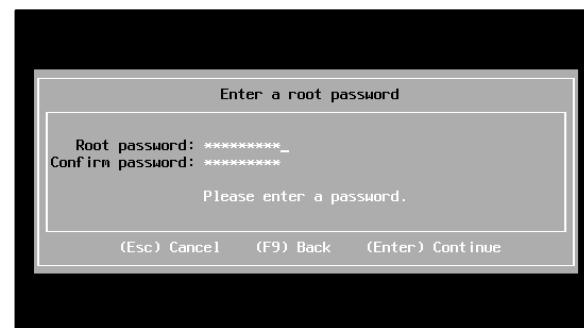


VMWare - Screenshot 5: Installation disk selection

We continue with the configuration of the keyboard and root password.

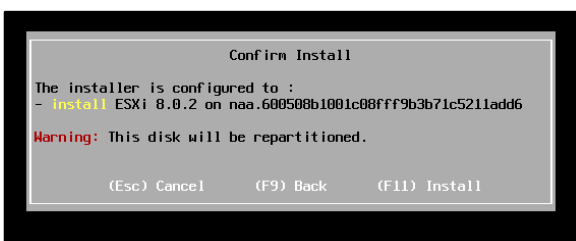


VMWare - Screenshot 6: Keyboard layout selection

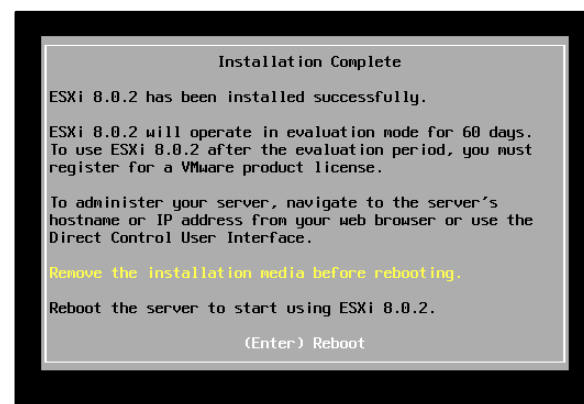


VMWare - Screenshot 7: setting root password

We confirm our installation and remove the installation media before rebooting.



VMWare - Screenshot 8: Install confirmation



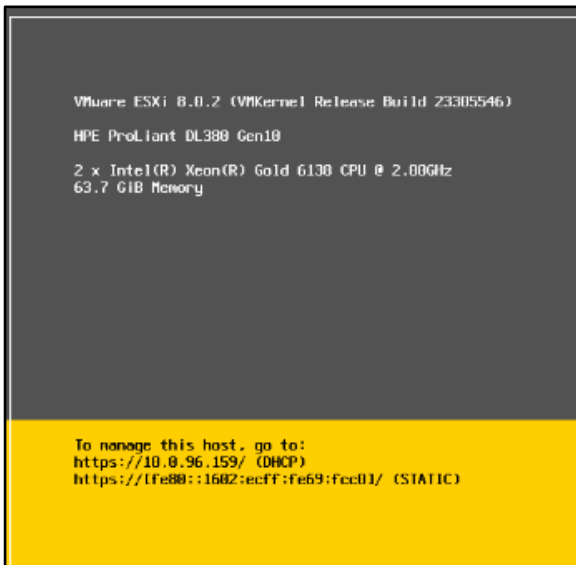
VMWare - Screenshot 9: Installation complete screen

One issue we encountered was that the CPU on the server may not be supported for future ESXi releases, which for this lab is quite irrelevant.

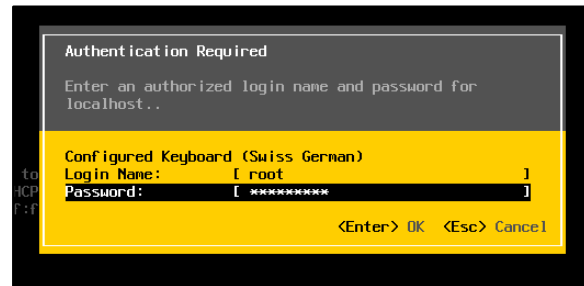
SETTING UP THE ESXi FOR THE VCENTER SERVER

vCenter is the main management interface for VMWare servers, vCenter is a Virtual machine running on the ESXi host itself, so before deploying it you need to configure some stuff on the ESXi itself.

Looking at the console view in the ILO, notice that the IP address is one from the DHCP, which is not what we wanted, so go into the system configuration by pressing F2 and log in as root.

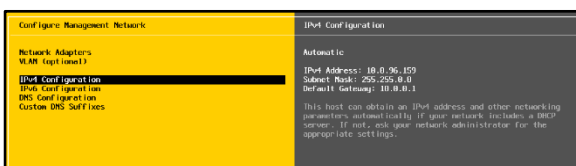


VMWare - Screenshot 10: ESXi Console

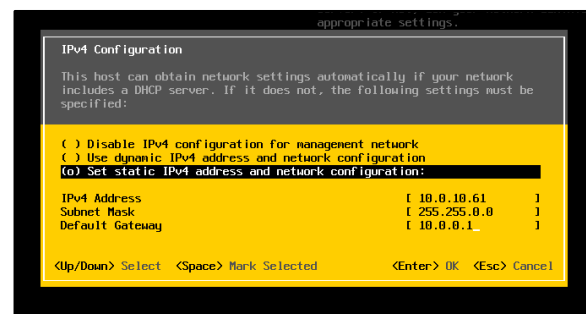


VMWare - Screenshot 11: Root Login

Once logged in, select “Configure Management Network” and then set the IP address to a static one, we were given 10.0.10.61 as our IP address for the server.

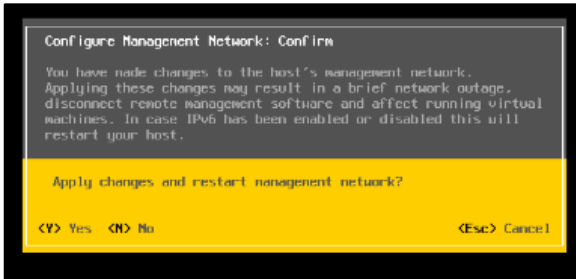


VMWare - Screenshot 12: Configure Network Management screen

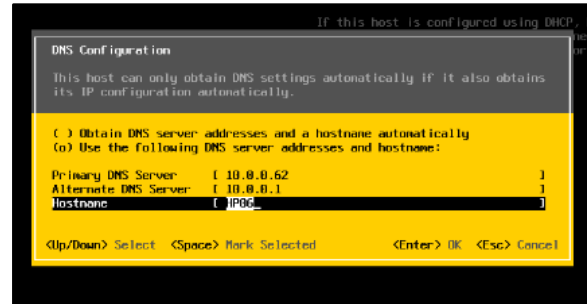


VMWare - Screenshot 13: Setting static IP address

Next confirm the configuration and to ensure the ESXi can resolve DNS Names, set the DNS Server. The reason we had two is because this primary one was set automatically and the alternate was set manually, we left it like this in case the primary one had to be set like that.



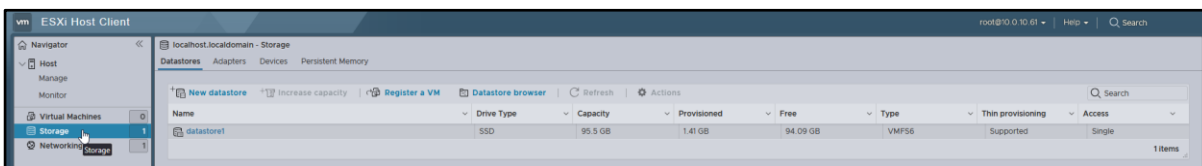
VMWare - Screenshot 14: Network Management Confirmation



VMWare - Screenshot 15: DNS Configuration

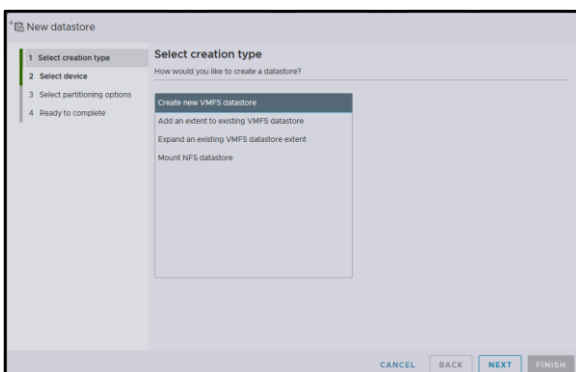
Next we need to configure our storage for the installing the VCenter Server onto.

Enter the IP address we just configured into a new browser tab and log in with our root account. On the host screen, select the storage tab and click on “New Datastore”.

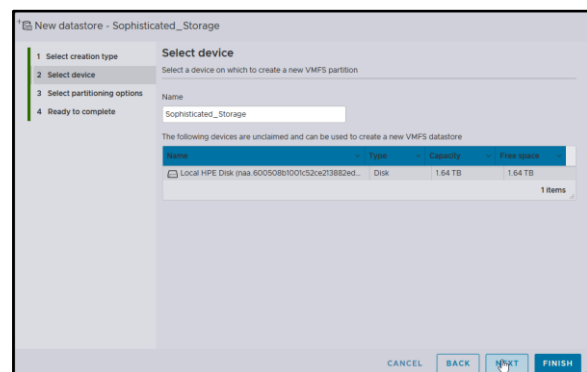


VMWare - Screenshot 16: ESXi Datastore screen

On the creation screen, select “Create new VMFS datastore”. Then select the HDD array that was setup on the server and give it the name, we chose “Sophisticated_Storage” in accordance with a Minecraft storage mod, during the partitioning step, select to use the full disk.



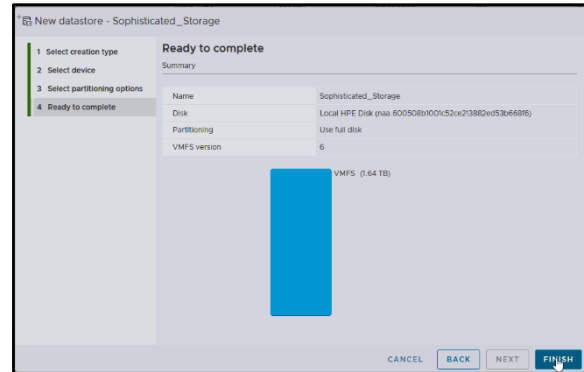
VMWare - Screenshot 17: Datastore creation type selection



VMWare - Screenshot 18: Datastore device selection



VMWare - Screenshot 19: Datastore partitioning options



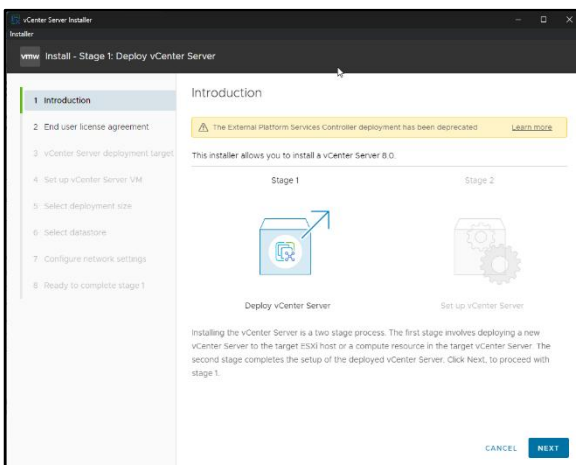
VMWare - Screenshot 20: Datastore completion screen

Now we're ready to install the VCenter Server, for managing VMs, onto the ESXi.

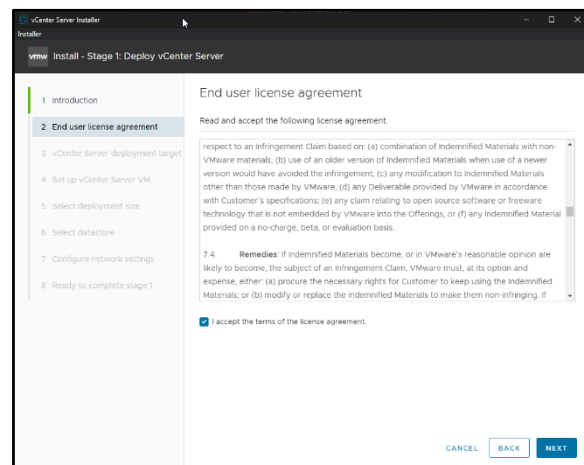
VCENTER INSTALLATION STAGE 1

Installing VCenter requires an external installation software that is mounted/exported onto your system.

Start the installer, the installation is a 2 part installation with first the deployment then the set up. We continue reading and accepting the EULA.



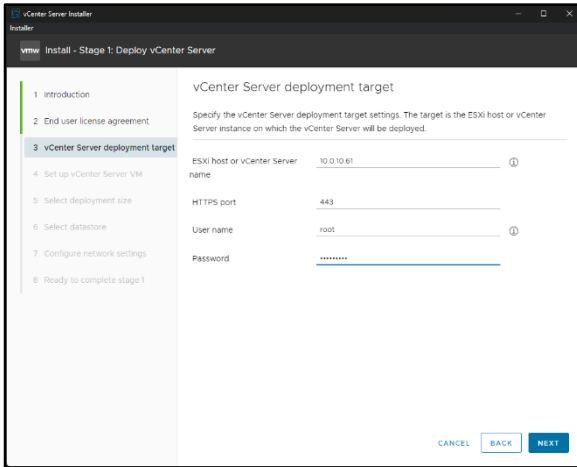
VMWare - Screenshot 21: VCenter installation start screen



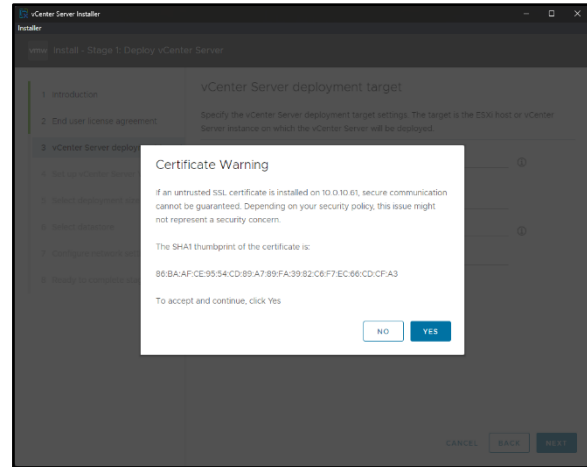
VMWare - Screenshot 22: VCenter EULA screen

Continuing onto step 3 we configure where the VCenter Server will get deployed, here we input the IP address our ESXi Server and use the root login.

You may get an SSL Certificate warning on clicking "Next", since we are in a lab environment we're aware that we don't have a certificate for HTTPS we can accept the warning.



VMWare - Screenshot 23: VCenter deployment target

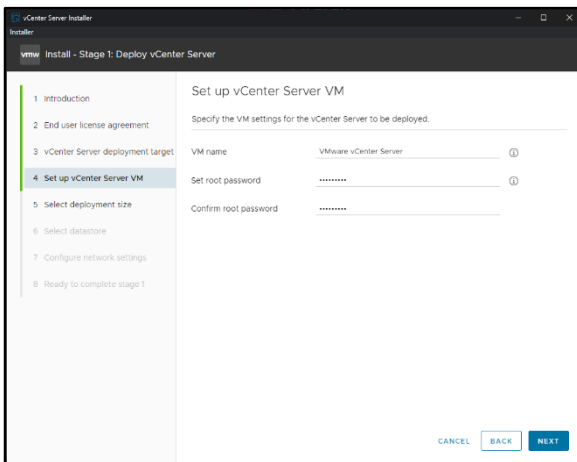


VMWare - Screenshot 24: SSL Certificate Warning

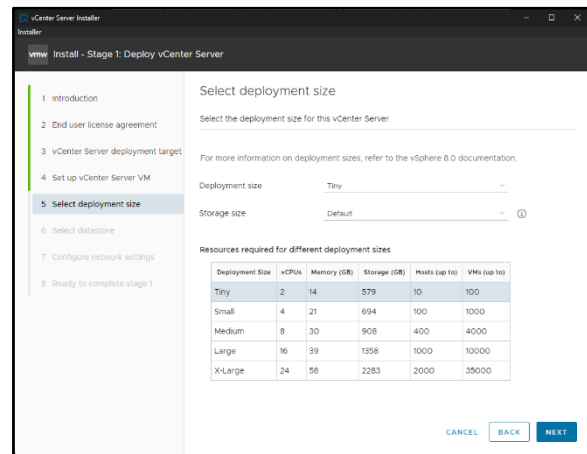
Next configure the rest of the installation, the VM Name with the root password, we chose tiny as our deployment size since that corresponded to our needs.

For the storage location select the datastore that was created before and enable Thin Disk mode, for saving up on space.

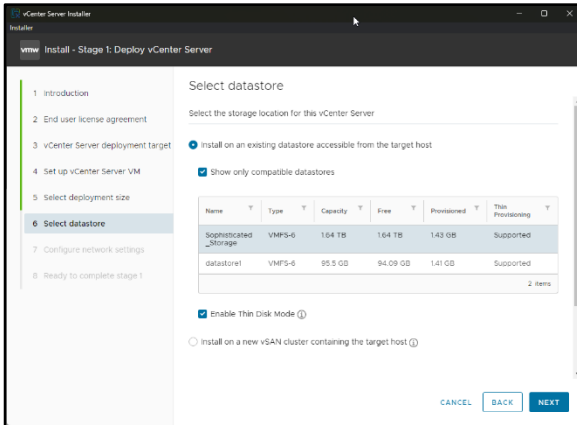
For the network settings, give it an IP address, we gave one from our given range for this project, and give the same default gateway and DNS servers as you did for the ESXi



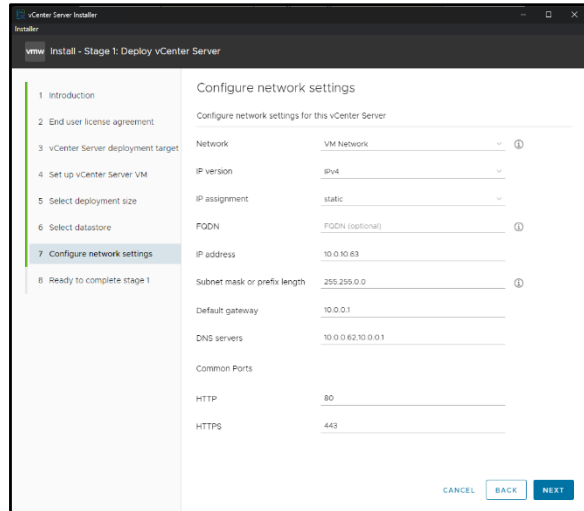
VMWare - Screenshot 25: vCenter VM settings



VMWare - Screenshot 26: vCenter deployment size

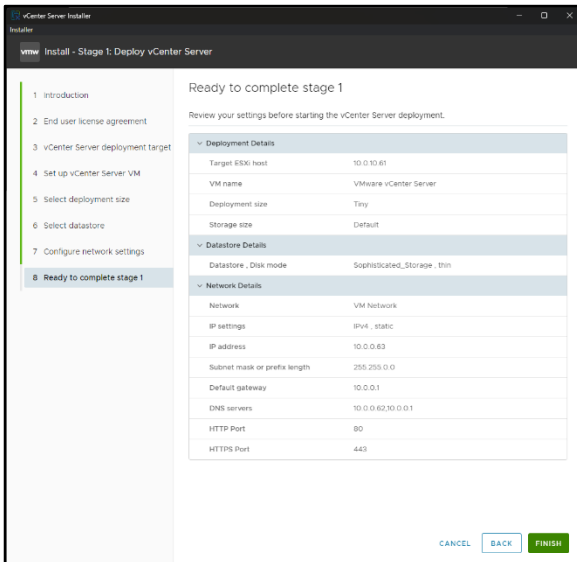


VMWare - Screenshot 27: vCenter storage location

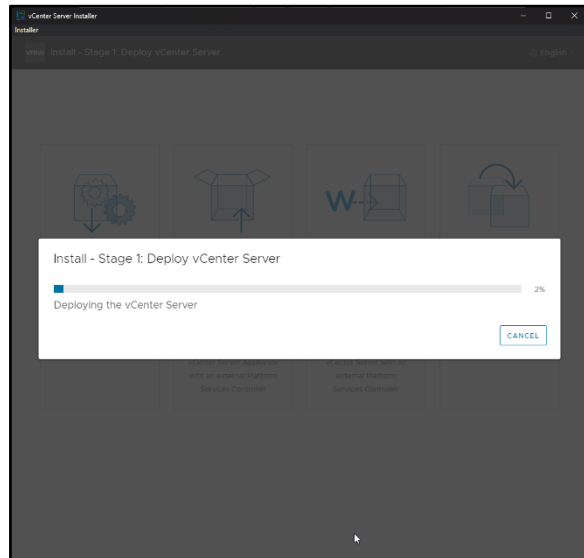


VMWare - Screenshot 28: vCenter network settings

Once that we're done we confirm that our configuration is correct and enter a long installation process.

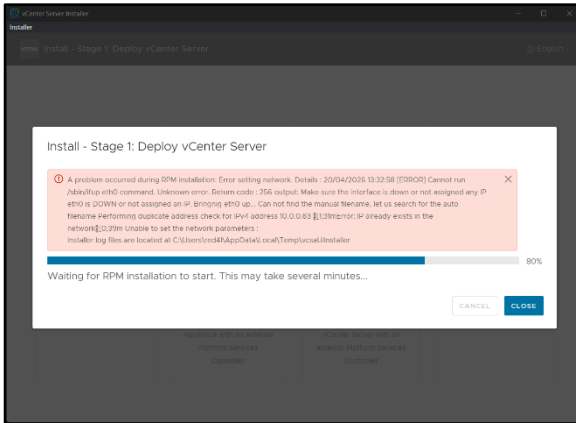


VMWare - Screenshot 29: vCenter deployment confirmation



VMWare - Screenshot 30: vCenter deployment loading screen

This step had to be redone 3 times, first time we selected the wrong deployment size, the second time, we encountered an error of the assigned IP already being in use somehow.

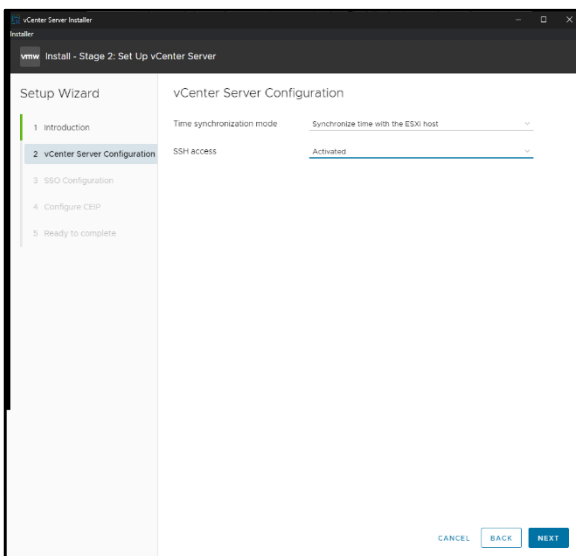
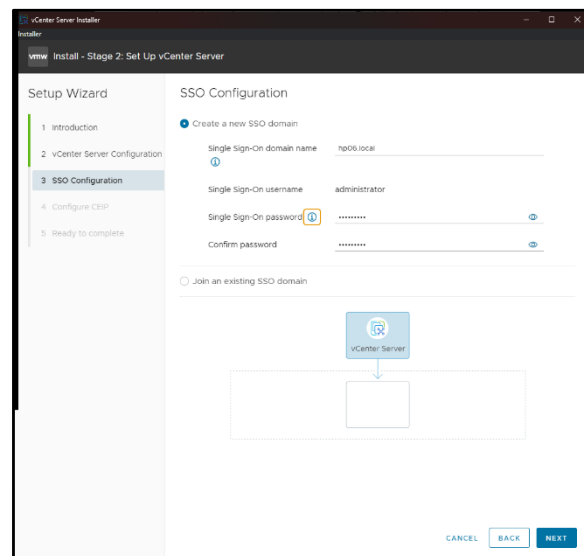
**VMWare - Screenshot 31: vCenter deployment error**

Once the deployment is completed we can move to the second stage of the vCenter setup.

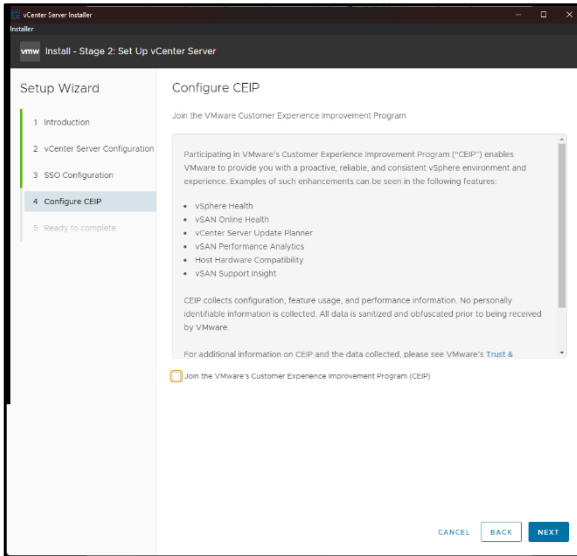
VCENTER INSTALLATION STAGE 2

The second stage is a lot quicker and less configurations to do, mainly just determining SSH access, a Single Sign-On user. So continue past the introduction screen to those screens.

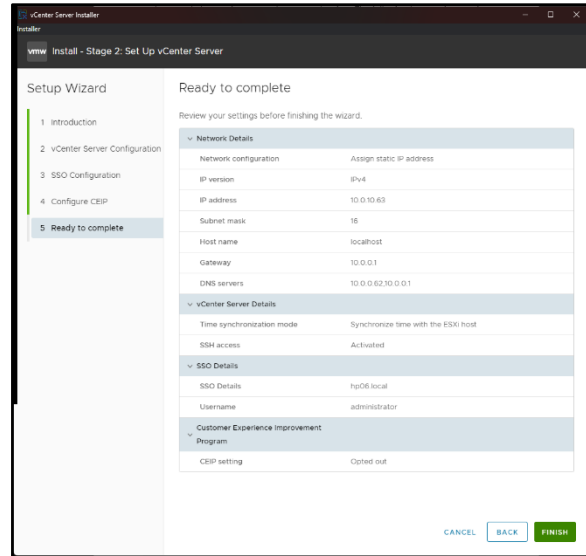
We wanted SSH access for the VCenter in case we need to do configurations within the CLI and for the SSO we used our servers name with a .local domain for simplicity.

**VMWare - Screenshot 32: vCenter Server configuration****VMWare - Screenshot 33: vCenter SSO configuration**

Ending the deployment it asks if you want to join the VMWare's Customer Experience, which we unchecked and a final confirmation screen to finish the setup.

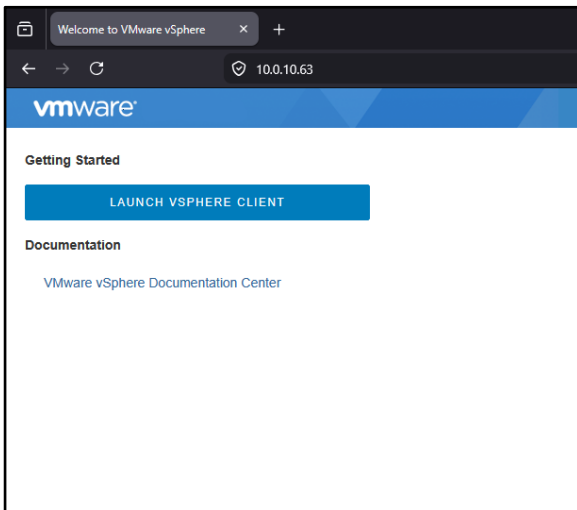


VMWare - Screenshot 34: vCenter Configure CEIP

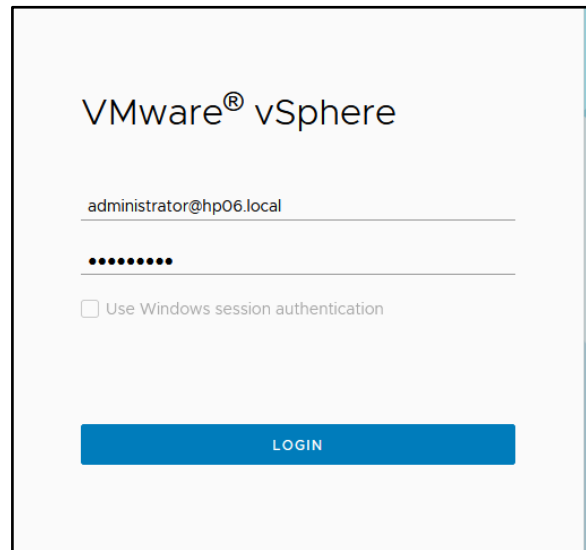


VMWare - Screenshot 35: vCenter final configuration screen

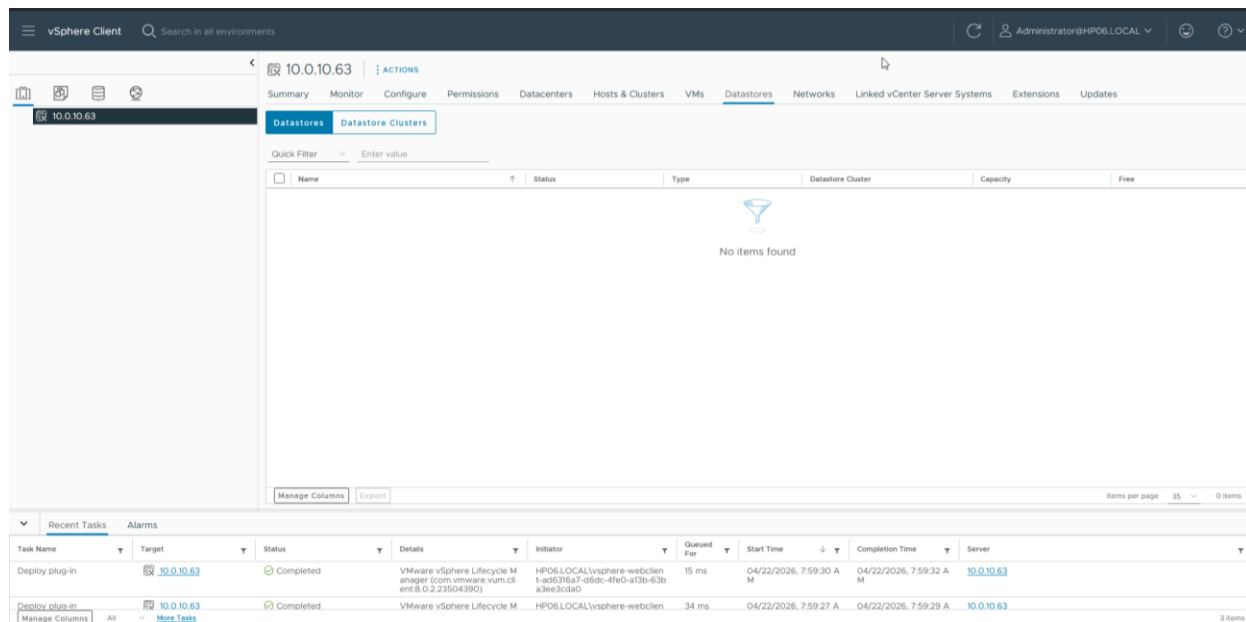
Once the server has completely installed use the IP address assigned to it (for us it was 10.0.10.63) to connect to the vSphere client and login using the SSO. And finally we'll see the main dashboard.



VMWare - Screenshot 36: vSphere getting started screen



VMWare - Screenshot 37: vSphere login



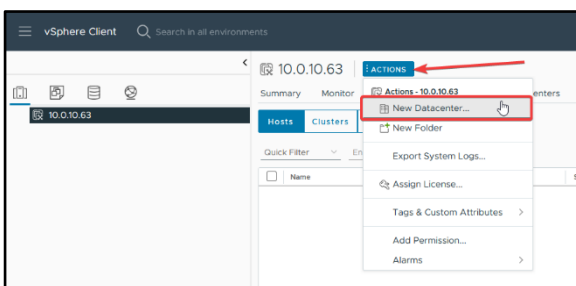
VMWare - Screenshot 38: Front page of vSphere

However this is only the vCenter server we see right now, so what still needs to be done is adding our main ESXi host for VM management.

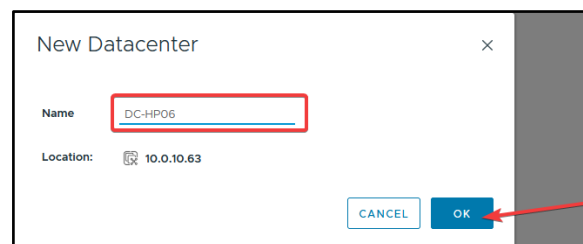
ADDING THE HOST FOR VSPHERE

First a datacentre needs to be added for the ESXi host to be put on. On the vSphere client, click on “Actions” and “New Datacenter”.

Note: For the VMWare documentation the context menu when Right-clicking the item in the inventory list is the same as the action button next to the name of the item you want to configure.

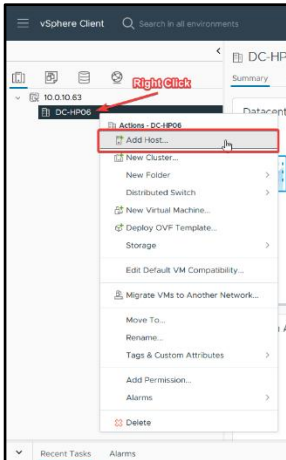


VMWare - Screenshot 39: Adding datacentre to VMWare

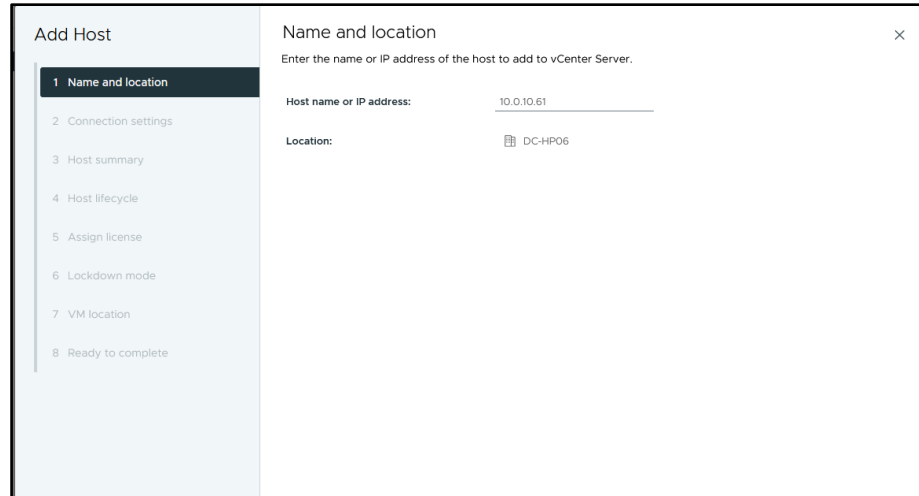


VMWare - Screenshot 40: Setting name of datacentre

Then we can add our host to the datacentre, we go into the context menu of the datacentre and click “Add Host”, on the screen we give the IP address of the host we configured the ESXi on

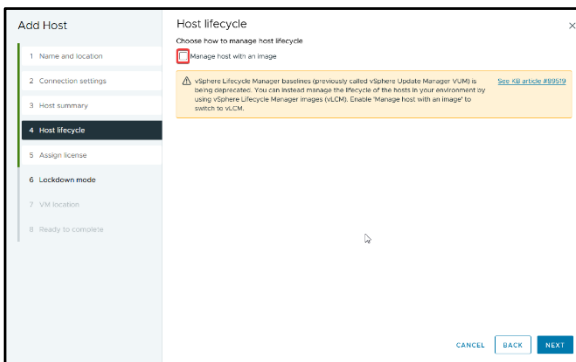


VMWare - Screenshot 41:
Adding Host

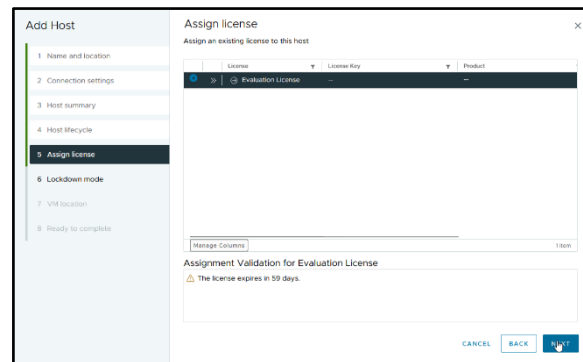


VMWare - Screenshot 42: Adding Host - Name and location

Continuing the steps of using the root login of the host and confirming the host summary, we don't want to manage the host with an image and the license is something we'll configure later, so we use the evaluation license for now.

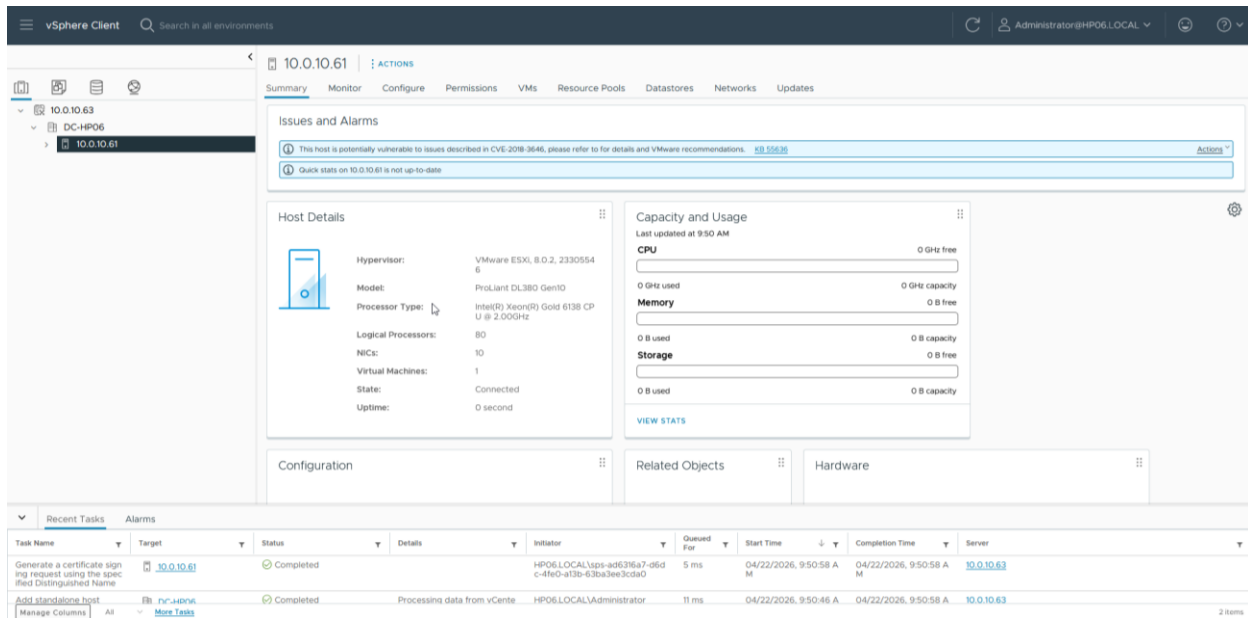


VMWare - Screenshot 43: Adding Host - Host lifecycle



VMWare - Screenshot 44: Adding Host - Assign license

Disable lockdown mode and set its location on onto the Datacentre that was created prior, confirm the last information and finish the host setup. Now we're able to create and manage VMs from the vSphere client on our ESXi hosts.

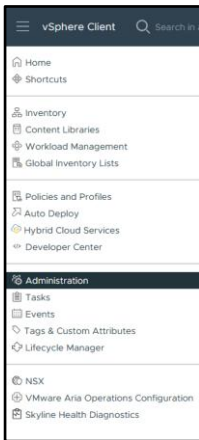


VMWare - Screenshot 45: Host screen

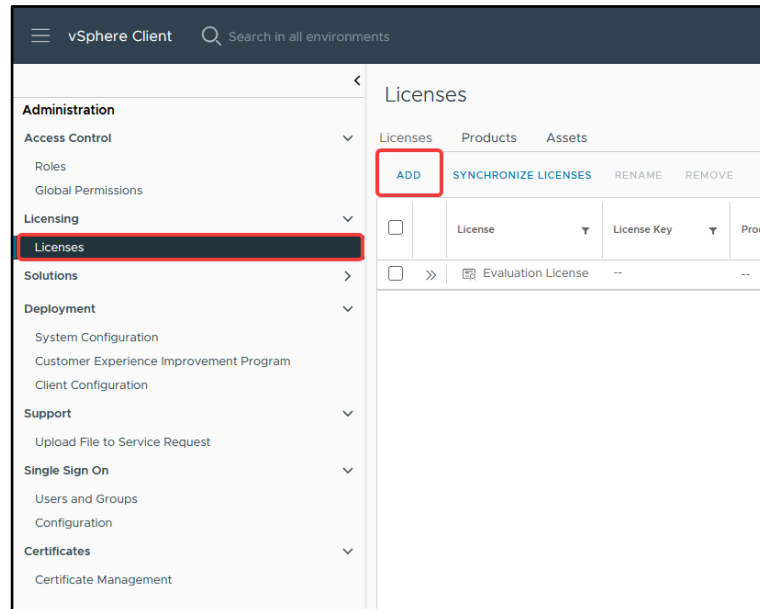
Since in our lab we have 2 ESXi hosts, to add the second host all that needs to be done is simply repeat the steps for adding a new host with our IP address ,10.0.10.81, no need for a new datacentre a vCenter Server.

ASSIGNING LICENSES

In the main page of the vSphere client go into the “Administration” tab and select “Licenses” under the Licensing tab. From there we add our licenses for the ESXi server and the vCenter server.

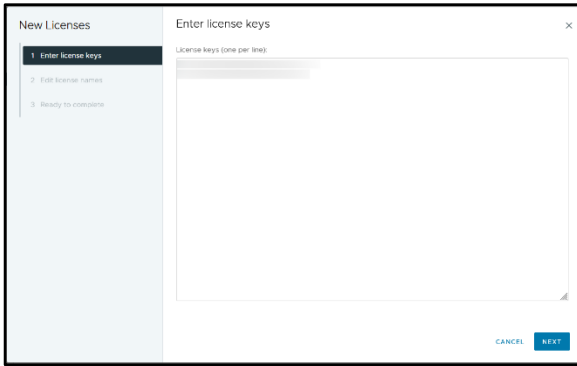


VMWare - Screenshot 46: Going to Administration for licensing

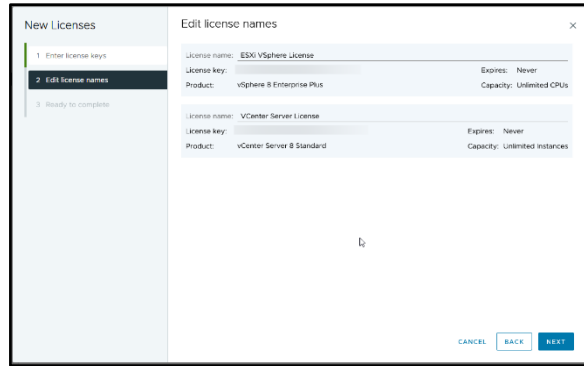


VMWare - Screenshot 47: Adding licenses

In the next screen, insert your 2 licenses and continue to naming them for better overview of the licenses.

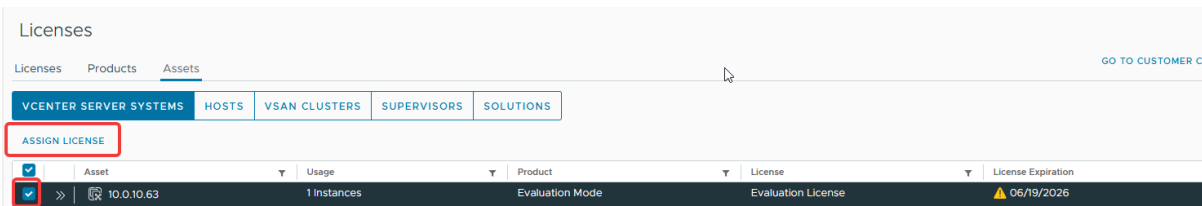


VMWare - Screenshot 48: Entering license keys

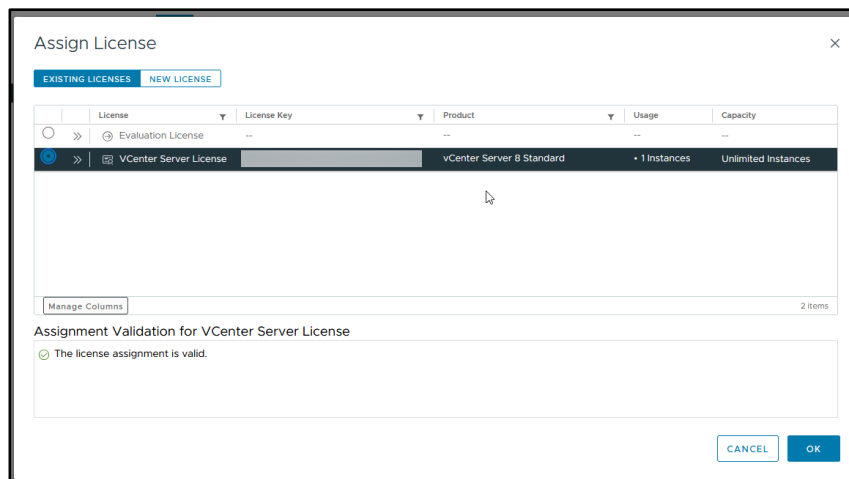


VMWare - Screenshot 49: Editing license names

First we'll assign the license for the vCenter server, for that in the Licenses tab, select your vCenter server and assign the license.

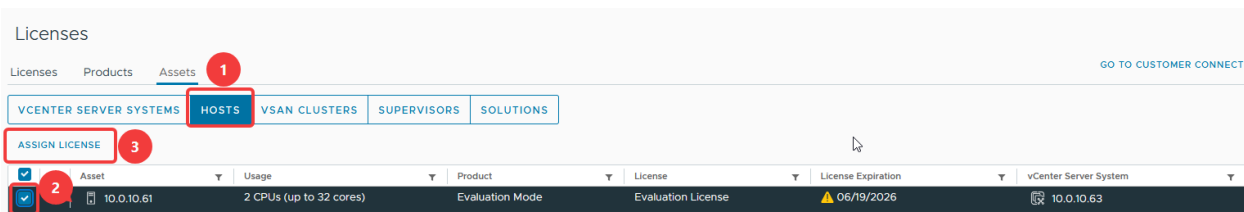


VMWare - Screenshot 50: Selecting vCenter server for licensing



VMWare - Screenshot 51: Assigning vCenter server license

Assigning the ESXi host license works the same just now in the Hosts subtab of the Licenses.



VMWare - Screenshot 52: Selecting ESXi host for licensing

With all of that done, the setup of VMWare and all its components is finished and we can finally start creating Virtual Machines and managing them.

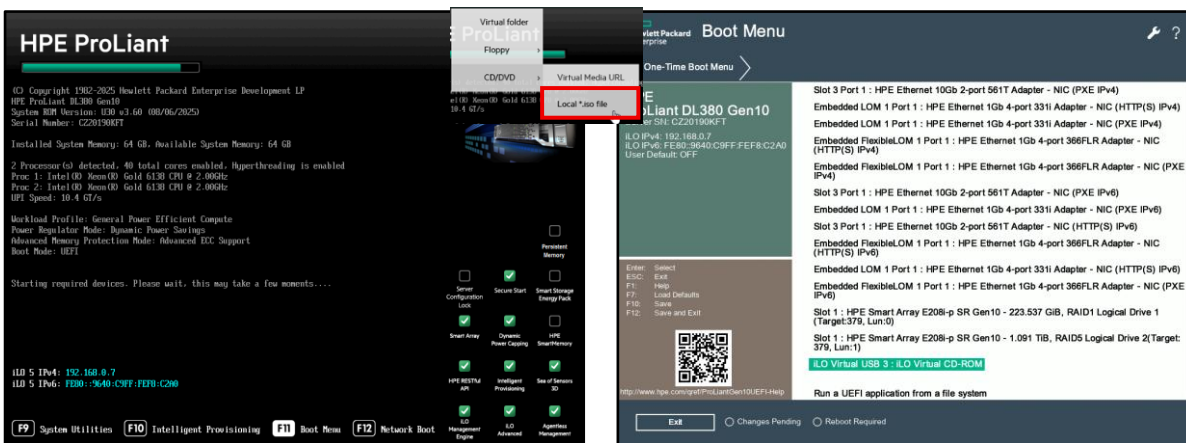
PROXMOX INSTALLATION



Here we will install Proxmox also via ILO. Make sure that you have the Proxmox ISO ready, it can be downloaded here: <https://www.proxmox.com/en/downloads>

In this document we will be using “Proxmox VE 9.1 ISO Installer”. It is worth noting that there may be future changes and that the documentation may not match one to one.

Go into the virtual console and mount the ISO. Start the server and go into the one time boot menu. Boot from the ILO Virtual Media.



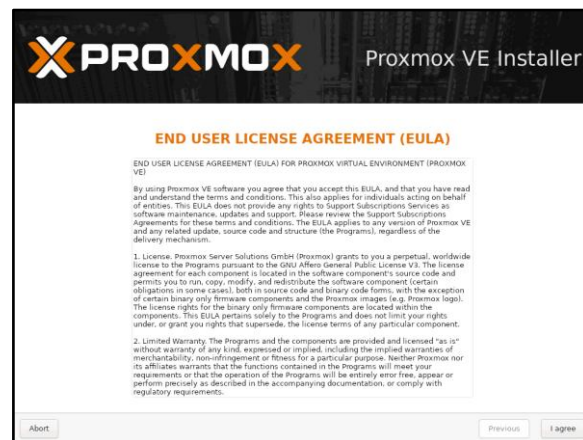
Proxmox - Screenshot 1 Startup screen

Proxmox - Screenshot 2 Boot menu with media selected

After you booted the ISO, you will see a menu with multiple boot options. Select graphical. After that you will be redirected to the EULA, read it and agree.



Proxmox - Screenshot 3 Install options

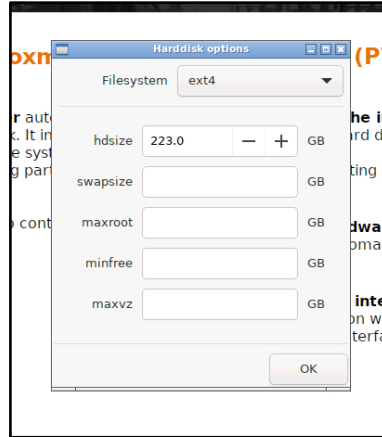


Proxmox - Screenshot 4 End User License Agreement

Next it will ask where the OS should be installed. Choose the SSD RAID. In the options make sure that all the space is being used and that ext4 is used as the filesystem.



Proxmox - Screenshot 5 Storage selection



Proxmox - Screenshot 6 Storage options

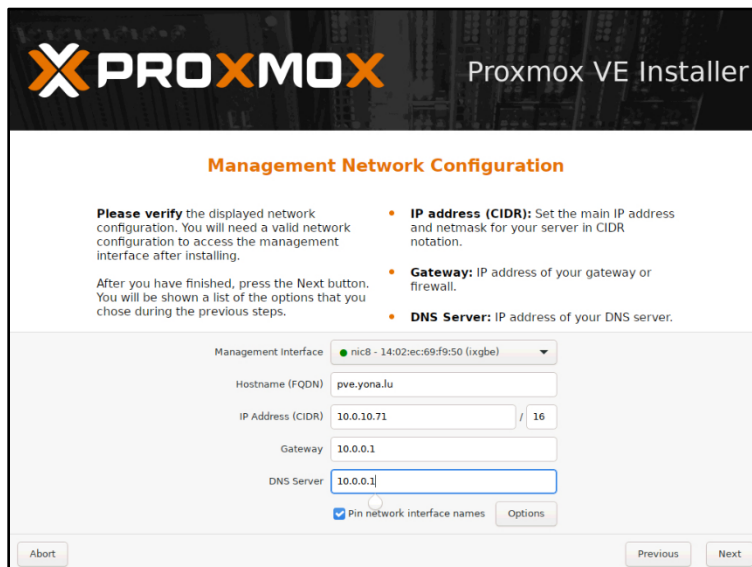
On the next screens it will ask for a few locale settings. Make sure that at least the keyboard is correct, as later we will only have a CLI and the password might be entered wrong as well. On the last screen before the summary it will ask for root credentials.



Proxmox - Screenshot 7 Locale settings



Proxmox - Screenshot 8 Root user details



Proxmox - Screenshot 9 Static network settings

Now it is time to enter all the networking details including: The network interface, hostname, IP address, Gateway and DNS Server. These are static settings, Proxmox does not work with DHCP.

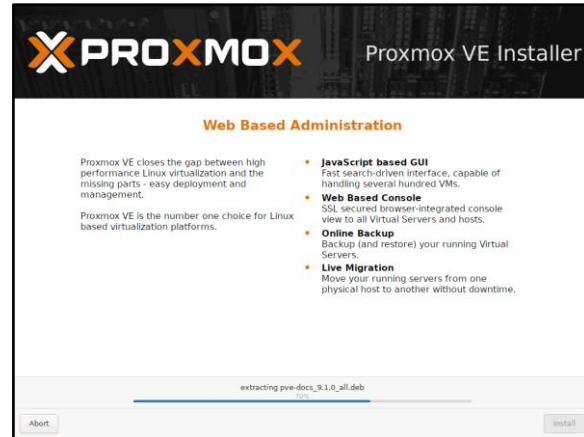
Make sure to use the correct network interface. If a network connection has been detected it should display a green dot.

I prefer to have the network interface name pinned.

Once all the details have been specified, it will show the summary page. If everything is correct, click "Install". This will take a while.

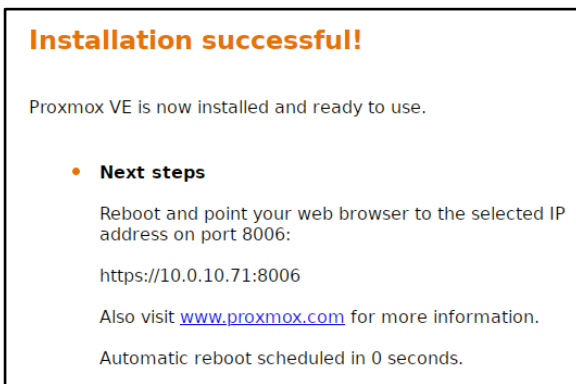


Proxmox - Screenshot 10 Setup Summary

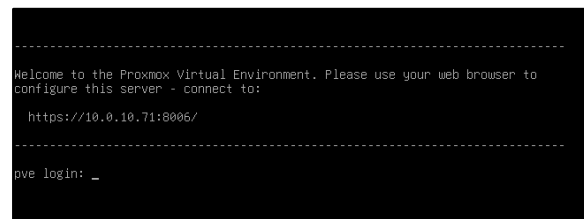


Proxmox - Screenshot 11 Installing and copying to disk

Now the installation should be successful, after a reboot it will directly boot into Proxmox and display the CLI login page

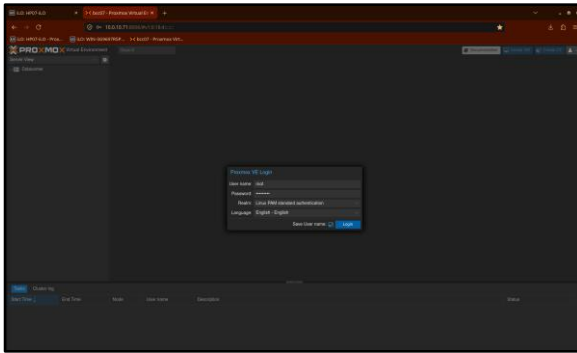


Proxmox - Screenshot 12 Success screen

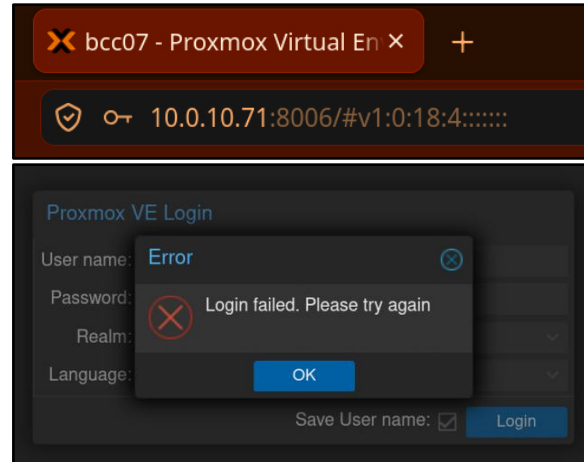


Proxmox - Screenshot 13 CLI after boot

After Proxmox has finished booting, we can navigate to it via the URL provided in the CLI. Now you can login. We had a "Login failed. Please try again" message, as someone else took our assigned IP address and we were not connected to the correct machine. You can see that the hostname is different. We quickly changed the IP address in the **/etc/network/interfaces** file.

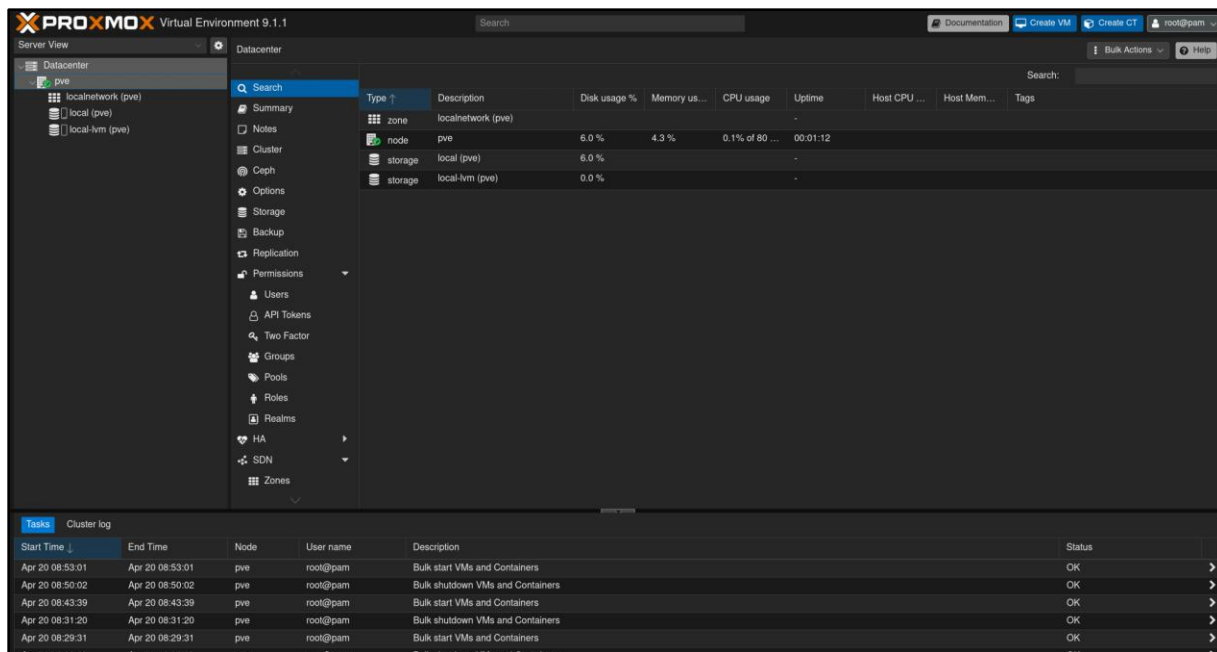


Proxmox - Screenshot 14 Login Page



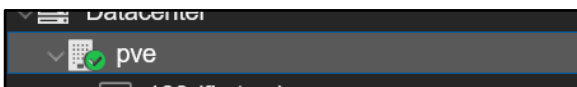
Proxmox - Screenshot 15 Login failed due to IP being taken

This is Proxmox's Web Interface.

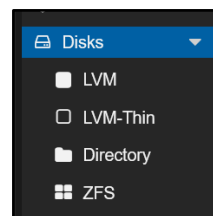


Proxmox - Screenshot 16 Front page of the panel

Now we should configure the HDD array so that we can store data on it as only the SSD is shown. Click on the server and go to disks.



Proxmox - Screenshot 17 Server selected



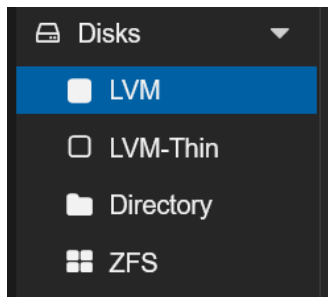
Proxmox - Screenshot 18 Server Disks selected

Now we can see our HDD array listed among all connected disks.

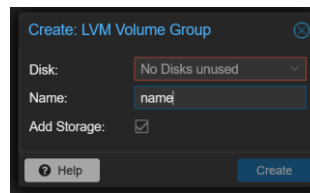
Device	Type	Usage	Size	GPT	Model	Serial	S.M.A.R.T.	M...	Wearout
/dev/sda	SSD	partitions	240.02 GB	Yes	LOGICAL_VOLUME	600508b1001c7ce91b08e2f...	OK	No	N/A
/dev/sda1	partition	BIOS boot	1.03 MB	Yes				No	N/A
/dev/sda2	partition	EFI	1.07 GB	Yes				Yes	N/A
/dev/sda3	partition	LVM	238.37 GB	Yes				No	N/A
/dev/sdc	unknown	LVM	1.20 TB	No	LOGICAL_VOLUME	600508b1001c4c605bf1882...	OK	No	N/A

Proxmox - Screenshot 19 All Drives listed for the Proxmox server

Then go to LVM and create a new LVM volume group for the HDDs. We named it “hdd” on the first Proxmox server.



Proxmox - Screenshot 20 LVM selected under Disks



Proxmox - Screenshot 21 Creating a new LVM Group

Here you can see that we have the newly created volume group.

Name	Number of LVs	Assigned to LVs	Size	Free
hdd	2	10%	1.20 TB	1.08 TB
/dev/sdc		10%	1.20 TB	1.08 TB
pve	8	93%	238.37 GB	17.18 GB
/dev/sda3		93%	238.37 GB	17.18 GB

Proxmox - Screenshot 22 All normal LVM groups

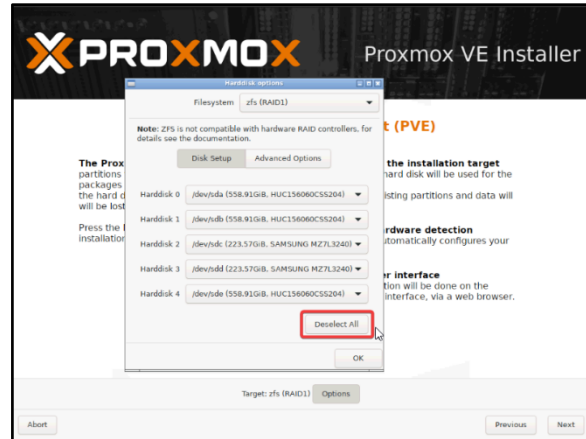
ADDING THE 2ND HOST

For the installation of the second host, on our server itself, we wiped away the existing RAID Array that was set on top of the Disks, this is different from server to server, so we won't cover it here, but now in the installation of the second host, we create a RAID 1 across the 2 SSDs we have.

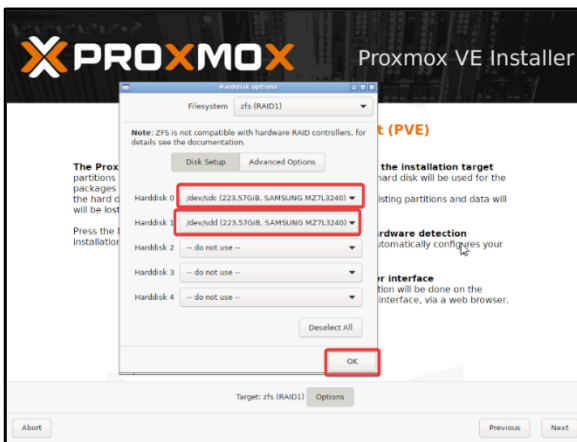
So in the installation of Proxmox, for the installation location, we choose to store it in a ZFS RAID 1. In the Options deselect the drives and only select the SSD drives, then you can continue on with the installation as normal.



Proxmox - Screenshot 23: Selecting harddisk installation option

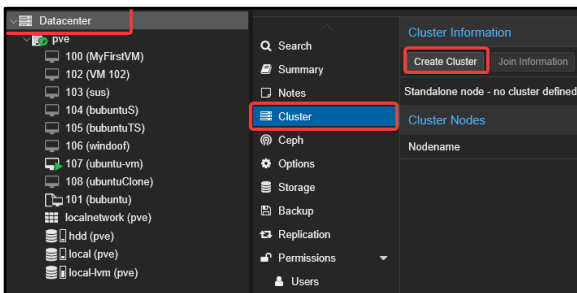


Proxmox - Screenshot 24: Deselecting harddisks for RAID 1 installation

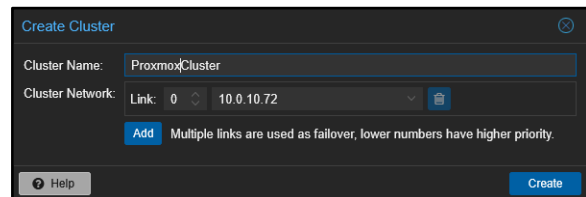


Proxmox - Screenshot 25: Selecting SSDs for RAID 1 installation

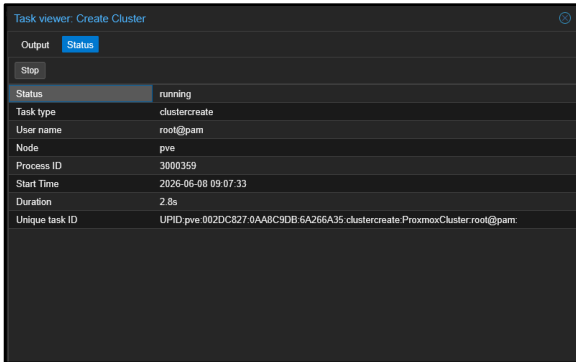
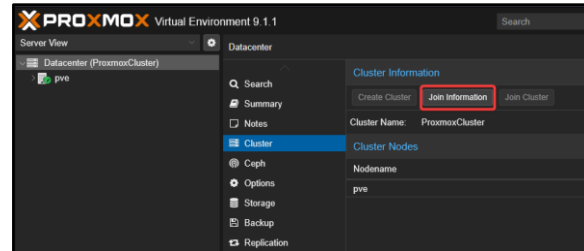
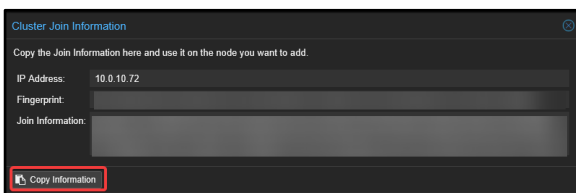
With Proxmox installed on the second host, we join it to the cluster. The cluster is created on the first node, from Datacenter → Cluster → “Create Cluster”; this sets up corosync, the service that keeps the nodes in sync. For our lab corosync runs over the single existing LAN, which is fine here – a dedicated cluster network is something the Clustering/HA teams go into, not us. For our second host to join, on the first node we open Datacenter → Cluster → “Join Information” and copy the join string it shows.



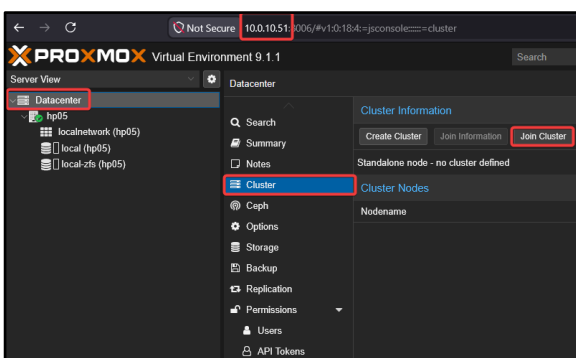
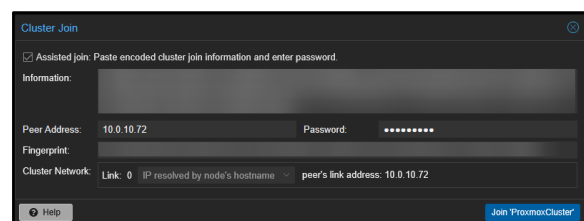
Proxmox - Screenshot 26: Creating the Cluster

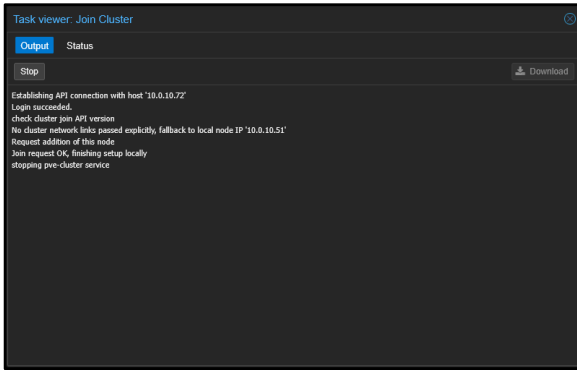


Proxmox - Screenshot 27: Naming and adding the cluster network

**Proxmox - Screenshot 28: Cluster creation status****Proxmox - Screenshot 29: Getting the Join information****Proxmox - Screenshot 30: Copying the join information**

On the second host, we go to Datacenter → Cluster → “Join Cluster”, paste that information, enter the first node’s root password (the existing cluster node), and confirm. It’s important that the joining host is empty, the join replaces the node’s /etc/pve contents with the cluster’s, so any VMs or storage defined on it beforehand would be lost. After a moment HP05 appears in the cluster and both nodes are manageable from a single web interface.

**Proxmox - Screenshot 31: Joining the Cluster****Proxmox - Screenshot 32: Entering the join information**



Proxmox - Screenshot 33: Joining Cluster status

Now you'll be able to see both hosts in the cluster, our first host PVE (HP07) and our second host HP05.

COMPARISON



Both installs start the same with booting the installer. VMWare asks less questions and takes the entire disk. It also does not ask for networking information, hostname or E-Mail, which may seem simpler at first, but they still have to be configured later. This makes Proxmox faster as all the necessary settings are configured before first boot.

VMWare has an interactive TUI, where Proxmox only has a CLI.

VMWare's ISO file is also smaller compared to Proxmox (ignoring vSphere).

Looking at both Webinterfaces it is clear that Proxmox has more features than VMWare out of the Box. VMWare still needs vSphere for that.

VIRTUAL MACHINE CREATION THROUGH GUI AND CLI

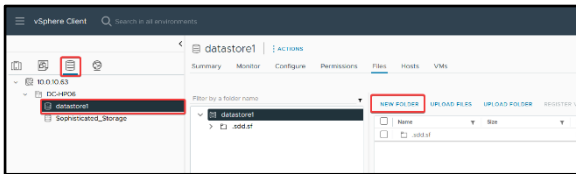
VMWARE



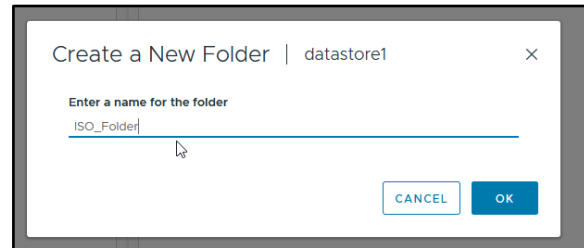
The first VM created with the GUI on VMWare will be an Ubuntu Server. After that we'll create a Windows 10 Client with the CLI.

CREATING A VM THROUGH GUI

Before creating the VMs, the ISOs for the OS needs to be uploaded first. For that head to the datastore on the vSphere client and create a new Folder for storing the ISO Files.

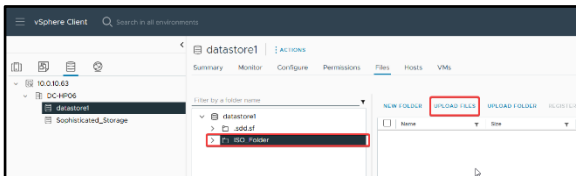


VMWare - Screenshot 53: ISO Folder creation page

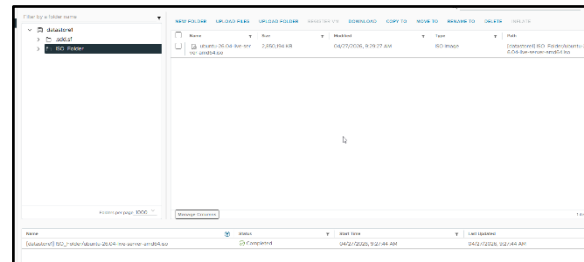


VMWare - Screenshot 54: ISO Folder name creation page

Next, within the new folder, select “Upload Files” and from our local machine select the ISO file we want to upload. After the upload we get a confirmation that it was uploaded underneath.

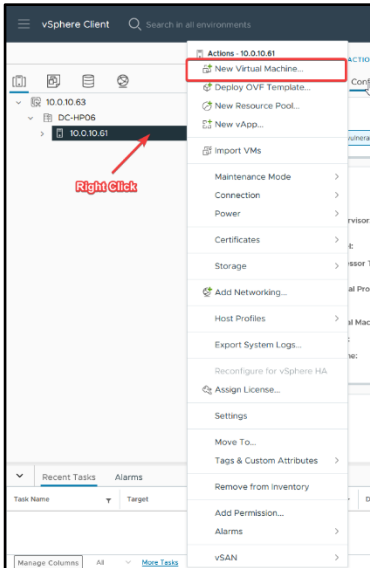


VMWare - Screenshot 55: Uploading the ISO into the ISO Folder

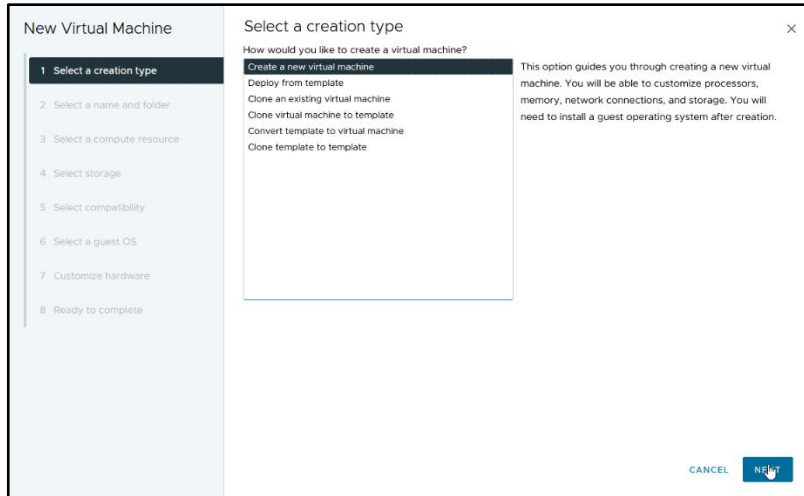


VMWare - Screenshot 56: After ISO upload

Next head back to the VM Management screen to create the VM. Use the context menu to enter a new screen for creating a new Virtual machine on the host.

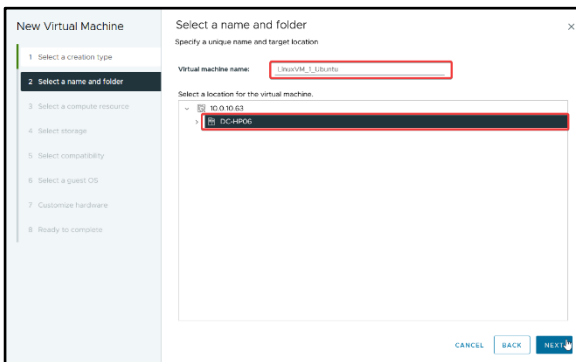


VMWare - Screenshot 57: New Virtual machine

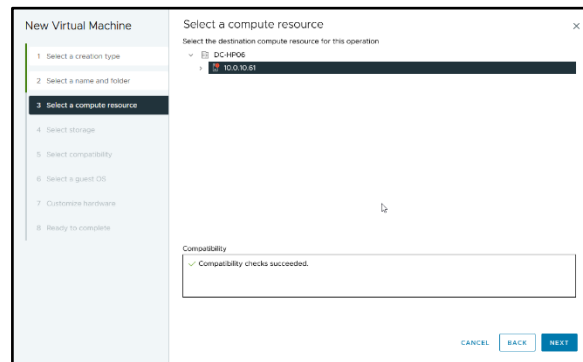


VMWare - Screenshot 58: VM creation - selecting creation type

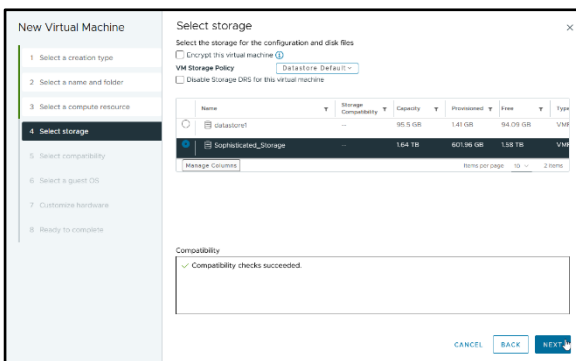
Next set the name of the VM and under which folder, so datacentre, it should be put in along with the compute resource, so the ESXi host. Afterwards select the storage where the VM should be stored onto.



VMWare - Screenshot 59: VM creation - selecting datacentre



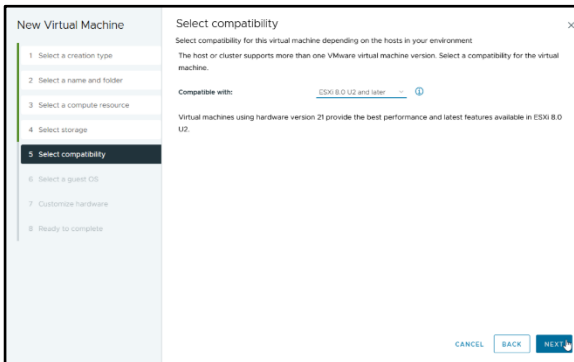
VMWare - Screenshot 60: VM Creation – selecting compute resource



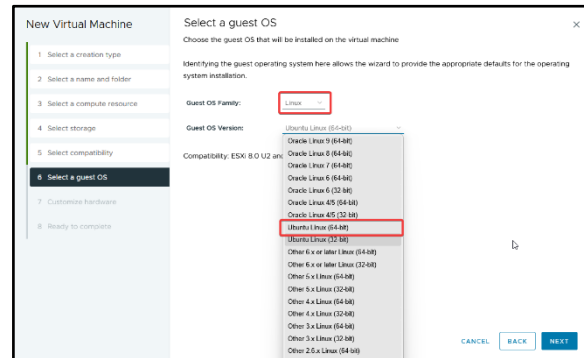
VMWare - Screenshot 61: VM Creation - selecting storage

Next are the options for the VM itself, like compatibility, guest OS and virtual hardware. The guest OS isn't the image itself but sets defaults for setting up the VM. For the hardware, we chose 2 vCPUs, 2 GB memory and a

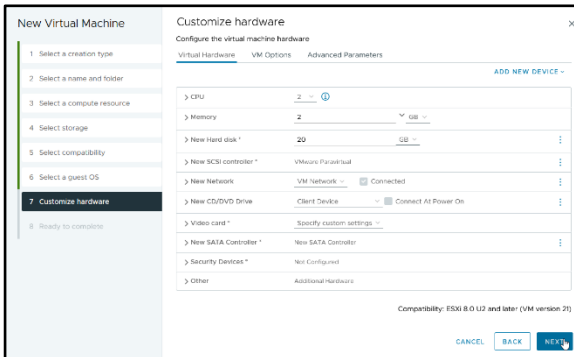
20 GB hard disk, an important setting was forgotten, which was setting the disk drive to use our datastore ISO file, so we'll change that later.



VMWare - Screenshot 62: VM Creation - selecting compatibility



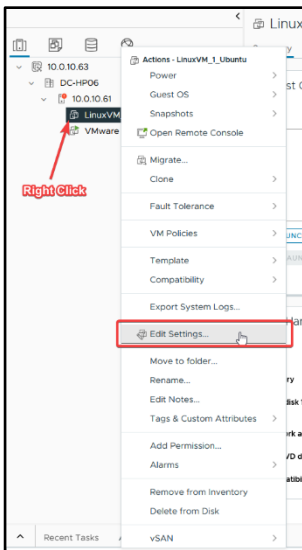
VMWare - Screenshot 63: VM Creation - selecting Guest OS



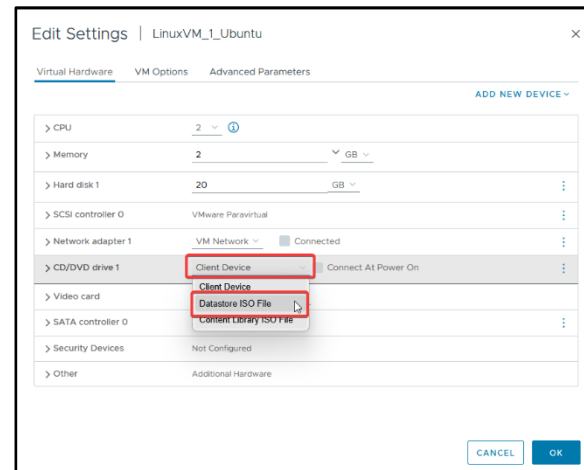
VMWare - Screenshot 64: VM Creation - customising hardware

To change the used OS, head into the actions of the VM and choose “Edit Settings” from there navigate to the CD/DVD drive element of the VM and change the setting from Client Device to Datastore ISO file, then choose the ISO necessary for installing an Ubuntu Server.

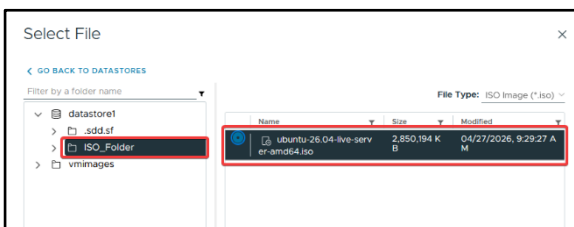
Make sure the setting “Connect at Power On” is selected for actually booting into the installation media.



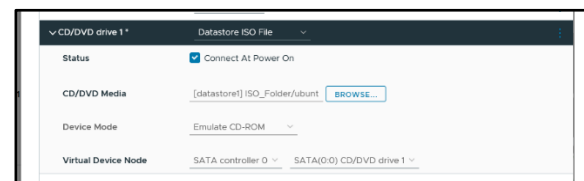
VMWare - Screenshot 65: LinuxVM_1 edit settings for ISO



VMWare - Screenshot 66: Changing ISO source for LinuxVM_1



VMWare - Screenshot 67: Selecting ISO file



VMWare - Screenshot 68: Connect at Power On

With that the creation of a Virtual Machine is finished for ESXi through the GUI is finished and we can move onto creating a VM with the CLI of ESXi.

CREATING A VM THROUGH CLI

First SSH onto the ESXi host itself, not the vSphere client. We'll be using the ESXi's vim-cmd for creating the VM. This configuration through this way is fairly unintuitive and may require a few reconfigurations to get right the first time.

vim-cmd is a set of commands for managing ESXi through cli. Commands for managing VMs is under vmxvc/.

```

1 root@10.0.10.61 + 
root@10.0.10.61:22

[root@HP06:~] vim-cmd vmsvc/
Commands available under vmsvc/:
acquiremsticket          get.snapshotinfo
acquireticket            get.spaceNeededForConsolidation
createdummyvm            get.summary
destroy                  get.tasklist
device.connection        getallvms
device.connusbdev        gethostconstraints
device.ctrladd           message
device.ctrlremove       power.getstate
device.disconnectusbdev  power.hibernatem
device.diskadd           power.off
device.diskaddexisting   power.on
device.diskextend        power.reboot
device.diskremove        power.reset
device.getdevices        power.shutdown
device.nvdimmmadd        power.suspend
device.nvdimmmremove     power.suspendResume
device.toolsSyncSet      queryftcompat
devices.createencl       reload
get.capability           setscreenres
get.config               snapshot.create
get.config.cpuidmask     snapshot.dumpoption
get.configoption         snapshot.get
get.datastores           snapshot.remove
get.disabledmethods      snapshot.removeall
get.environment          snapshot.revert
get.filelayout           snapshot.setoption
get.filelayoutex         tools.cancelinstall
get.guest                tools.install
get.guestheartbeatStatus tools.upgrade
get.managedentitystatus  unregister
get.networks             upgrade
get.runtime

[root@HP06:~] vim-cmd vmsvc/createdummyvm
Insufficient arguments.
Usage: createdummyvm vm_name datastore_path [hw_version]

Create a pre-configured dummy vm.

[root@HP06:~]

```

VMWare - Screenshot 69: vim-cmd commands for managing VMs

First create a dummy VM, so basically an empty VM with no configurations yet, using the command.

`vim-cmd vmsvc/createdummyvm "Windows_VM_1" "[Sophisticated Storage]"`

This will return the id of the VM. Here it was 10.

Now to manage the configurations of the VM, all the configurations are within vmx configuration file, which needs to be edited manually. Use the command:

`vi /vmfs/volumes/Sophisticated_Storage/Windows_VM_1/Windows_VM_1.vmx`

Insert/Edit following configurations:

```

bios.bootOrder = "cdrom,hdd"
ethernet0.virtualDev = "vmxnet3"
ethernet0.networkName = "VM Network"
ethernet0.addressType = "generated"
ethernet0.wakeOnPckRcv = "FALSE"
ide0:0.present = "TRUE"
ide0:0.fileName = "Windows_VM_1.vmdk"
ide0:0.deviceType = "disk "
ide1:0.present = "TRUE"

```



```
ide1:0.deviceType = "cdrom-image"  
ide1:0.filename = "/vmfs/volumes/datastore1/ISO_Folder/WindowsClientEnterprise.iso"  
ide1:0.startconnected = "TRUE"  
numvcpus = "4"  
cupid.coresPerSocket = "2"  
memSize = "8192"  
guestOS = "windows11-64"  
scsi0.virtualDev = "pvscsi"
```

With the file configured like that, a few more commands are necessary.

```
vim-cmd vmsvc/reload 10
```

This will reload the VM with all of it's new configurations

Before being able to use the Windows 11 client we need to increase the size of the disk using:

```
vmkfstools -X 64G /vmfs/volumes/Sophisticated_Storage/Windows_VM_1.vmdk
```

Then now we can power on the VM with:

```
vim-cmd vmsvc/power.on 10
```

With that you'll have a running windows 11 client through the CLI

PROXMOX

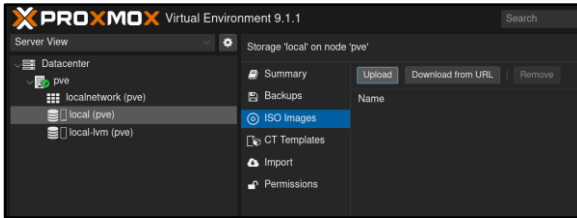


CREATING A VM WITH GUI

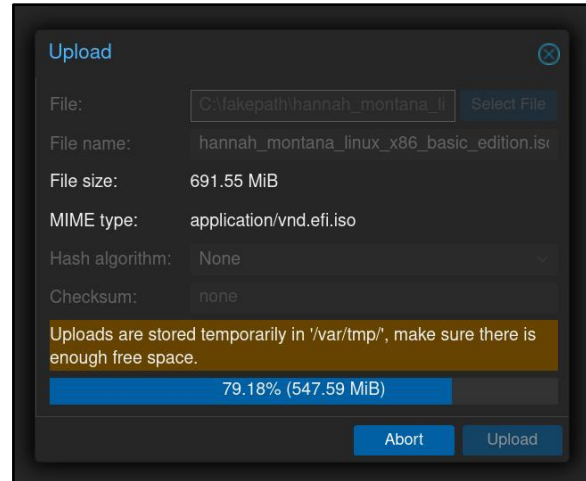
UBUNTU VM

The first VM will be running a **supported** OS(Ubuntu Server). This is a good test to see if Proxmox is able to run the OS with or without changing options. After that we will install a Tiny 10 Windows Client Server via the GUI. Then we will create a VM via the CLI.

First we have to select a storage device that is capable of storing ISOs. We then upload our legacy OS via the ISO. I would suggest to upload all ISO images needed as it simplifies the process later. You can also download it via URL and validate it via hash, which is recommended.

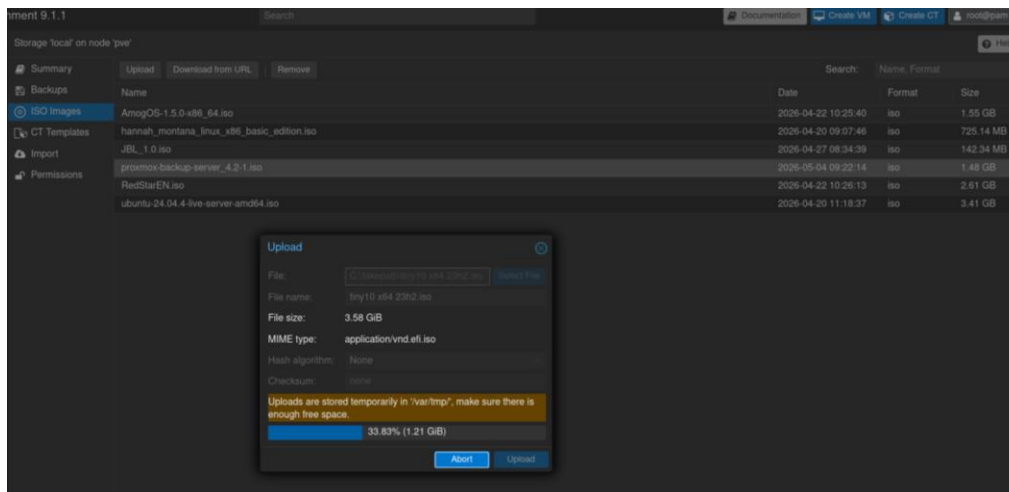


Proxmox - Screenshot 34 ISO Image store on SSD



Proxmox - Screenshot 35 Uploading first ISO

As previously said, we will upload all our ISOs for convenience later. It is worth noting the storage name and ISO name for later.



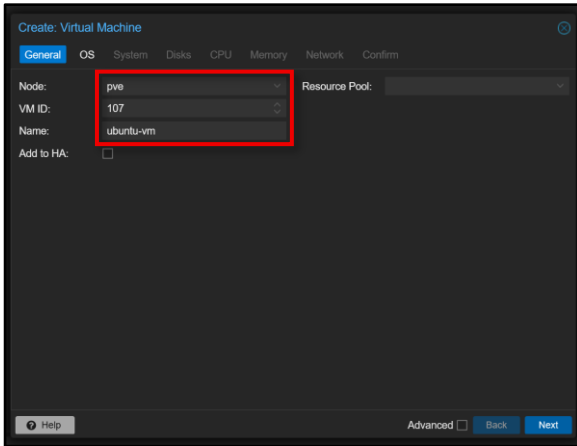
Proxmox - Screenshot 36 Uploading all needed ISOs to work with later

After the ISOs are uploaded, click on “Create VM at the top right corner of the interface.”

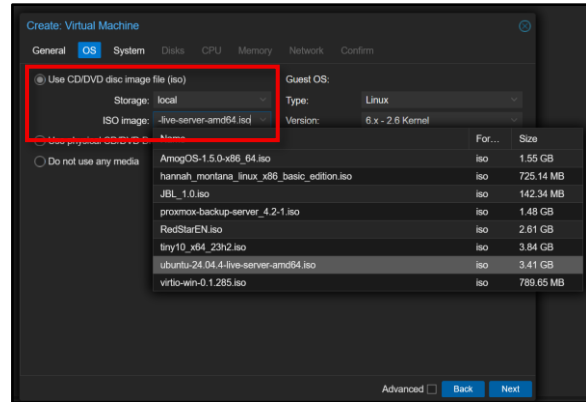


Proxmox - Screenshot 37 Create VM button on the top right

This will open a new dialog box that will ask all questions needed to create the VM. First it will ask where to physically install, the ID and the name. As we only have one server we will not add it to HA or a Resource Pool. After configuring the general settings click on next. On the OS page select the storage where the ISO has been stored and select the OS you want to install. Make sure the Type and version is correctly selected.

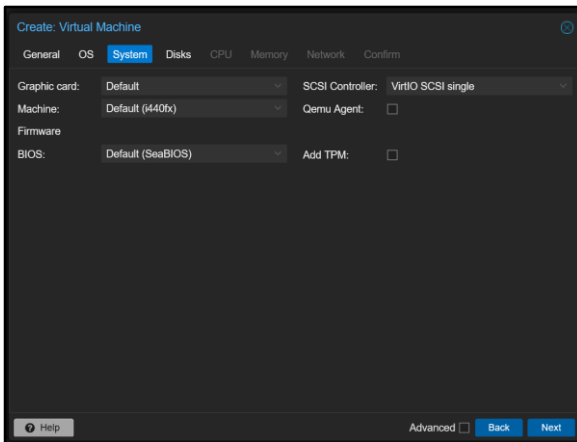


Proxmox - Screenshot 38 General settings for VM

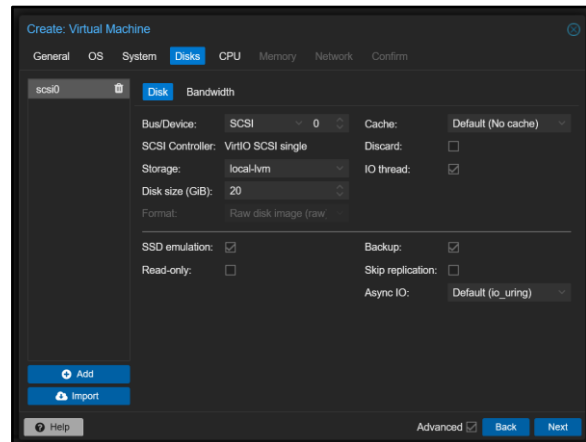


Proxmox - Screenshot 39 OS Selection for VM

On the system screen you can leave the defaults unless otherwise desired. On the disks screen at at least one boot disk. We will put it on the SSD. We will give it 20GB of space as we don't need more right now. I enabled the advanced settings to be able to enable SSD emulation so that the VM can schedule disk tasks better.

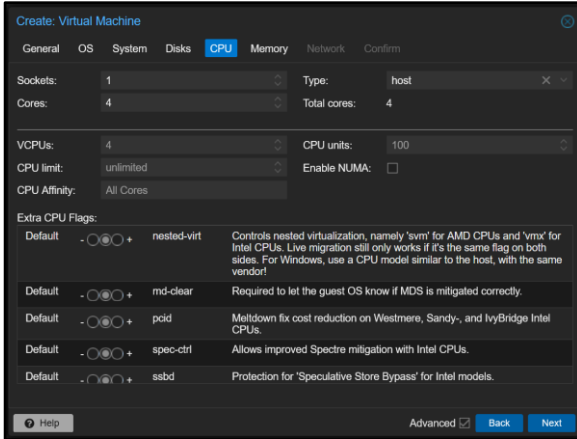


Proxmox - Screenshot 40 System settings for VM

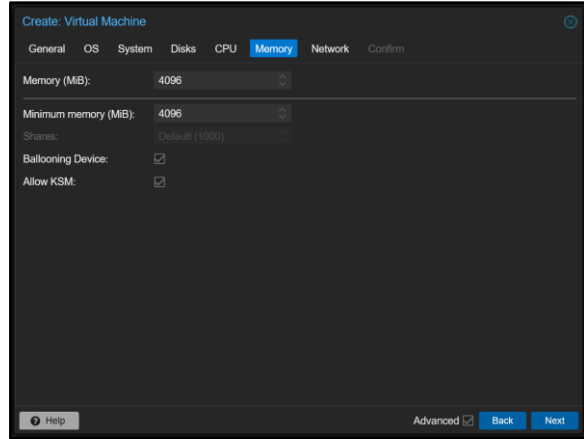


Proxmox - Screenshot 41 Disk settings for VM

On the CPU screen we chose the “host” type so that the VM can see all CPU features. This usually gives more performance. We will give it only four cores on one CPU, so that the VM doesn't have to spread over two sockets which help with latency. On the memory screen we will configure 4GB of RAM as it's all we need.

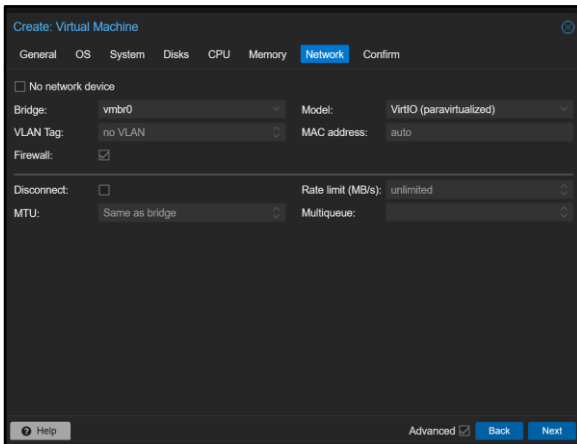


Proxmox - Screenshot 42 CPU settings for VM

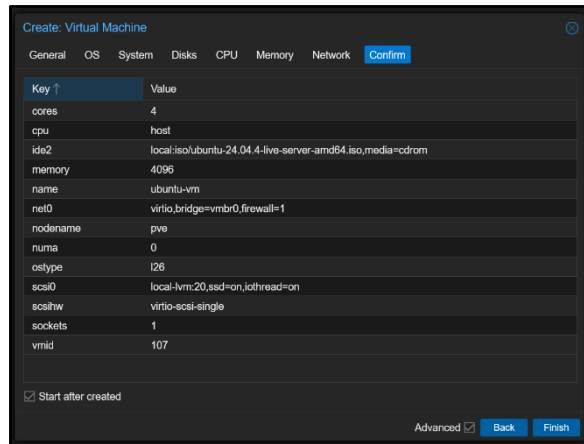


Proxmox - Screenshot 43 Memory settings for VM

On the network page we will choose the standard virtual bridge and leave the default settings. This makes it so we get a random MAC Address with a virtual interface that we can later configure the IP address in the VM. On the confirmation screen it lists all important details before creating the VM. We are happy with the config, so we can click start on create and click on finish. This starts the VM creation, shortly after it can be seen on the left hand side.



Proxmox - Screenshot 44 Network settings for VM

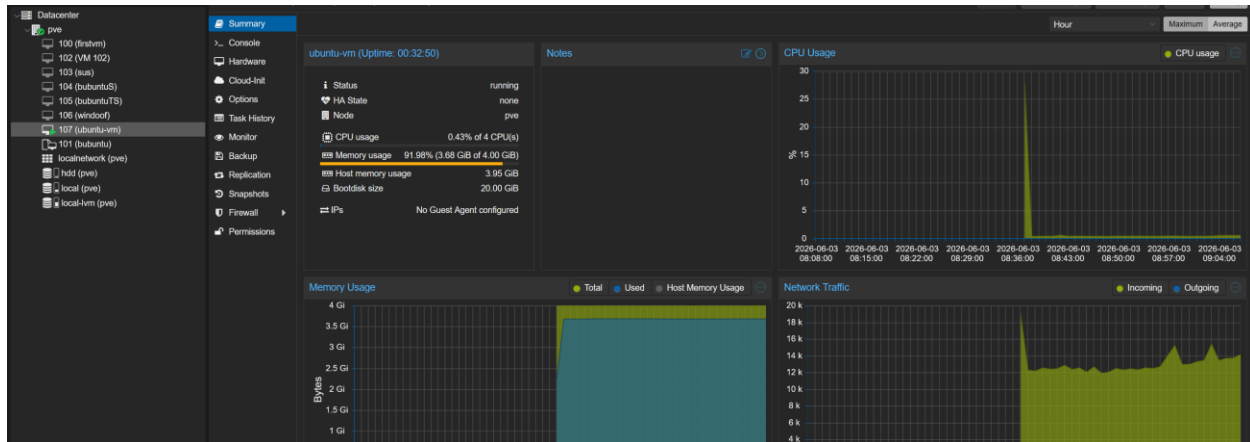


Proxmox - Screenshot 45 Confirmation screen for VM creation

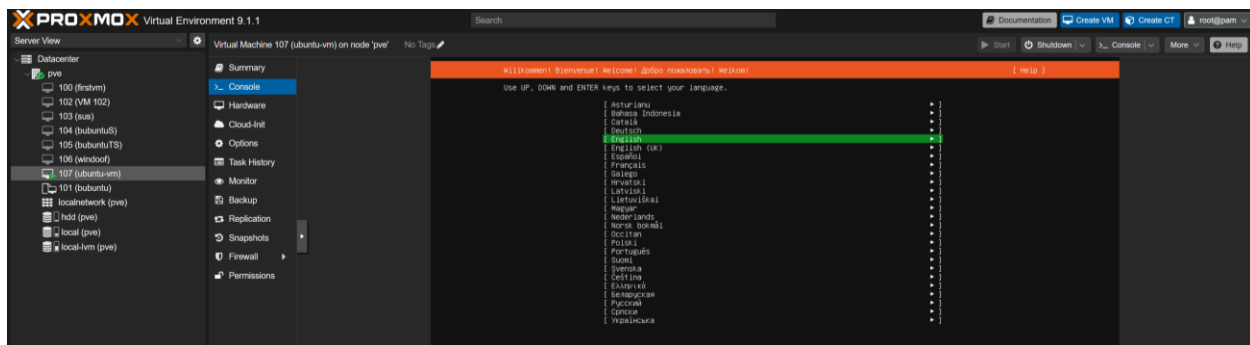


Proxmox - Screenshot 46 The Ubuntu VM

After clicking on the newly created VM we can see statistics, config pages and the console by clicking on the corresponding tabs.



Proxmox - Screenshot 47 Summary page of Ubuntu VM

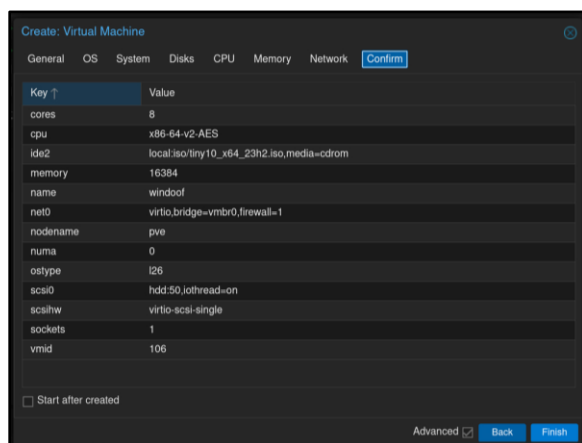


Proxmox - Screenshot 48 Console page of Ubuntu VM

Now you are able to setup Ubuntu Server like you are used to. The next steps should be familiar.

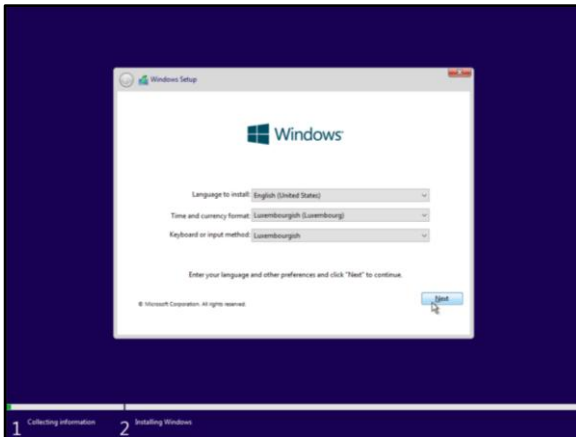
WINDOWS VM

Now we will create a windows VM. This is the summary page of all the settings we choose. We will it install it on the HDDs as we need more space and we will allocate more RAM.

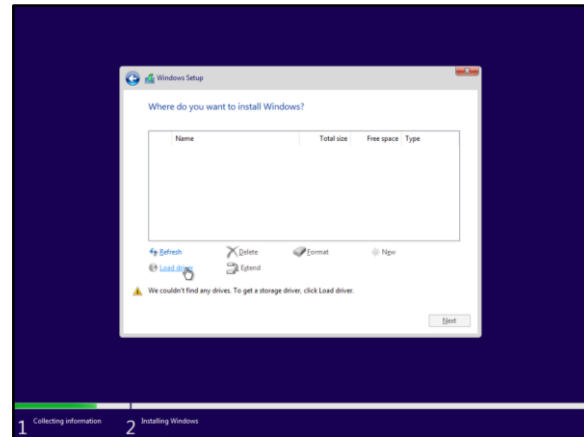


Proxmox - Screenshot 49 Windows 10 (Tiny10) VM creation summary

Then we will continue the installation. We will run into an issue where Windows might not find any storage device. This is due to a lack of VirtIO drivers. We will therefore need to download the VirtIO drivers and load them during install.



Proxmox - Screenshot 50 Installing Windows like normally



Proxmox - Screenshot 51 Running into a storage driver issue

Proxmox offers good documentation on this topic. If you need more information on the topic, read it.

Windows VirtIO Drivers

Contents [hide]

- 1 Introduction
- 2 virtio-win Releases
- 3 Windows OS Versions
- 4 Installation
 - 4.1 Using the ISO
 - 4.1.1 Wizard Installation
 - 4.1.2 Manual Installation
 - 4.2 Downloading the Wizard in the VM
- 5 Troubleshooting
 - 5.1 Known Issues
 - 5.1.1 0.1.285: VirtIO SCSI/VirtIO Block: Read errors and performance issues with IO-heavy Windows Server 2025 VMs
 - 5.1.2 0.1.215..0.1.262: VirtIO SCSI/VirtIO Block: "Reset to device was issued" events, blocked IO
- 6 Further Reading
- 7 See also

Introduction

VirtIO Drivers are paravirtualized drivers for **kvm**/Linux (see <http://www.linux-kvm.org/page/Virtio>). In short, they enable direct (paravirtualized) access to devices and peripherals for virtual machines using them, instead of slower, emulated, ones.

A quite extended explanation about VirtIO drivers can be found here: <http://www.ibm.com/developerworks/library/l-virtio/>.

At the moment these kinds of devices are supported:

- block (disk drives), see [Paravirtualized Block Drivers for Windows](#)
- network (ethernet cards), see [Paravirtualized Network Drivers for Windows](#)
- balloon (dynamic memory management), see [Dynamic Memory Management](#)

You can maximize performance by using VirtIO drivers. The availability and status of the VirtIO drivers depend on the guest OS and platform.

virtio-win Releases

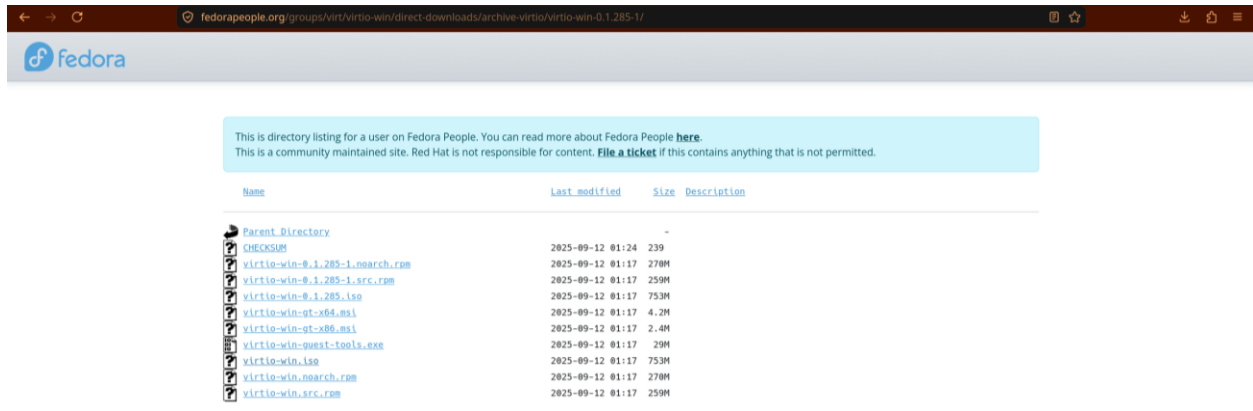
Windows does not have native support for VirtIO devices included. However, there is excellent external support through open-source drivers provided by the [virtio-win](#) project, which are available compiled and signed for Windows.

The virtio-win project periodically releases new versions of the virtio-win drivers. The following virtio-win repository provides the most recent as well as older versions of the virtio-win drivers:

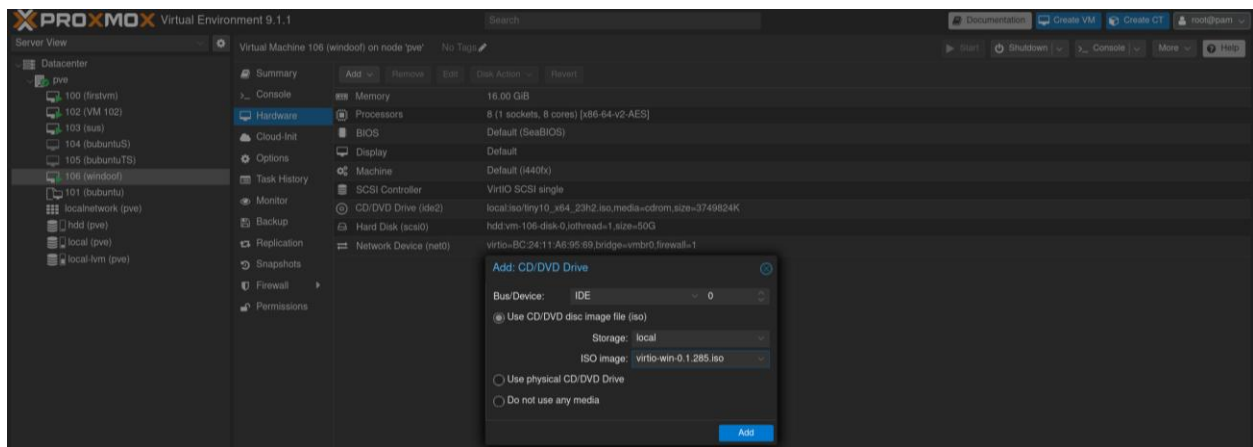
Proxmox - Screenshot 52 Documentation: Windows VirtIO Drivers

Download the virtio-win.iso and upload it to Proxmox. Once it has been uploaded, start the VM with the Windows ISO with the VirtIO ISO also mounted.

<https://fedorapeople.org/groups/virt/virtio-win/direct-downloads/archive-virtio/virtio-win-0.1.285-1/>

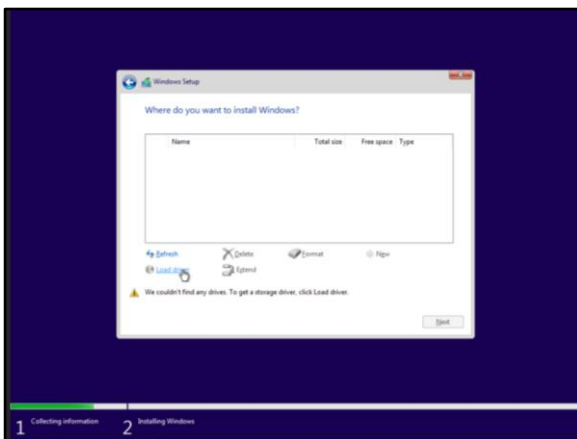


Proxmox - Screenshot 53 VirtIO Win download page on fedorapeople.org

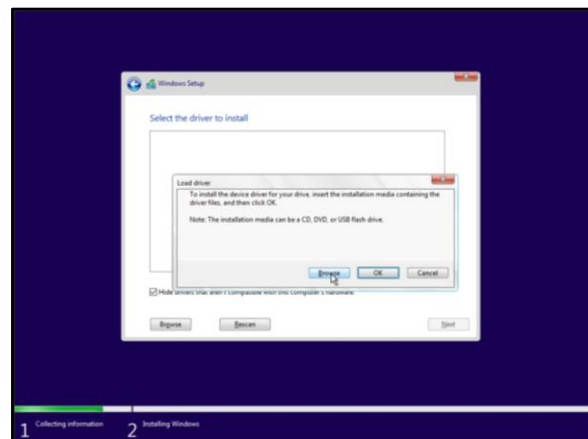


Proxmox - Screenshot 54 Adding the VirtIO Win ISO to the VM

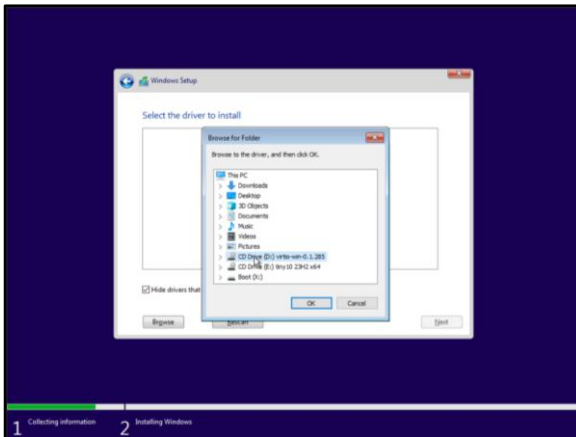
Then when on the storage selection screen click on Load driver, click on Browse and search for the VirtIO ISO. In there you will see a folder called “w10” for Windows 10 and in there choose the correct architecture (amd64). After clicking on okay it should show the VirtIO driver as installable. Install the driver and then you can continue to install Windows like usual.



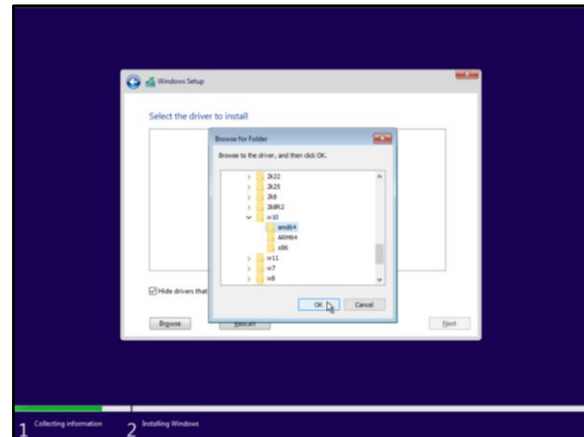
Proxmox - Screenshot 55 No device detected, Load driver



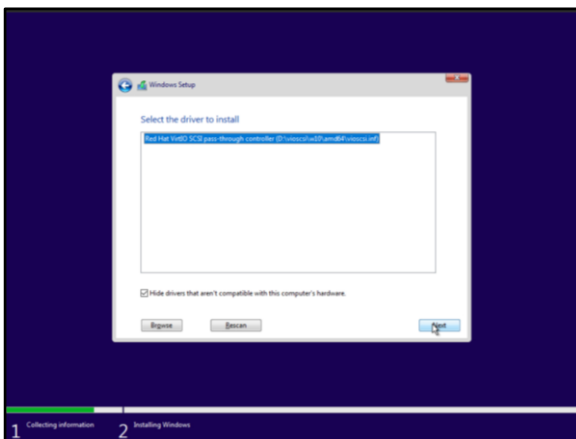
Proxmox - Screenshot 56 Browse for driver



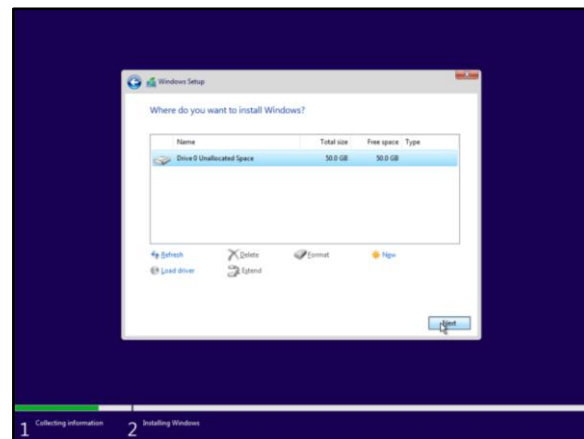
Proxmox - Screenshot 57 Select the VirtIO ISO



Proxmox - Screenshot 58 Select the correct folders for your install

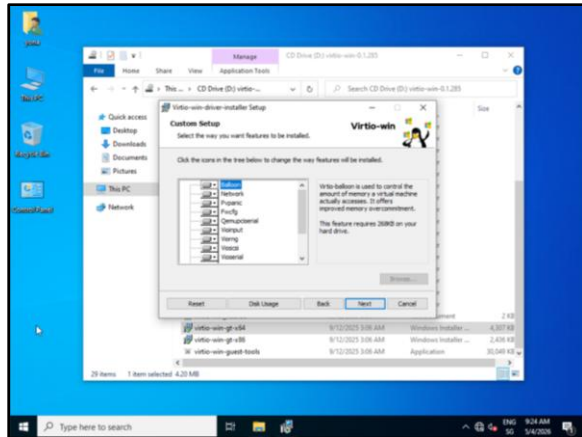


Proxmox - Screenshot 59 Install the VirtIO Driver



Proxmox - Screenshot 60 Continuing to install Windows

After installing Windows you can also install the guest additions for better VM management between Proxmox and Windows. Just execute the installer and install the guest additions.



Proxmox - Screenshot 61 Installing VirtIO guest additions for ethernet etc.

CREATING A VM WITH CLI

```
qm create 200 \  
--name Windows-CLI \  
--memory 4096 \  
--cores 4 \  
--sockets 1 \  
--cpu host \  
--net0 virtio,bridge=vbr0 \  
--ostype win10 \  
--scsihw virtio-scsi-pci
```

Proxmox can also create a VM entirely from the command line with the `qm` tool, which is handy for scripting or repeatable builds. The command above creates VM 200 named Windows-CLI with 4 GB of RAM, 4 cores on a single socket, the CPU type set to host, a VirtIO network card on the vbr0 bridge, the OS type set to Windows 10 and a VirtIO SCSI controller. This is the same thing the GUI wizard does, just expressed as one command – the equivalent of editing the `.vmx` file by hand on the VMware side. `qm create` only defines the VM though; we still attach the install ISO and a disk and then start it before installing the OS. Because the disk and NIC are VirtIO, the Windows installer won't see them until the VirtIO drivers are loaded, the same driver step covered in the VM creation section.

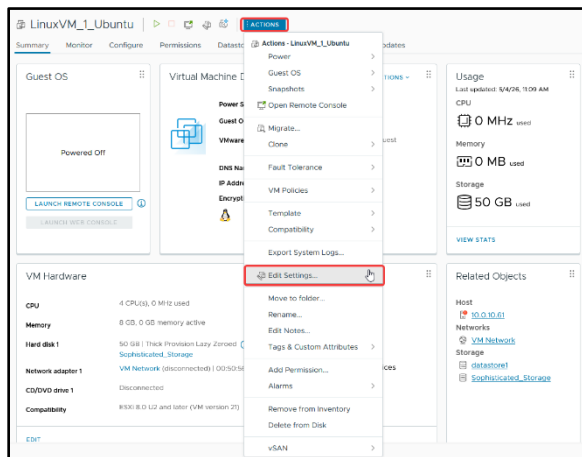
VIRTUAL MACHINE CONFIGURATION

VMWARE



Configuring the settings of a Virtual machine is fairly simple in the vSphere client. For this we'll be configuring the Linux VM we created before. Make sure it's powered off for being able to edit most settings.

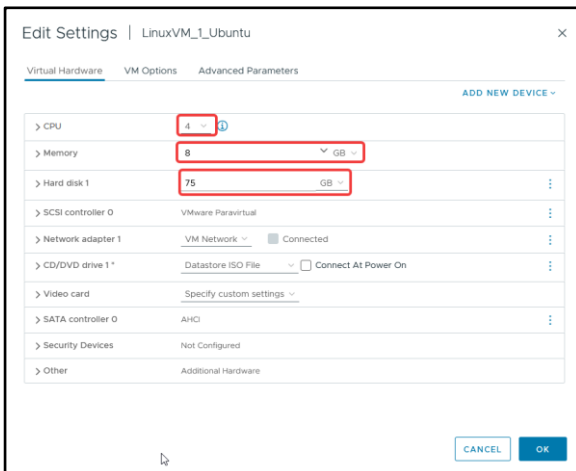
First in the action menu of the VM. Click on "Edit Settings".



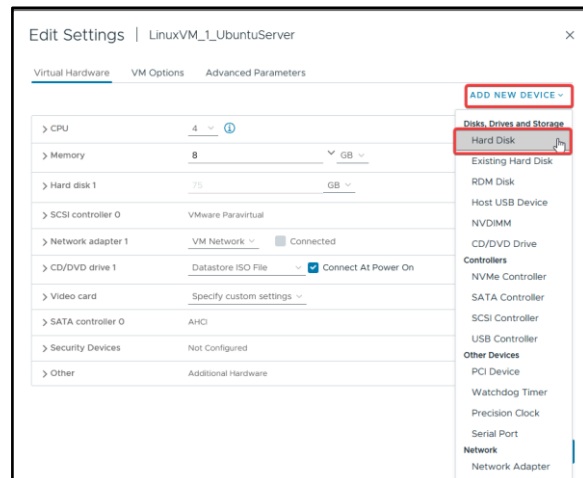
VMWare - Screenshot 70: Editing setting of LinuxVM_1

There are then 3 tabs for editing VMs, the first tab is for the hardware, here we can change the amount of vCPUs, Memory, Storage size a VM has. Along with adding more devices like additional hard disks and as shown before this is also where you can change the OS of the VM through the disk drive.

Here we change the CPUs, Memory and Storage of our VM and adding an additional hard disk.



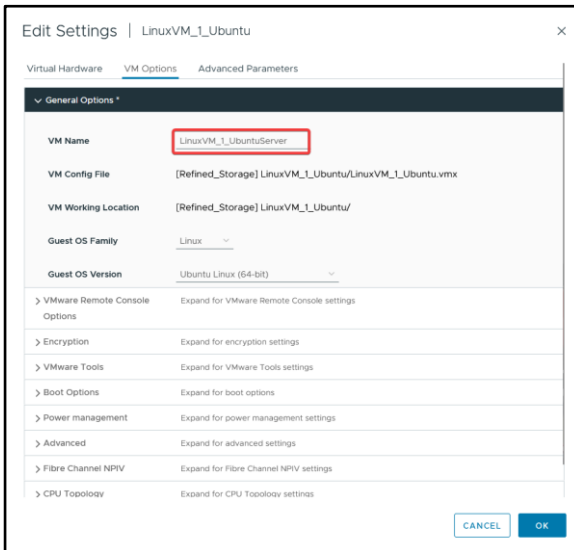
VMWare - Screenshot 71: Changing Hardware setting of VM



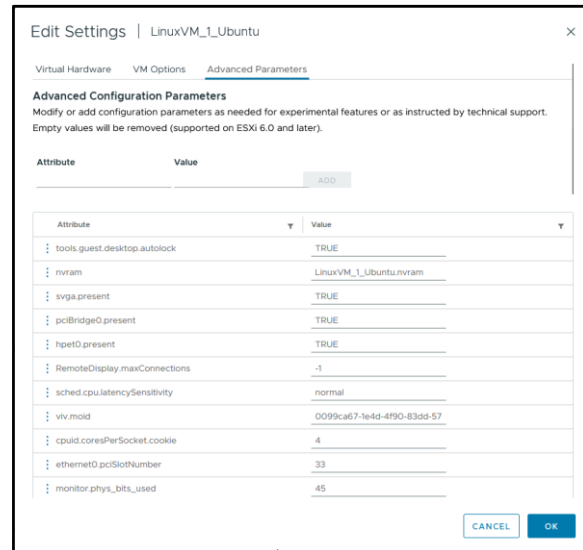
VMWare - Screenshot 72: Adding new Hard Disk to VM

In the VM Options there are further general configurations of a VM like its display name or the Guest OS. So settings usually outside of the VM and its virtual hardware. Here we're only changing the name to LinuxVM_1_UbuntuServer

Though we're not configuring anything here, note that in the "Advanced Parameters" is the same like the vmx file we edited manually through the CLI.



VMWare - Screenshot 73: Editing VM Name



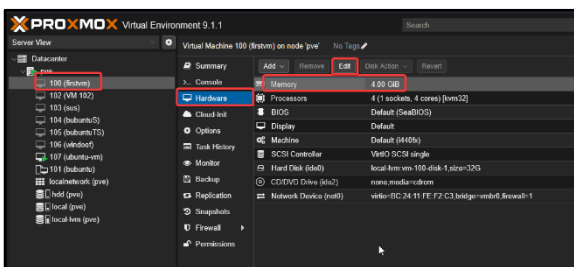
VMWare - Screenshot 74: VM Settings advanced parameters

PROXMOX



Unlike vSphere, where every setting sits behind a single "Edit Settings" dialog, Proxmox splits a VM's configuration across two tabs: the Hardware tab for the virtual hardware itself, and the Options tab for the more general settings. For this we'll be configuring the first VM (firstvm, ID 100) we created earlier.

The Hardware tab lists every virtual component of the VM: Memory, Processors, the SCSI controller, the hard disk, the CD/DVD drive and the network device. To change one of these, select the row and click "Edit", which opens a dialog specific to that component. Here we select Memory to adjust the RAM assigned to the VM.

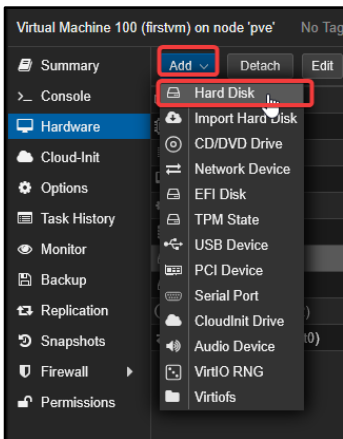


Proxmox - Screenshot 62: Editing Hardware settings of VM

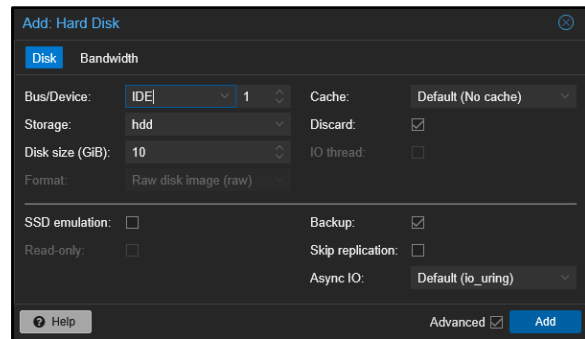
Worth noting is that Proxmox doesn't strictly require the VM to be powered off the way VMware does for most edits. With hotplug enabled (visible later in the Options tab), disks and network devices can be added to a running VM.

To add a new component rather than edit an existing one, use the "Add" dropdown. This lists everything that can be attached to the VM.

As an example we add a second hard disk. In the dialog we choose the bus/device type (IDE), the storage it should live on (we put this one on our “hdd” storage rather than the SSD) and the size, here 10 GiB. The disk is created and appears in the Hardware list as ide1.

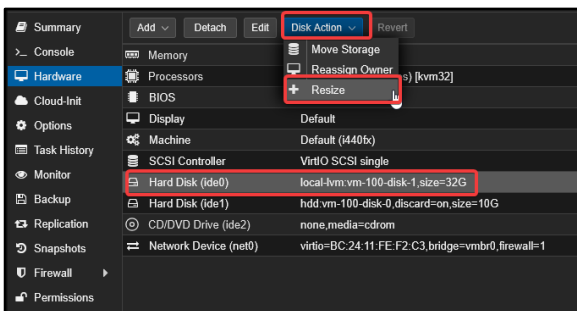


Proxmox - Screenshot 63: Creating a new Hard Disk for VM

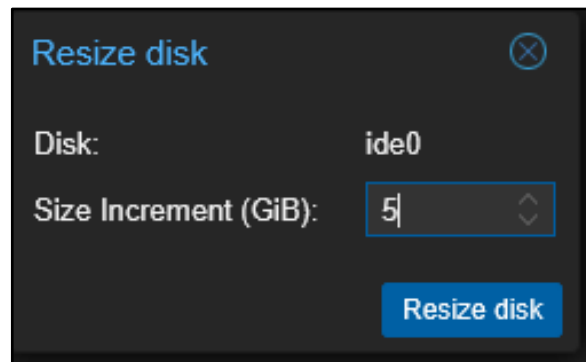


Proxmox - Screenshot 64: Entering Hard Disk information

Disks can also be resized after the fact. Select the disk and open the “Disk Action” dropdown, then choose “Resize”. One thing to be careful of here: Proxmox asks for a size increment, not a target size, so the value entered is added on top of the current disk. Our boot disk (ide0) was 32 GB, so an increment of 5 GiB brings it to 37 GB. Note also that a disk can only be grown this way, not shrunk.



Proxmox - Screenshot 65: Resizing Hard Disk

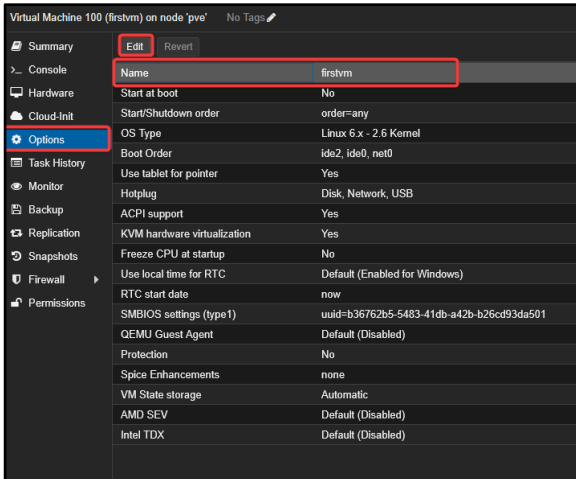


Proxmox - Screenshot 66: Growing Hard Disk by 5

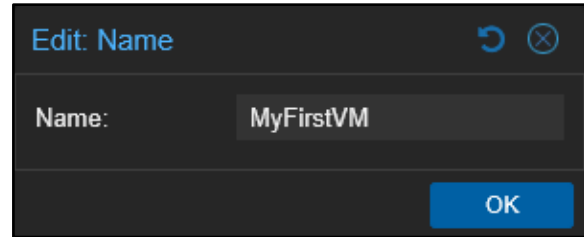


Proxmox - Screenshot 67: Final Hard Disk size

For the more general settings we switch to the Options tab. This holds everything that sits outside the virtual hardware, like the VM’s display name, OS type, boot order, whether it starts at boot, the hotplug settings mentioned earlier, and the QEMU guest agent. This is roughly the equivalent of VMware’s “VM Options” tab. Here we only change the display name: select “Name”, click “Edit”, and rename the VM to MyFirstVM.



Proxmox - Screenshot 68: Editing VM Name



Proxmox - Screenshot 69: Entering new VM Name



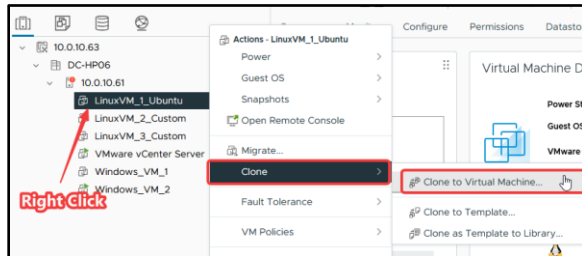
Proxmox - Screenshot 70: Confirmation of Name change

CLONING A VIRTUAL MACHINE

VMWARE

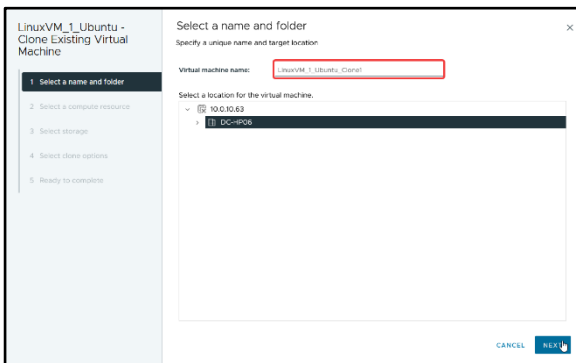


To clone a VM in vSphere, go to the action menu of the VM you want to clone, open the “Clone” dropdown and select “Clone to Virtual Machine”.

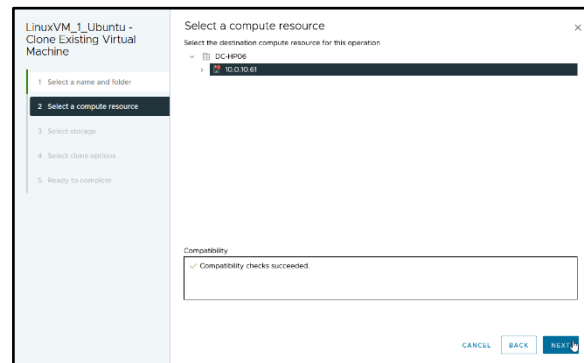


VMWare - Screenshot 75: Clone VM Action menu

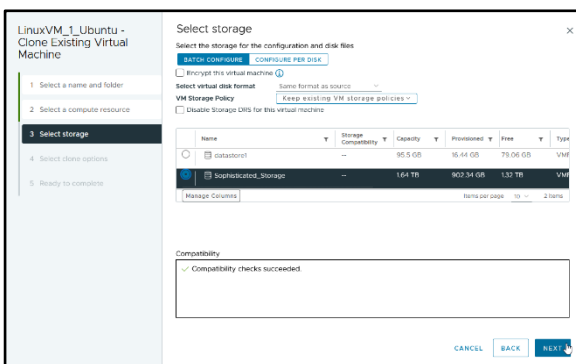
Similar to the creating a VM, we edit the name and select which datacentre will store it, and select which host will be used for compute and which storage.



VMWare - Screenshot 76: Cloning VM - selecting name and folder

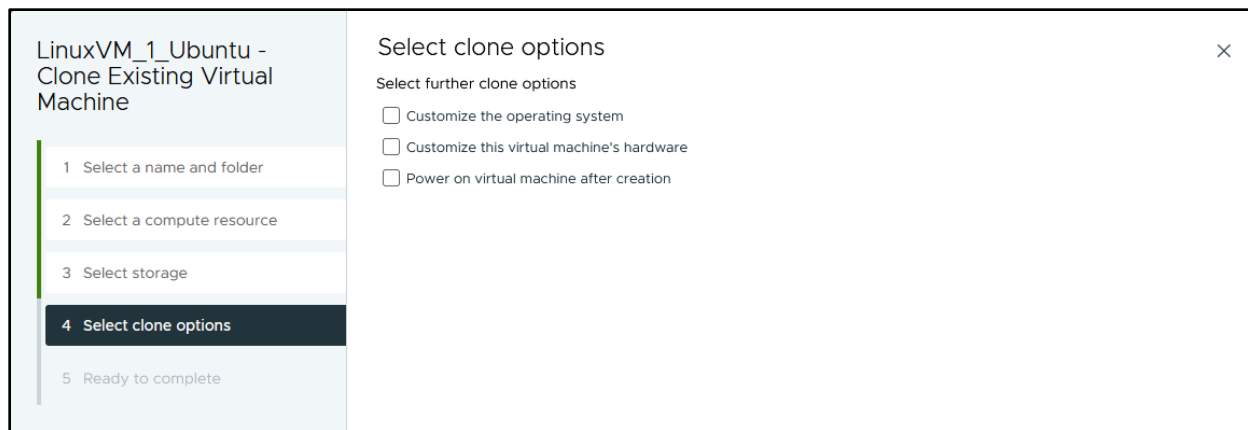


VMWare - Screenshot 77: Cloning VM - selecting compute resource

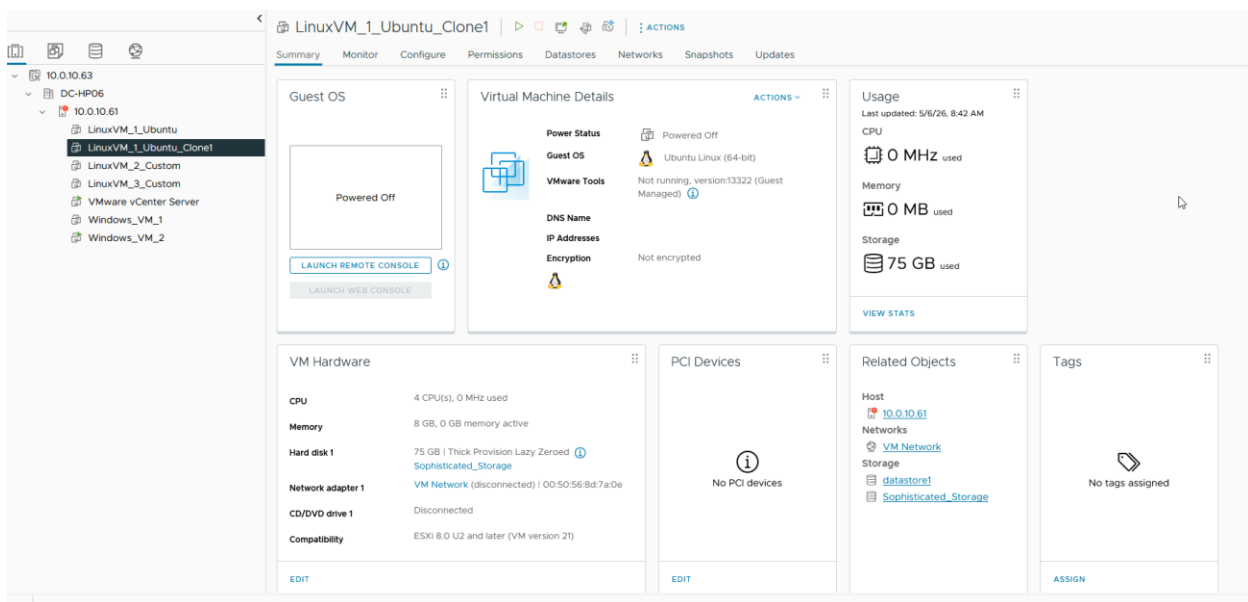


VMWare - Screenshot 78: Cloning VM - selecting storage

In this next step, it will allow you to configure some settings for the clone, like hardware or the Operating system, which we won't do now since we want an exact clone, but note that it's a possibility in case you don't have enough resources to support an exact clone. You can also optionally turn on the VM after cloning.

**VMWare - Screenshot 79: Cloning VM - select clone options**

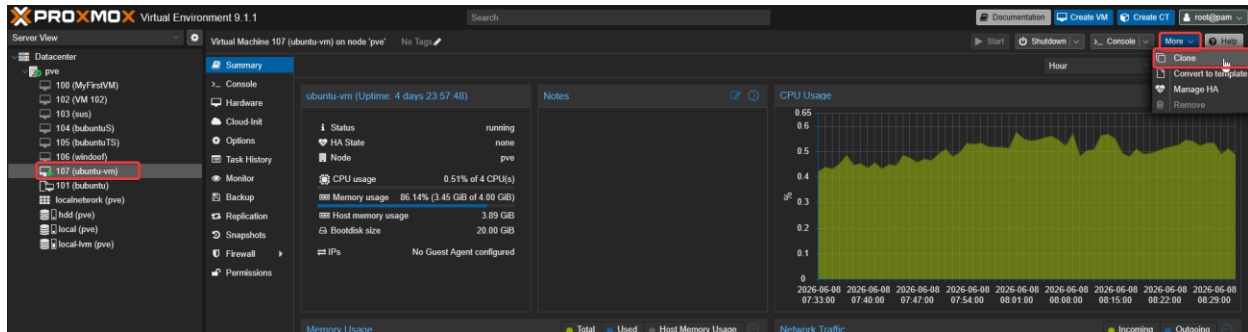
Confirm the settings and then we'll see that the clone was successfully created in the menu.

**VMWare - Screenshot 80: After cloning**

PROXMOX

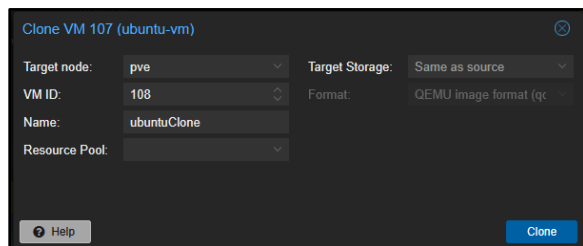


Cloning in Proxmox is reached from the VM itself rather than a separate menu. Select the VM you want to clone (here the running ubuntu-vm, ID 107), open the “More” dropdown in the top right and choose “Clone”.



Proxmox - Screenshot 71: Cloning a VM

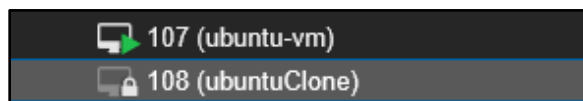
This opens the clone dialog. We pick the target node (here our node pve), and Proxmox suggests the next free VM ID, 108. We give it the name ubuntuClone and leave the resource pool empty.



Proxmox - Screenshot 72: Entering Clone information

Because the source VM is running, Proxmox performs an online (live) clone and the Target Storage and Format fields are locked to “Same as source”. Had the VM been powered off, we could have picked a different storage and disk format here instead. Cloning a regular VM like this always produces a Full Clone, meaning the new VM is a completely independent copy with its own disks, not tied to the original in any way. The lighter alternative, a Linked Clone that shares the original’s base image, is only offered when the source is a template, which we cover in the Templates section.

After clicking “Clone”, the new VM 108 appears in the inventory straight away, but with a padlock icon next to it. Proxmox locks the clone while the copy task runs so it cannot be started or edited mid-clone.



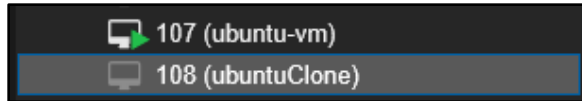
Proxmox - Screenshot 73: Locked clone

The clone task itself can be followed in the task log at the bottom of the interface.



Proxmox - Screenshot 74: Cloning running status

Once the task finishes, the padlock disappears and ubuntuClone is ready to use as a normal, independent VM.



Proxmox - Screenshot 75: After cloning finished

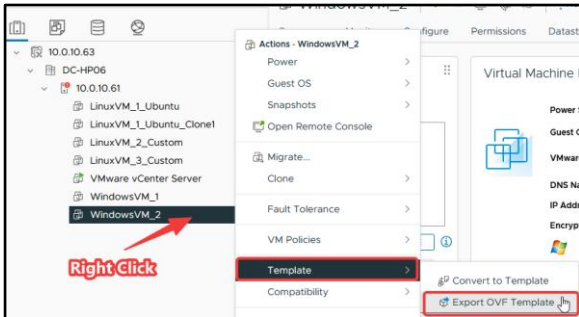
EXPORTING A VIRTUAL MACHINE

VMWARE

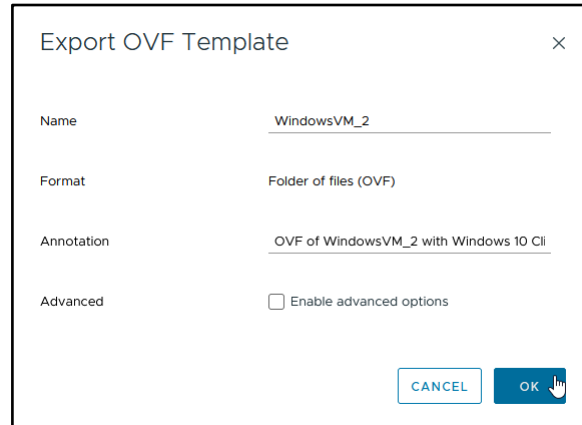


To export a Virtual Machine in vSphere, there isn't an explicit export button, but you actually turn the VM into a Template, specifically an OVF Template.

To do that, in the action menu of a VM, select "Clone" then "Export OVF Template", for this we're using a second Windows VM with Windows 10. So we keep the name and create an annotation if you want.



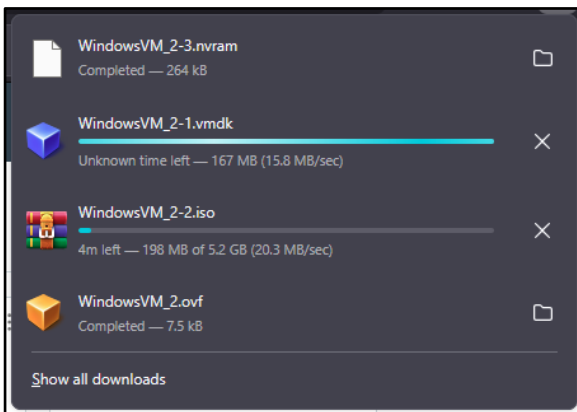
VMWare - Screenshot 81: Export OVF Template Action Menu



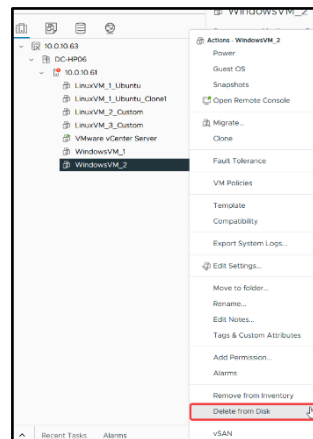
VMWare - Screenshot 82: Export VM as OVF

Confirming the box, will start a download of a few files, keep them in a place where you'll remember since we'll be needing them to redeploy.

Though unnecessary we delete, the existing second windows instance to then redeploy it with the OVF template, this is mainly to show that the same VM will be created that has the same properties as WindowVM_2.

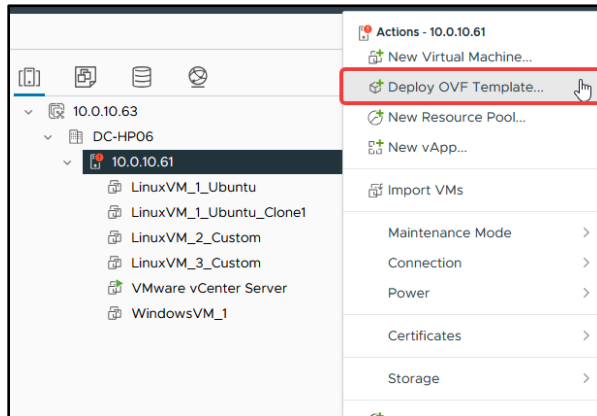


VMWare - Screenshot 83: OVF Downloaded files



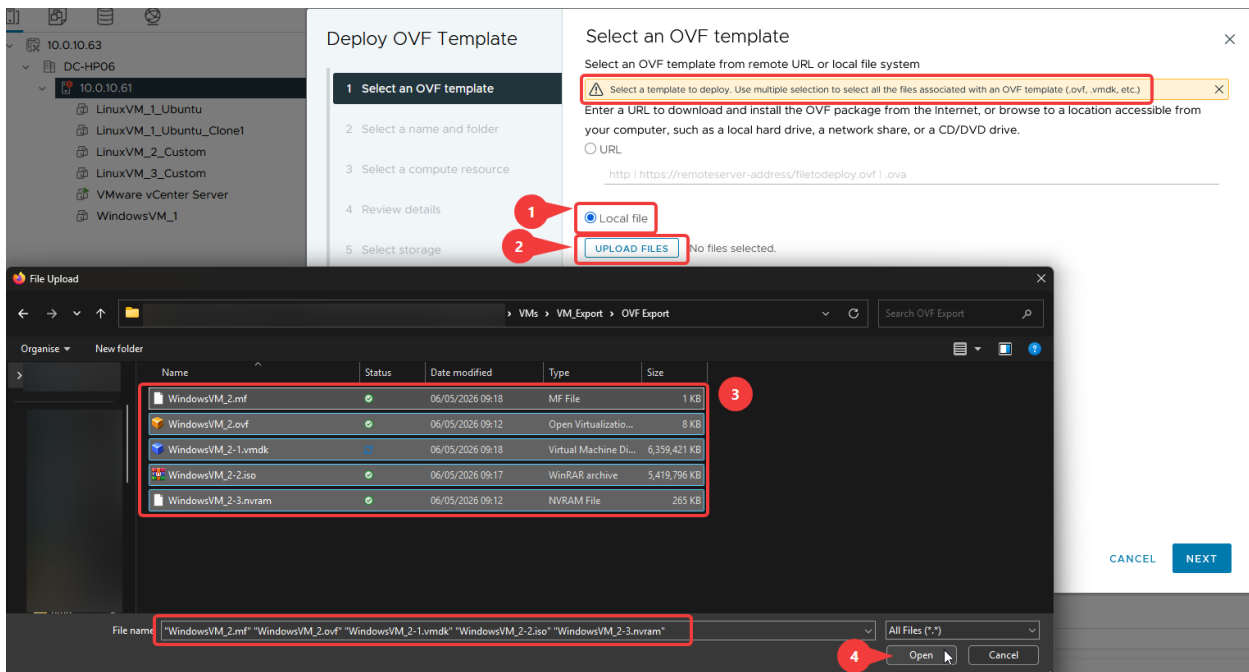
VMWare - Screenshot 84: Deletion of WindowsVM_2

To deploy an OVF template, in the action menu of the host, click on “Deploy OVF Template



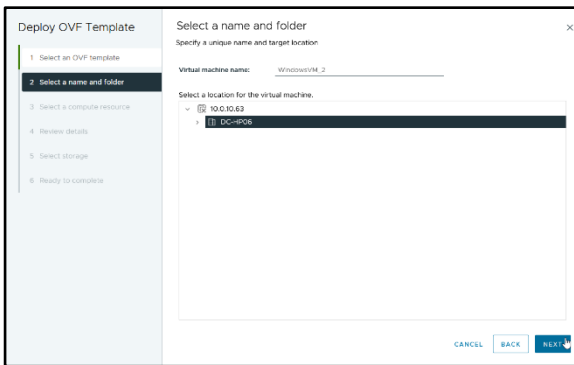
VMWare - Screenshot 85: Deploy OVF Template

In this next step we select our OVF Template, for this we choose to upload them locally, note the warning stating that we need to upload multiple files, so head to where you stored your downloaded files and select each one.

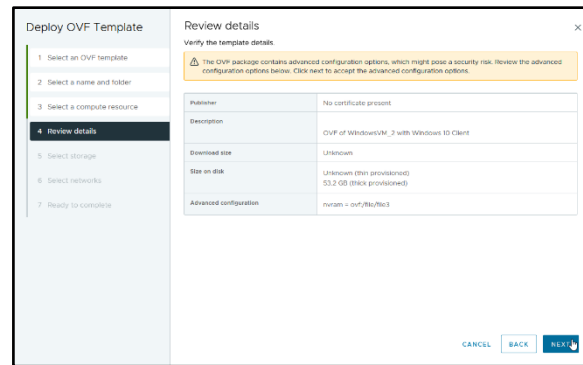


VMWare - Screenshot 86: Export VM - select an OVF template

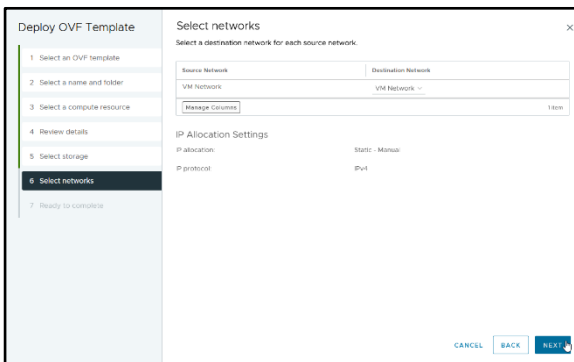
With that done, it's now very similar to creating a VM, with your folder selection, compute resource, except that you don't change any hardware properties. And new is that you have to select on which Network you want to deploy your VM onto, here we keep the default, but we can then specify how it will obtain its IP address to, which we changed to static.



VMWare - Screenshot 87: Export VM - selecting name and folder

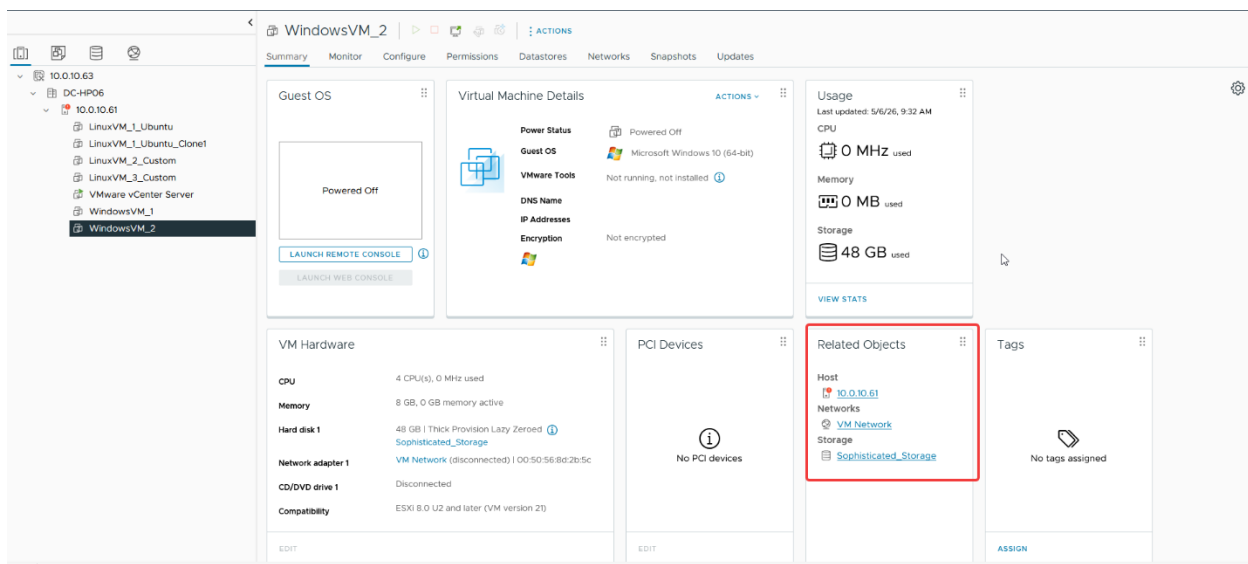


VMWare - Screenshot 88: Export VM - review details



VMWare - Screenshot 89: Export VM - select networks

Once the deployment is completed, we can review the newly created VM.



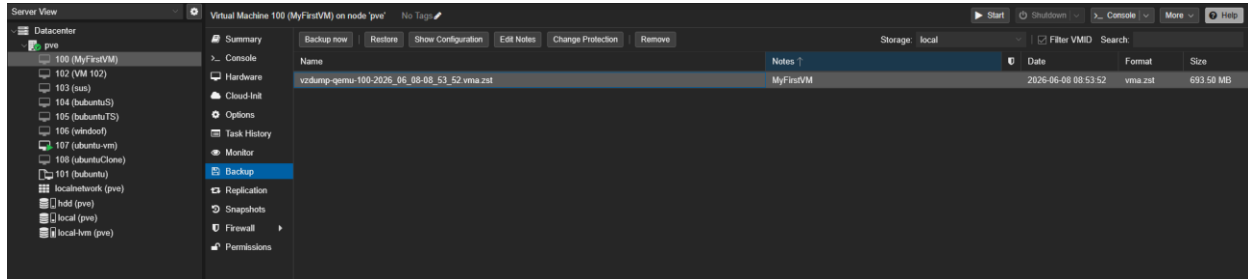
VMWare - Screenshot 90: After OVF deployment

PROXMOX



Proxmox doesn't have a dedicated "Export" button the way it has clone or backup. In practice, exporting a VM and backing one up are the same operation: both use `vzdump`, which produces a single portable archive

(a .vma.zst file). The only real difference is intent. To “export” a VM for moving it elsewhere, we run a backup in Stop mode so the VM is shut down first and the archive is clean and consistent, then copy the resulting .vma.zst off the host. That archive can be restored onto any Proxmox node to recreate the VM exactly.



Proxmox - Screenshot 76: Export / Backup of VM

This is actually close to the VMware side, where there is no plain “export” either – vSphere exports a VM as an OVF template, as documented above. The difference is the format: VMware produces the open OVF/OVA set of files, while Proxmox produces its own vzdump archive that’s restored back into Proxmox.

BACKUP CONFIGURATION & VIRTUAL MACHINE

VMWARE

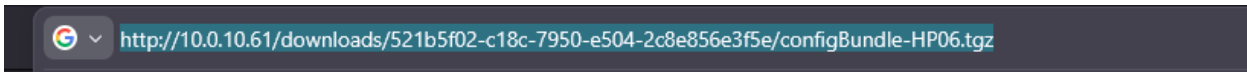


BACKING UP HYPERVISOR CONFIGURATION

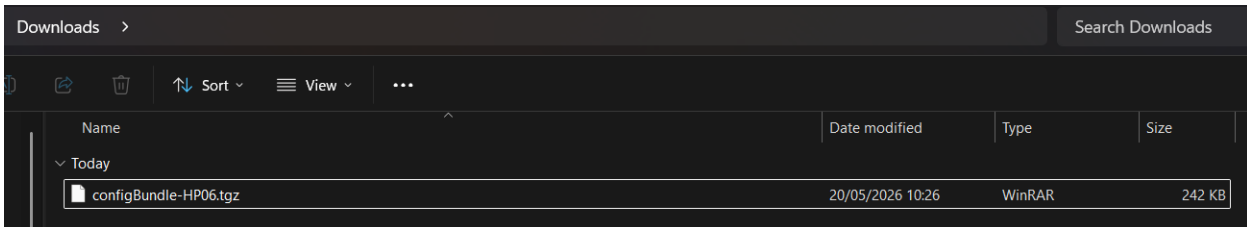
ESXi can back up its own host configuration from the command line, so no third-party tool is needed for this part. With SSH enabled on the host, we run `vim-cmd hostsvc/firmware/backup_config`, which bundles the host's configuration and returns a download URL for a `configBundle.tgz` file. We download that bundle and keep it safe; if the host ever needs rebuilding, the matching restore command puts the configuration back. This is worth contrasting with Proxmox, which has no equivalent built-in button and expects you to script the backup yourself, as shown on the Proxmox side below.

```
[root@HP06:~] vim-cmd hostsvc/firmware/sync_config
[root@HP06:~] vim-cmd hostsvc/firmware/backup_config
Bundle can be downloaded at : http://*/downloads/521b5f02-c18c-7950-e504-2c8e856e3f5e/configBundle-HP06.tgz
```

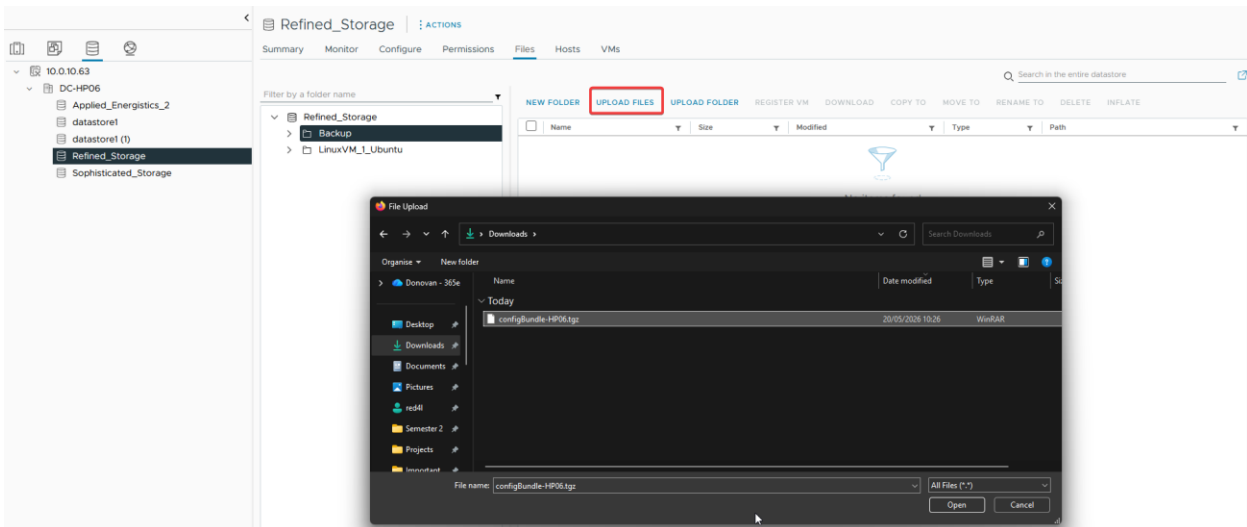
VMWare - Screenshot 91: Creating backup of configuration



VMWare - Screenshot 92: Entering backup download in browser



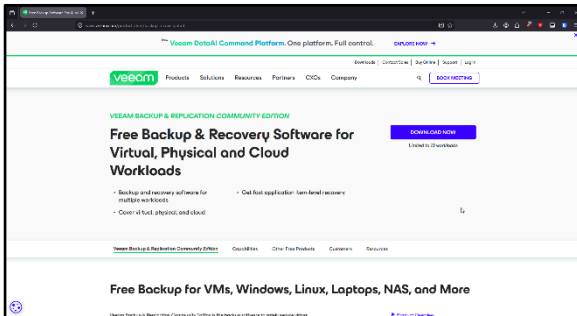
VMWare - Screenshot 93: Configuration backup file in file explorer



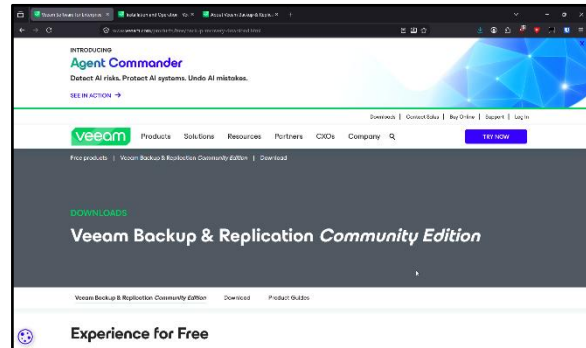
VMWare - Screenshot 94: Uploading backup file for storage

BACKING UP VIRTUAL MACHINE

To backup a Virtual Machine in VMWare a 3rd party tool is necessary. For this lab, we used the Veeam Backup & Replication Community Edition, since it's free with VMWare support.



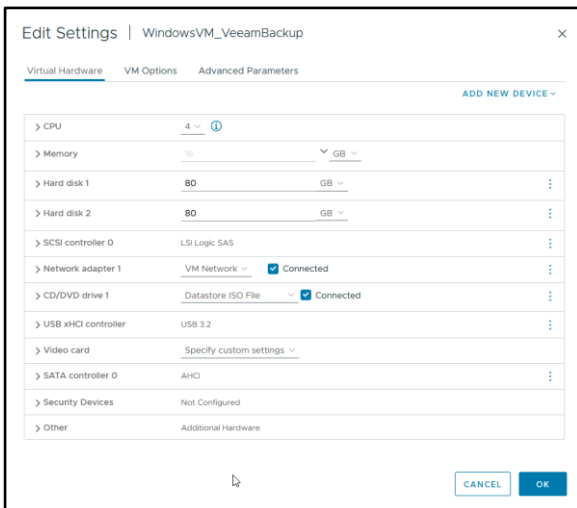
VMWare - Screenshot 95: Veeam home page



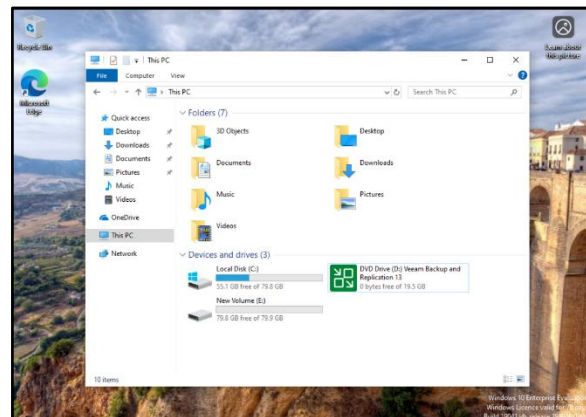
VMWare - Screenshot 96: Veeam Backup & Community Edition page

For the installation of Veeam, we created a Windows Virtual Machine on the server with 2 storage disks, one for the software itself and one for storing the backups onto.

On the Virtual Machine itself, load the installed ISO onto it and begin the installation of Veeam



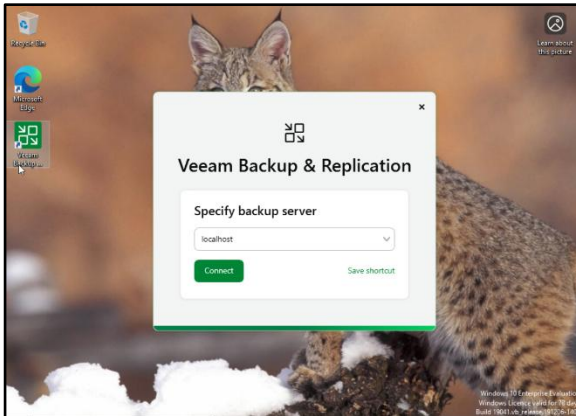
VMWare - Screenshot 97: Veeam Backup client settings



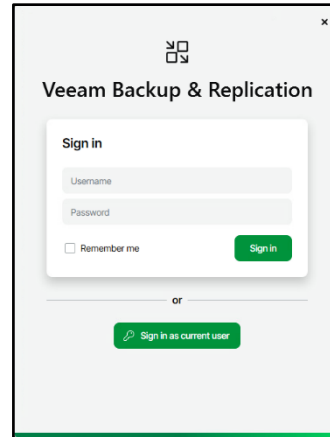
VMWare - Screenshot 98: Veeam Backup client

The installation of Veeam is simple and just going through confirmation screens, so this documentation won't cover it. However as note, the installation will take a while.

Once the installation is done, it will ask you to specify the backup server, which will be the localhost, and which user to login with, just log in with the current user. You may get a message saying that localhost has an untrusted certificate which is normal, so we can trust it.

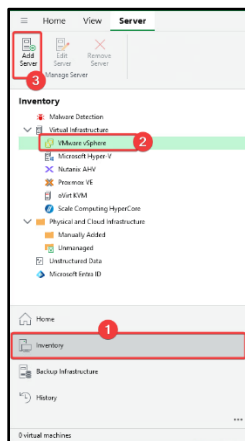


VMWare - Screenshot 99: Specify backup server



VMWare - Screenshot 100: Veeam login screen

Once logged in, you'll see the main home screen of Veeam, from here head to your inventory to add the vCenter server into it.



VMWare - Screenshot 101: Adding the vSphere to the inventory

Now to specify our server and used the administrator login to add it, you may again get an untrusted certificate which we already are aware of, so you can safely ignore it for the lab. Next is to apply the settings and wait for Veeam to get the information from the vCenter server.

New VMware Server

Name

Specify DNS name or IP address of VMware server.

DNS name or IP address:
10.0.10.63

Description:
Created by .\DAMN at 01/06/2026 10:13.

< Previous Next > Finish Cancel

VMWare - Screenshot 102: Specify VMWare server IP for Veeam

New VMware Server

Credentials

Select server administrator's credentials. If required, specify additional connection settings including web service port number.

Select an account with local administrator privileges on the server you are adding. Use DOMAIN\USER format for the domain account name, and HOST\USER for the local account name.

Credentials:
Select existing credentials or add new

Add...

Manage accounts

possible port customization in

OK Cancel

< Previous Apply Finish Cancel

VMWare - Screenshot 103: Log in with administrator credentials to add the server to Veeam

New VMware Server

Apply

Please wait while required operations are being performed. This may take a few minutes...

Message	Duration
Starting infrastructure item update process	0:00:02
Creating database records for server	0:01:47
Collecting disks and volumes info	
VMware server saved successfully	

< Previous Next > Finish Cancel

VMWare - Screenshot 104: Applying VMWare server to Veeam

New VMware Server

Summary

You can copy the configuration information below for future reference.

VMware vCenter server '10.0.10.63' was successfully saved.
Host info: VMware vCenter Server 8.0.2 build-23504390
Connection options:
User: administrator@hp06.local
Port: 443

< Previous Next > Finish Cancel

VMWare - Screenshot 105: VMWare addition to Veeam summary

Once Veeam has finished, we'll be able to see the datacentre, hosts and all of our VMs in the Inventory.

Inventory

Malware Detection

Virtual Infrastructure

VMware vSphere

vCenter Servers

10.0.10.63

DC-HP06

10.0.10.61

10.0.10.81

Microsoft Hyper-V

Nutanix AHV

Proxmox VE

oVirt KVM

Scale Computing HyperCore

Physical and Cloud Infrastructure

Manually Added

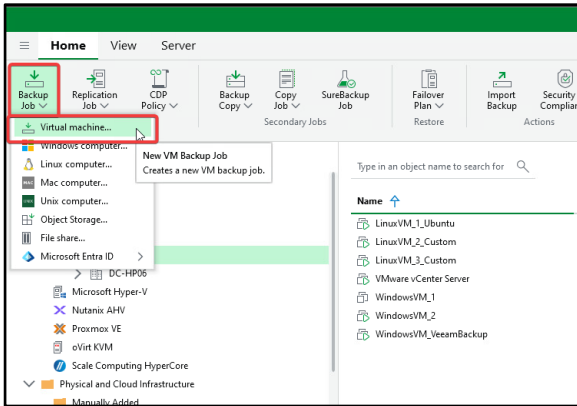
Type in an object name to search for

Name	Used Size	Host	Guest OS	Last Backup	Malware Status
LinuxVM_1_Ubuntu	3.4 GB	10.0.10.61	Ubuntu Linux (64-bit)	never	Clean
LinuxVM_1_Ubuntu_Clone1	75 GB	10.0.10.61	Ubuntu	never	Clean
LinuxVM_2_Custom	40 GB	10.0.10.61	Linux	never	Clean
LinuxVM_3_Custom	40 GB	10.0.10.61	Linux	never	Clean
VMware vCenter Server	69.9 GB	10.0.10.61	VMware Photon OS (64-bit)	never	Clean
WindowsVM_1	64 GB	10.0.10.61	Microsoft Windows 11	never	Clean
WindowsVM_2	56.1 GB	10.0.10.81	Microsoft Windows 10 (1507)	never	Clean
WindowsVM_VeeamBackup	96.2 GB	10.0.10.81	Microsoft Windows 10 (64-bit)	never	Clean

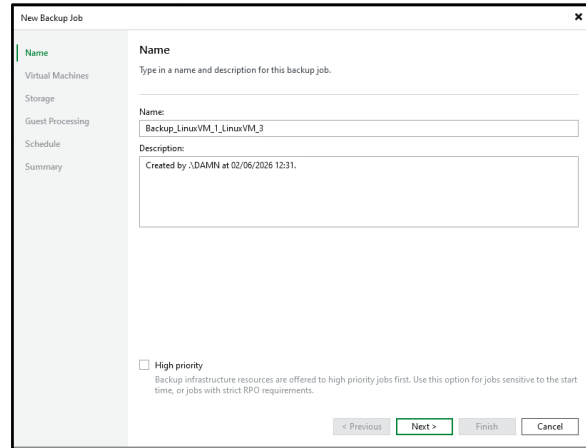
VMWare - Screenshot 106: Veeam VMWare inventory

Now we can create backups for our VMs, for this demonstration we'll be backing up LinuxVM_1 and LinuxVM_3 on a monthly basis.

First create a new Backup Job for Virtual machines and give it a name.



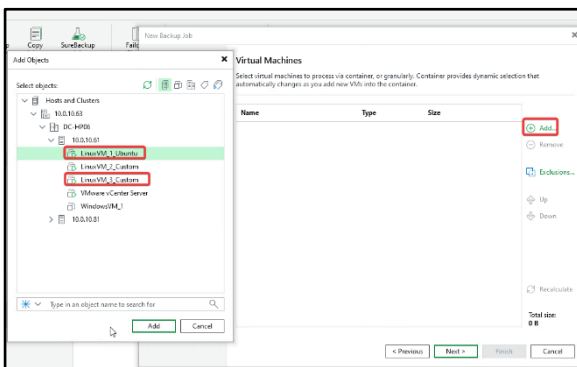
VMWare - Screenshot 107: Creating a new Backup Job



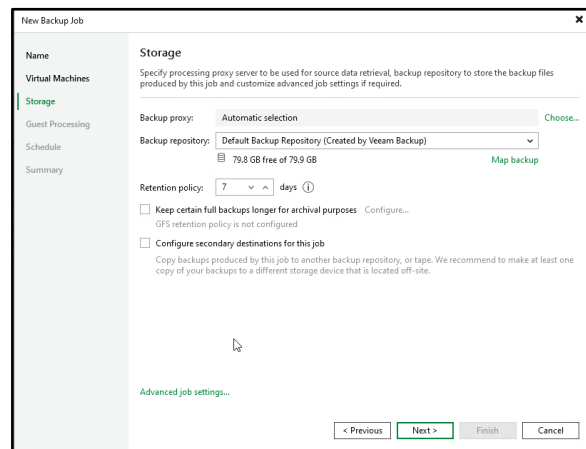
VMWare - Screenshot 108: Naming the Backup Job

Next we'll add our 2 VMs to the job, specify the storage of it and set the schedule.

Leave the default settings in the Guest Processing step. In the summary make sure to enable "Run the job when I click Finish" to automatically start the Backup Job.



VMWare - Screenshot 109: Selecting VMs for Backup Job

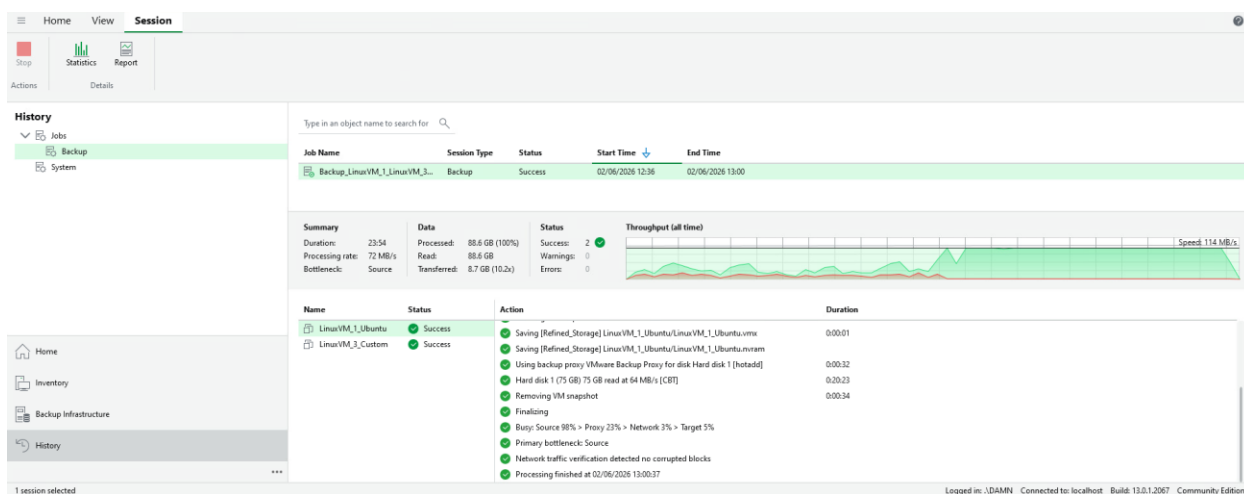


VMWare - Screenshot 110: Selecting Storage for Backup job

VMWare - Screenshot 111: Setting the schedule for the Backup Job

VMWare - Screenshot 112: Backup Job creation summary

Once you click “Finish”, the backup job starts and is viewable in the History section under Session. Wait for the backup to finish, and you’ll be able to see what it did, so the actions it took and if there were any errors.



VMWare - Screenshot 113: Backup Job session

PROXMOX



BACKING UP HYPERVISOR CONFIGURATION

Proxmox has no GUI button for backing up the host’s own configuration, and that absence is itself the comparison point: where ESXi ships the vim-cmd backup_config command, Proxmox expects you to script it, which fits the “script” backup option the brief allows. The host configuration lives in a handful of directories, so we back those up with a simple tar to our remote NAS mount. The main one is /etc/pve, which is where Proxmox keeps its cluster and VM configuration; alongside it we include /etc/network, /etc/hosts, /etc/hostname and /etc/resolv.conf so the networking comes back too.

`tar -czf /mnt/pve/rmStorage/pve-config-backup.tar.gz /etc/pve /etc/network /etc/hosts /etc/hostname /etc/resolv.conf`

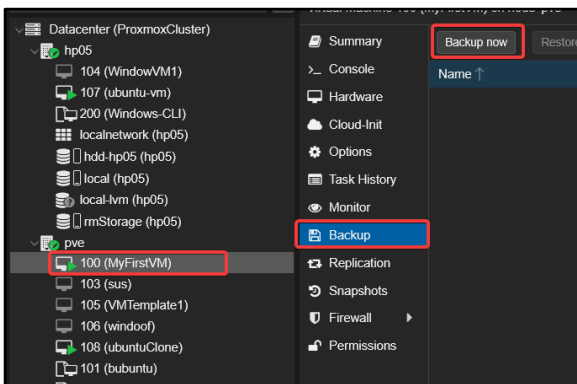
tar prints a “removing leading ‘/’ from member names” message while it runs; this is normal and expected, not an error – it just means the paths are stored relative inside the archive. The archive lands on the remote storage at /mnt/pve/rmStorage/. One thing worth knowing is that /etc/pve isn’t an ordinary folder: it’s the pmxcfs virtual filesystem, backed by a SQLite database at /var/lib/pve-cluster/config.db and synced across the cluster, which is why backing up its contents captures the node’s VM and cluster configuration.

```
root@hp05:~# tar czf /mnt/pve/rmStorage/pve-config-$(hostname)-$(date +%F).tar.gz /etc/pve /etc/network /etc/hosts /etc/hostname /etc/resolv.conf
tar: Removing leading `/' from member names
tar: Removing leading `/' from hard link targets
root@hp05:~# ls -lh /mnt/pve/rmStorage/pve-config-*.tar.gz
-rwxr-xr-x 1 root root 25K Jun  8 10:41 /mnt/pve/rmStorage/pve-config-hp05-2026-06-08.tar.gz
root@hp05:~#
```

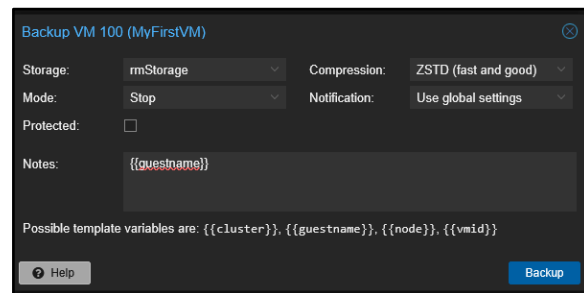
Proxmox - Screenshot 77: Backing up Hypervisor configuration

BACKING UP VIRTUAL MACHINE

Backing up a VM in Proxmox is built in and uses the same vzdump engine as the export above. The quickest way is per VM: select the VM, open its “Backup” tab, click “Backup now”, choose the target storage and a mode, and it runs once.

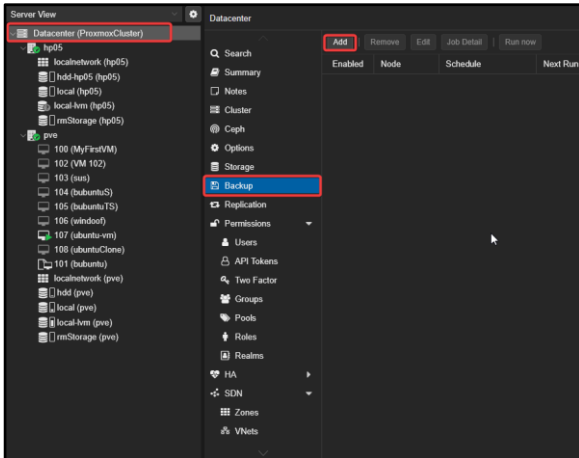


Proxmox - Screenshot 78: Creating Backup of VM

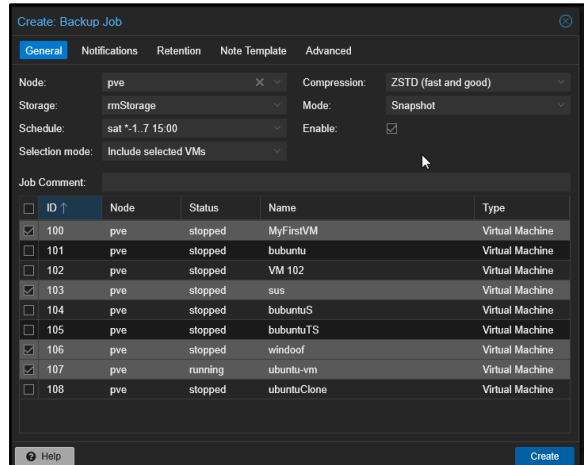


Proxmox - Screenshot 79: Entering information of VM Backup

Since backups should always be automated we can set up a recurring job at the Datacentre level, under Datacentre → Backup → Add, which lets us pick the VMs, a schedule, the storage and the retention all in one place



Proxmox - Screenshot 80: Creating Recurring Backup job on datacentre



Proxmox - Screenshot 81: Entering Backup job information

Note that, the target storage only shows up in the dropdown if it has the “VZDump backup file” content type enabled, so we had to make sure our backup storage was allowed to hold backups first. For the mode, Stop gives the cleanest backup (the VM is shut down briefly), while Snapshot keeps the VM running by backing up from a temporary snapshot; we used ZSTD compression for a good size-to-speed balance. Worth noting too is that a raw iSCSI block device can’t hold vmdump files directly, backups need a file-level target like a directory, NFS, SMB or a Proxmox Backup Server.

The contrast with VMware is clear: VMware needed a separate third-party tool (Veeam) to back up a VM, whereas in Proxmox the backup scheduler is part of the base product, with no extra software to install.

SNAPSHOTS

DEFINITION

A snapshot preserves the state of a virtual machine at a specific point in time, including its virtual disk, its settings and, optionally, the contents of its memory (the running state). It works as a delta/differencing layer on top of the base disk: once a snapshot is taken, all new writes go into a separate delta file while the original disk is frozen, which is what allows you to roll back to the exact captured moment. This makes snapshots ideal right before risky operations like patching, updates or configuration changes.

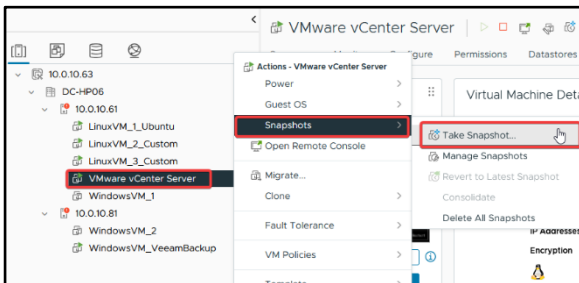
It is important to note that a snapshot is **not a backup**. Because it depends on the original base disk, if the base VM is lost or corrupted the snapshot is worthless too. Snapshots are meant to be short-lived, as a long-running snapshot keeps growing and degrades VM performance.

VMWARE

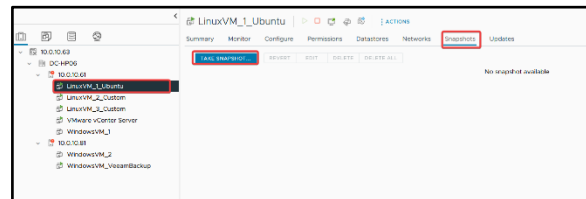


There are 2 ways to create snapshots in the vSphere client, one way is to use the action menu of the VM you want to take a snapshot of, or to manage them through the “Snapshot” tab.

For this lab we’ll test Snapshots on our vCenter Server and our Ubuntu Server.

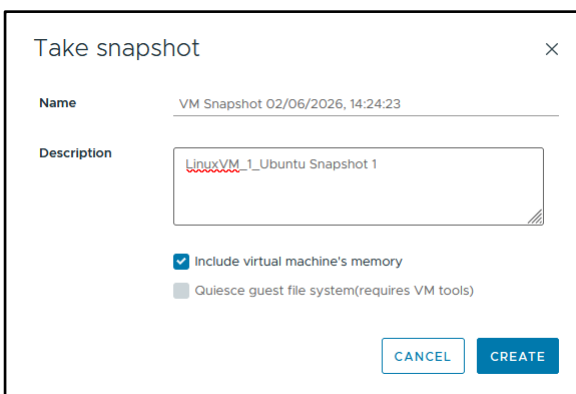


VMWare - Screenshot 114: Taking snapshot via action menu

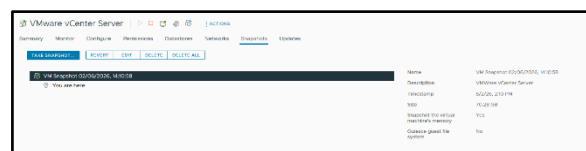


VMWare - Screenshot 115: Taking snapshot via Snapshot Tab

This is where the VM will create that state where the VM writes it into the delta layer. Now give the Snapshots a name and select to also save their memory.



VMWare - Screenshot 116: Snapshot naming of Ubuntu Server Snapshot

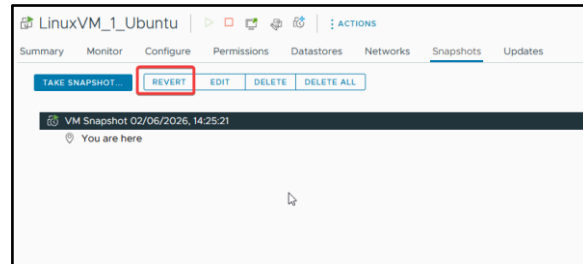


VMWare - Screenshot 117: Snapshot manager of vCenter Server

Now to actually test out the Snapshots, head into the Ubuntu Server and create a random file. Next head into the Snapshot management tab of the Server, select the Snapshot that was just created and click revert. This will set the state back to when we took the snapshot, so before creating the file. Meaning the storage of that file was within that delta data, meaning when we reverted back, all we did was remove the delta layer from the disk.

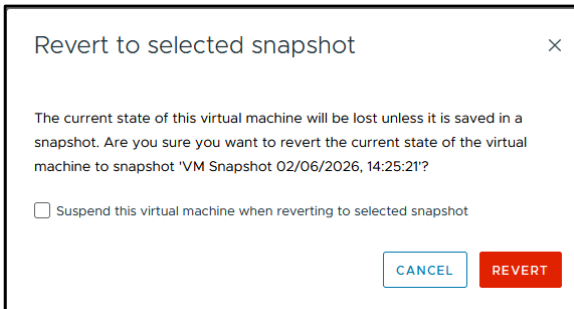
```
damn@damn:~$ mkdir snapshottest
damn@damn:~$ cd snapshottest/
damn@damn:~/snapshottest$ touch Test
damn@damn:~/snapshottest$ ls
Test
```

VMWare - Screenshot 118: Creating a folder and file for testing the snapshot

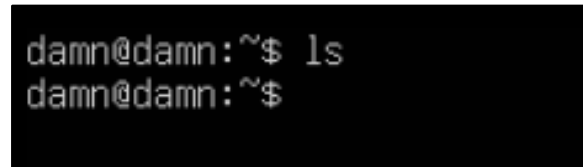


VMWare - Screenshot 119: Reverting back to a Snapshot

It will ask you to confirm the reversion, once you confirm it, head back into the interface of the VM and you'll see that the folder and file aren't there anymore.



VMWare - Screenshot 120: Snapshot reversion confirmation



VMWare - Screenshot 121: Confirmation of absence of created file

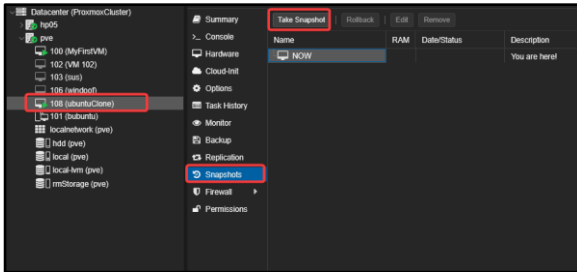
To delete Snapshots, in the Snapshot tab, select your Snapshot and click "Delete", or if you want to delete all of them select "Delete All"

It's worth understanding what these actually do. "Delete" doesn't throw the snapshot's data away – it commits that snapshot's delta back into the base disk and then removes the delta file, so the changes are kept and the chain is shortened. "Delete All" does the same for the whole chain at once, collapsing every snapshot back into the base disk. There's also a "Consolidate" option, which cleans up orphaned delta files that can be left behind when a delete didn't finish properly; vCenter usually warns you when a consolidation is needed. The classic mistake is to leave snapshots sitting around forever – the delta files keep growing and hurt performance – so they should be committed and removed once they're no longer needed.

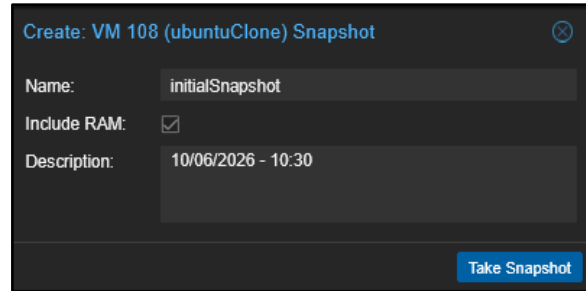
PROXMOX



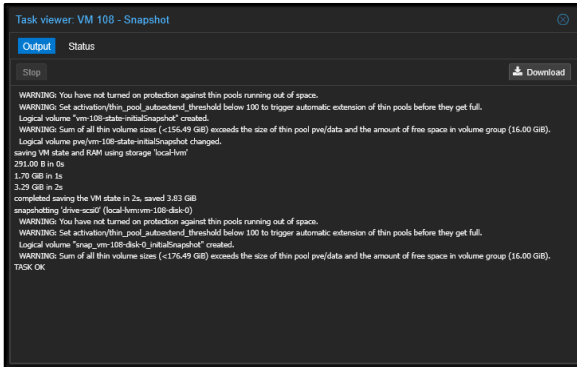
Snapshots in Proxmox are handled per VM, under the VM's own "Snapshots" tab, so there's no central manager like vSphere's. For this we'll snapshot our ubuntu-vm. Click "Take Snapshot", give it a name, and optionally tick "Include RAM" to also capture the running memory state – the same idea as saving the memory in VMware. With RAM included the snapshot can be rolled back to a fully running machine; without it, only the disk state is captured.



Proxmox - Screenshot 82: Creating initial snapshot

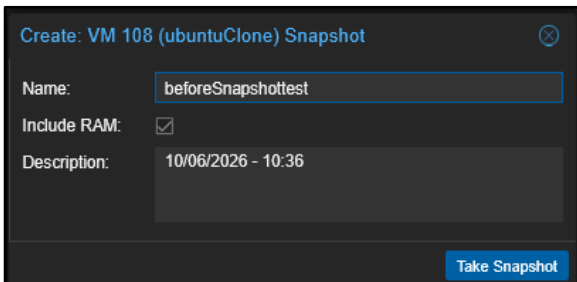


Proxmox - Screenshot 83: Entering initial Snapshot information

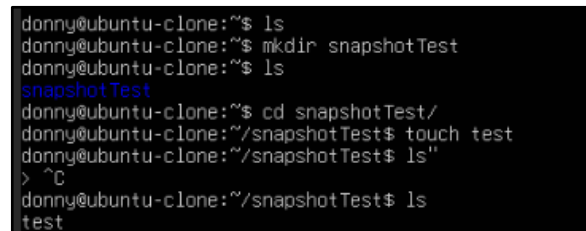


Proxmox - Screenshot 84: initial Snapshot status

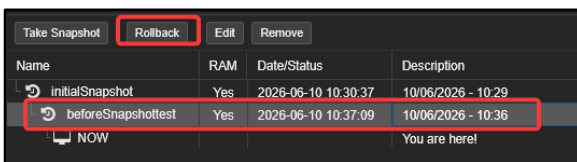
To test it we do the same as on the VMware side: take the snapshot, then inside the VM create a random file. Back in the Snapshots tab, select the snapshot and click “Rollback”. Proxmox asks to confirm, and after rolling back the file is gone – the VM is exactly as it was when the snapshot was taken.



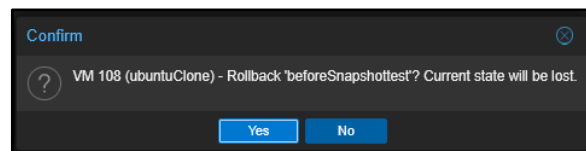
Proxmox - Screenshot 85: Creating snapshot before test



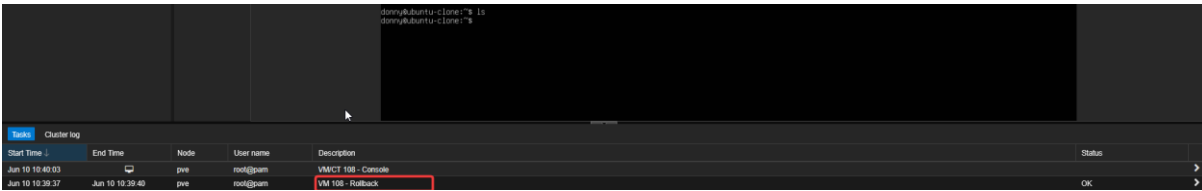
Proxmox - Screenshot 86: Creating a folder and file on VM



Proxmox - Screenshot 87: Rolling back to created Snapshot



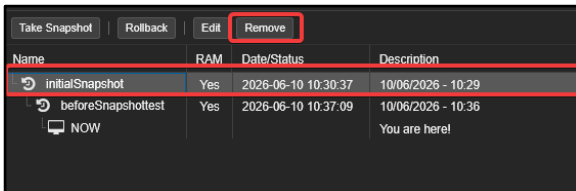
Proxmox - Screenshot 88: Snapshot rollback confirmation



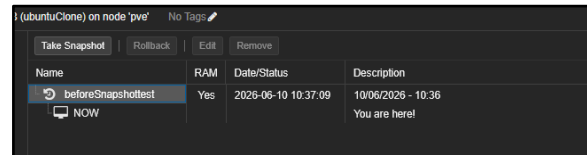
Proxmox - Screenshot 89: VM after rolling back

One important difference from VMware is that Proxmox ties snapshot support to the underlying storage type. Snapshots only work on storage that supports them: qcow2 (file-based), LVM-thin, ZFS or Ceph. A VM whose disk sits on plain LVM (our “hdd” volume group) cannot be snapshotted at all – the “Take Snapshot” button is simply unavailable. This is why it matters which storage a VM’s disk lives on, so we made sure our demo VM’s disk was on snapshot-capable storage before this step.

Deleting Snapshots will remove the state of the delta layer and add it to the base disk. For this we remove our initial snapshot, since we already have a snapshot past it to rollback to.



Proxmox - Screenshot 90: Removing Snapshot of VM



Proxmox - Screenshot 91: After Removing snapshot

COMPARISON



Compared to VMware, the snapshot feature itself is simpler to reach in Proxmox: it’s built straight into the VM’s panel with no vCenter required. The trade-off is the storage constraint above, where VMware will snapshot a VM regardless of the datastore type. The other practical difference is size: on the VMware side our example snapshot came out at 70.28 GB because the memory was included by default, while Proxmox only captures the RAM when you explicitly tick the box, which keeps snapshots smaller unless you actually need the running state.

A few do’s and don’ts apply to snapshots on either hypervisor. Do take a snapshot before any risky change – patching, updates or a configuration you might need to undo – and do delete it again once the change is confirmed good. Don’t treat a snapshot as a backup: it depends on the base disk, so if the base VM is lost or corrupted the snapshot goes with it. Don’t leave snapshots lying around for more than a day or two either, as a long-running snapshot keeps growing as the delta fills up and steadily degrades the VM’s performance. And only include the memory/RAM when you genuinely need the running state, otherwise you’re just wasting storage on something you won’t use.

TEMPLATES

DEFINITION

When a VM is created from a template it uses that template as a baseline for deployment. The template is not a VM itself, but acts as a master image for deploying new VMs for repeatability.

Templates rely on generalisation, so that VMs deployed from a template get their own identity and don't conflict with each other. For this to happen, the VM being converted into the template has its unique identifiers removed, like the hostname, MAC address and SSH keys.

This is different from cloning. A clone is a one-to-one copy of a single VM, and comes in two forms: a full clone, which is an exact and independent copy, and a linked clone, which shares the base disk with the source. A template, by contrast, is meant for mass, consistent deployment rather than a single copy.

A distinction between templates in ESXi and Proxmox, is that Proxmox allows for VMs deployed from templates to share and be linked to a base disk, while ESXi deploys independent VMs from templates by default.

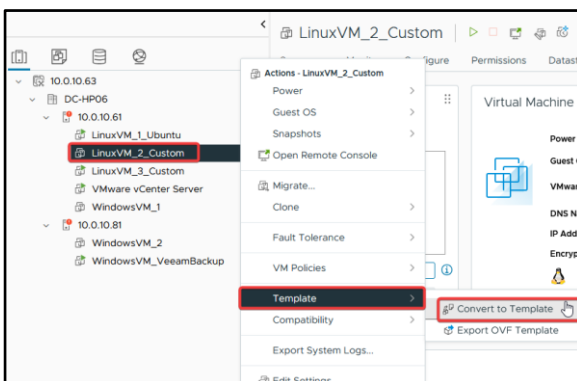
The main difference between cloning and using a template, is on the needs of the user, if they only need a one-time exact copy of a VM and its configurations through cloning, or they need a consistent way of deploying multiple of the same VMs.

VMWARE

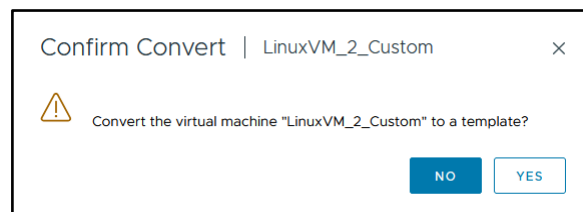


To convert a VM into a template in VMWare, head into the action menu of a VM and under the “Template” dropdown menu select “Convert to Template”. This will give you a confirmation screen, since after confirming, it will remove your VM from the list. For this we'll be using a custom Linux VM.

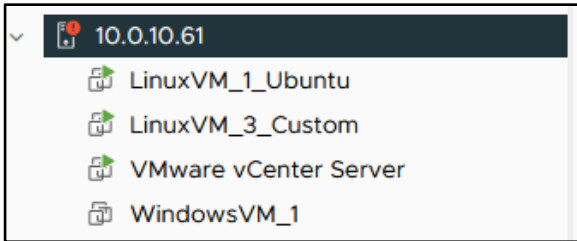
Unlike Proxmox, VMware doesn't treat this as permanent. A template can be turned straight back into a runnable VM at any time through the action menu, under “Template” → “Convert to Virtual Machine”, which is useful when you need to patch or update the master image and then convert it back. This is a clear difference from Proxmox, where the conversion only goes one way.



VMWare - Screenshot 122: Converting VM to Template

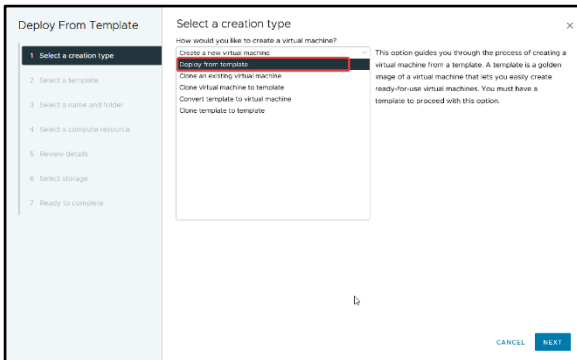


VMWare - Screenshot 123: Confirming conversion to template

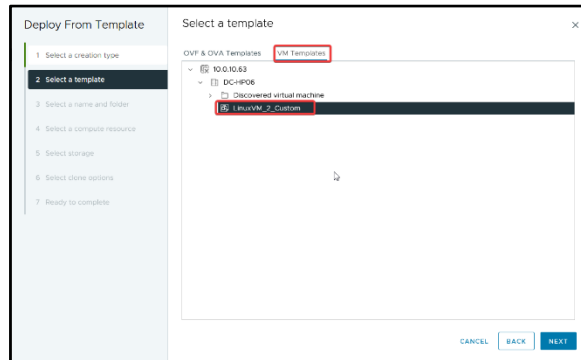


VMWare - Screenshot 124: Seeing that the converted VM has been removed

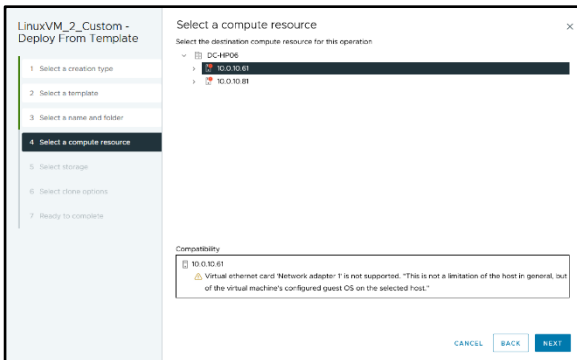
To then create a VM from the template, as usual deploy a new Virtual machine from the action menu of the host, this time for option we select “Deploy from template” and we select our Template we just created. As with every VM, we select our compute and storage.



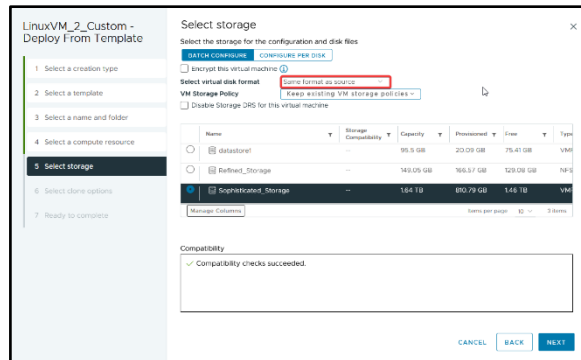
VMWare - Screenshot 125: Selecting template deployment as creation type



VMWare - Screenshot 126: Selecting template to deploy VM from



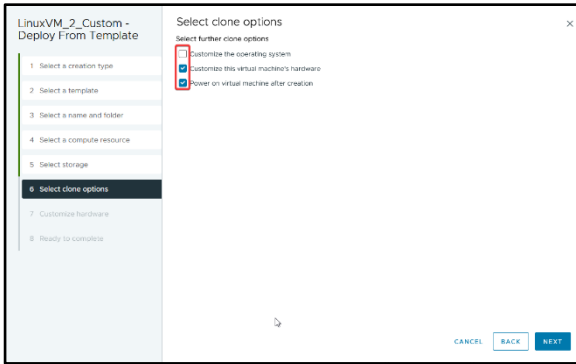
VMWare - Screenshot 127: Selecting compute for template VM



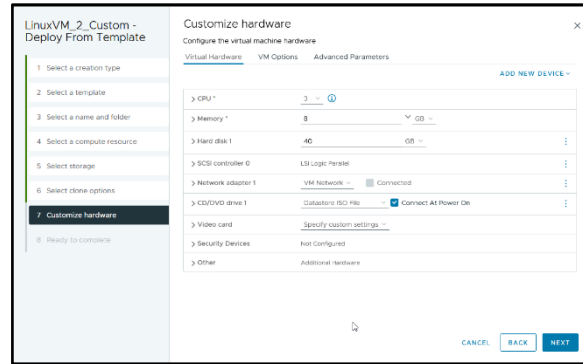
VMWare - Screenshot 128: Selecting storage for template VM

Next is the real purpose of templates, and that is the customisation options, first select the options you would like to customise, since we needed a specialised image of an operating system for the first option, we left it out here, but here you would make all the necessary changes to make each deployment unique for avoiding colliding information.

But here we change some of the hardware of the VM for deploying.



VMWare - Screenshot 129: Selecting clone options for template VM



VMWare - Screenshot 130: Customising hardware for template VM

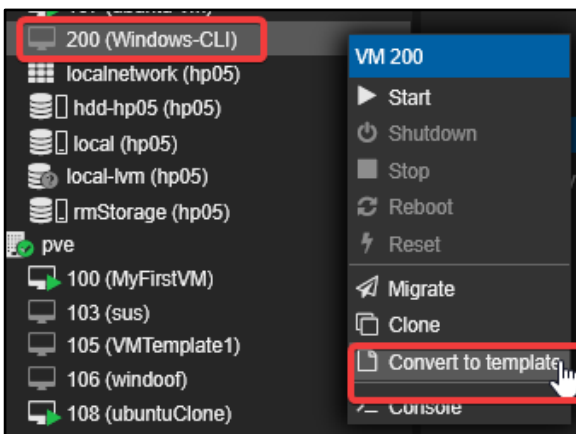
With that the VM can be deployed and runs without issue, if we were to create more we would need to change things within the VM or use the “Customise the operating system” option for the clone options, before deployment.

This is also where the “Customize the operating system” option matters for generalisation: although our template’s definition relies on each deployment getting its own identity, in our screenshots that box was left unchecked, so the deployed VMs actually came up sharing the source’s identity. To deploy several from the same template without conflicts, that option (backed by a guest customisation specification) is what strips and regenerates the hostname, SID/MAC and other unique identifiers per clone – the direct equivalent of Proxmox’s Cloud-Init step.

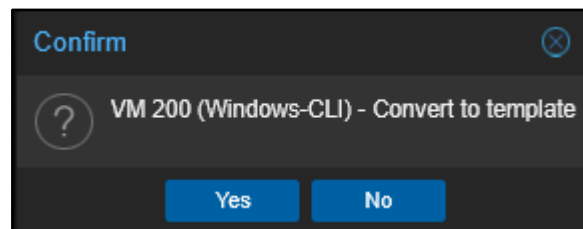
PROXMOX



Creating a template in Proxmox starts from an existing VM. We select the VM we want to base it on, open the “More” dropdown and choose “Convert to template” (the same can be done from the CLI with `qm template <vmid>`). Proxmox warns that this is permanent, and that’s the key point: converting to a template is one-way. Once a VM becomes a template it can no longer be started, and unlike VMWare there is no option to convert it back into a normal VM. So we always keep the original VM, or a clone of it, if we still need something we can boot.



Proxmox - Screenshot 92: Converting a VM into template

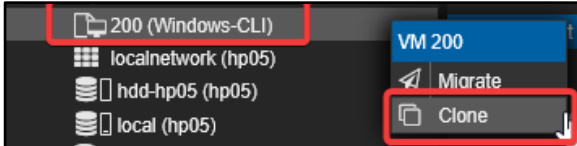


Proxmox - Screenshot 93: Confirming conversion to template

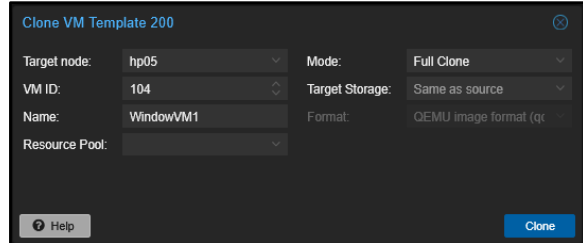


Proxmox - Screenshot 94: VM Template appearance

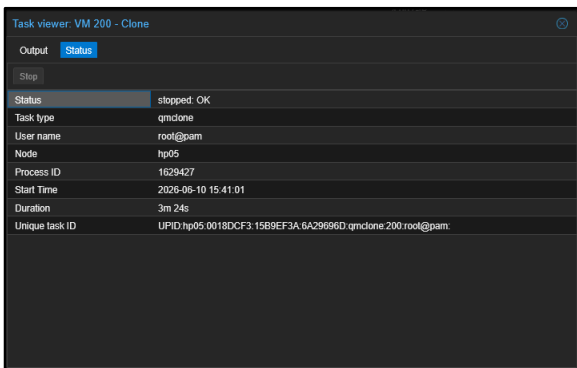
Deploying from a template is just cloning from it. We right-click the template and choose “Clone”, which is where the second clone mode finally becomes available. Because the source is now a template, the Mode dropdown offers both Full Clone and Linked Clone.



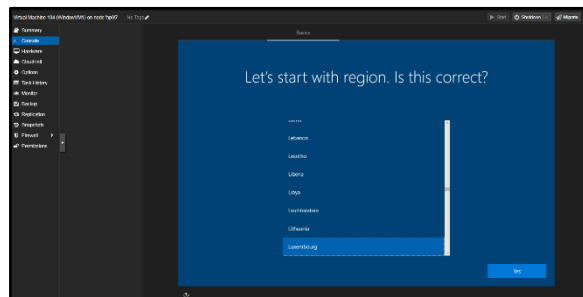
Proxmox - Screenshot 95: Creating clone of template



Proxmox - Screenshot 96: Entering template information



Proxmox - Screenshot 97: Template cloning status



Proxmox - Screenshot 98: Booted up VM created by template

A Full Clone copies the template’s disks into a brand-new independent VM. It works on any storage type and the result has no further link to the template, but it’s slower to create and uses the full disk space each time. A Linked Clone instead writes only its changes as a delta on top of the template’s base image, which makes it near-instant to create and very space-efficient – the trade-off being that the template can’t be deleted while any linked clone still depends on it. As with snapshots, linked clones need snapshot-capable storage (qcow2, LVM-thin, ZFS or Ceph) and are not possible on plain LVM.

There is one thing templates don’t solve on their own: generalisation. When Proxmox clones from a template it generates a new MAC address for each clone automatically, but everything inside the guest – the hostname, the machine-id, the SSH host keys and any static IP – is copied verbatim. So out of the box several clones of the same template will collide on the network and share identity. Proxmox’s answer to this is the same as VMware’s “Customize the operating system” step: a Cloud-Init drive (for Linux) or sysprep (for Windows). With a Cloud-Init disk attached to the template, each clone is given its own IP, hostname, user and SSH keys from the Cloud-Init tab (or via qm set) when it first boots, which is what makes a template genuinely reusable for mass deployment.

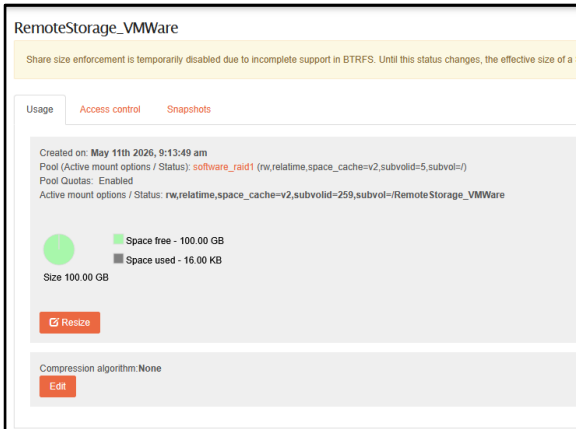
To pull the two together: cloning and templating solve different problems. A clone is the right tool when you want a one-off copy of a specific VM, identity and all – for example to duplicate a machine you've set up so you can experiment on the copy. A template is the right tool when you want to deploy many consistent machines from one known-good baseline over time. The template stays as an untouched master and you stamp out new VMs from it, ideally with Cloud-Init or guest customisation so each one comes up with its own identity. In short: clone for a single exact copy now, use a template for repeatable, generalised deployment later.

REMOTE STORAGE

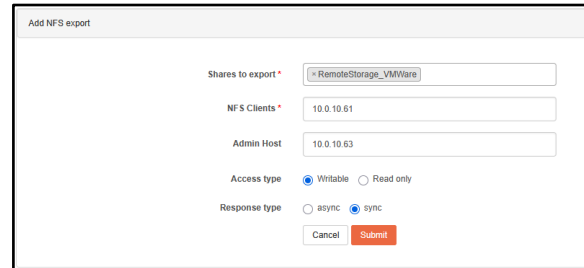
VMWARE

Before adding the Remote Storage for the ESXi hosts, we must first create an export on our NAS Server first. For us it's Rockstor.

So we first created a Share on the NAS and made 2 NFS exports of it, one for each ESXi host, with the vCenter Server as administrative unit.



VMWare - Screenshot 131: Remote Storage Share

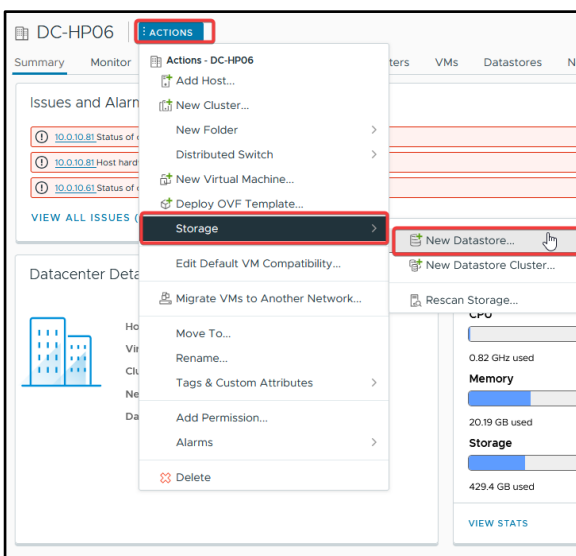


VMWare - Screenshot 132: Creating NFS Export

RemoteStorage_VMWare	10.0.10.81	no	sync	 
RemoteStorage_VMWare	10.0.10.61	no	sync	 

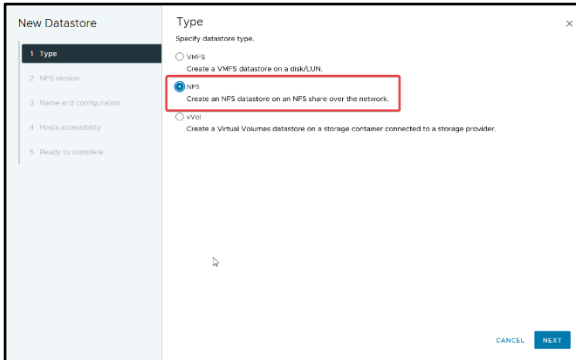
VMWare - Screenshot 133: Both NFS exports for ESXi hosts

Now to add our Remote NAS Storage in vSphere, open the action menu of our datacentre we created and under the “Storage” dropdown, select “New Datastore”.

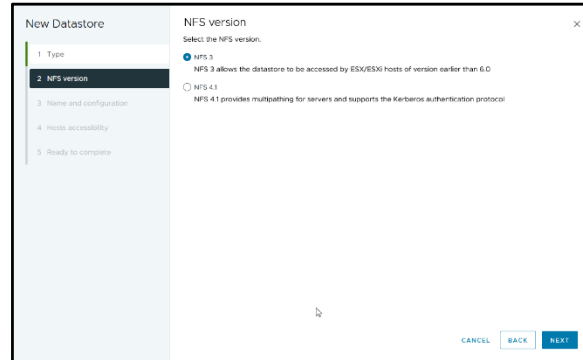


VMWare - Screenshot 134: Creating New Datastore

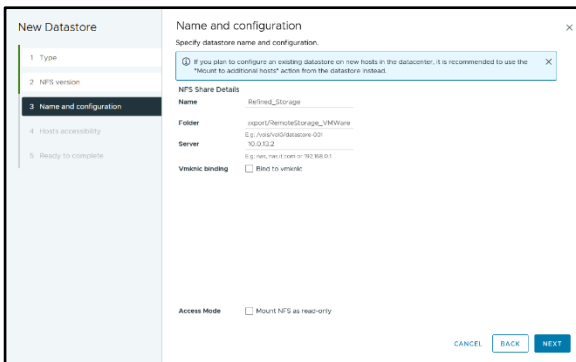
In the setup screen, select NFS as the datastore type and use NFS version 3. Then give your datastore a name and indicate the folder location of where the export is on the server, which you also indicate. Afterwards select both hosts for the accessibility. Finally verify your configuration and that the servers successfully created the NAS datastore.



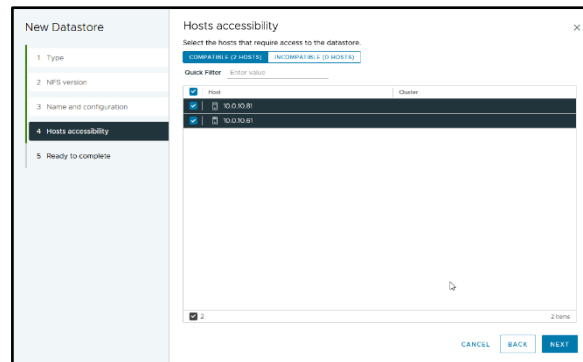
VMWare - Screenshot 135: Remote Storage - selecting datastore type



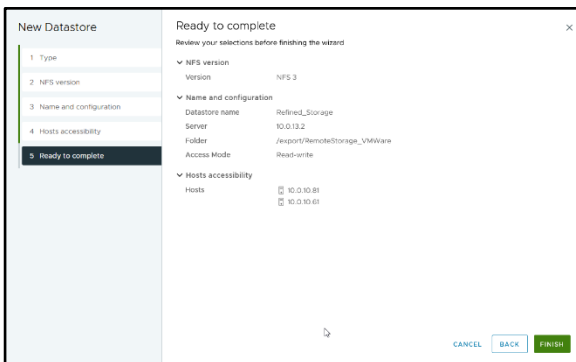
VMWare - Screenshot 136: Remote Storage - selecting NFS version



VMWare - Screenshot 137: Remote Storage - name and configuration



VMWare - Screenshot 138: Remote Storage - Host accessibility



VMWare - Screenshot 139: Verification of configuration

Create NAS datastore	10.0.10.63	Completed	HP06LOCAL\Administrator	20 ms	05/11/2026, 9:19:42 AM	05/11/2026, 9:19:42 AM	10.0.10.63
Create NAS datastore	10.0.10.81	Completed	HP06LOCAL\Administrator	14 ms	05/11/2026, 9:19:42 AM	05/11/2026, 9:19:42 AM	10.0.10.63

VMWare - Screenshot 140: Successful creation of NAS datastore

One error that occurred was a denial of the mount request from the NFS server, this was due to mistake of using the wrong folder during step 3, the name and configuration step, so the ESXi hosts never found the correct mount and the NFS server just denied them.

Create NAS datastore	10.0.10.61	● An error occurred during host configuration: . Operation failed, diagnostics report: Mount failed: Unable to complete Sysinfo operation. Please see the VMkernel log file for more details.: The NFS server denied the mount request	HP06.LOCAL\Administrator	3 ms	05/11/2026, 9:08:13 AM	05/11/2026, 9:08:13 AM	10.0.10.63
Create NAS datastore	10.0.10.61	● An error occurred during host configuration: . Operation failed, diagnostics report: Mount failed: Unable to complete Sysinfo operation. Please see the VMkernel log file for more details.: The NFS server denied the mount request	HP06.LOCAL\Administrator	12 ms	05/11/2026, 9:08:13 AM	05/11/2026, 9:08:13 AM	10.0.10.63

VMWare - Screenshot 141: Remote Storage Error

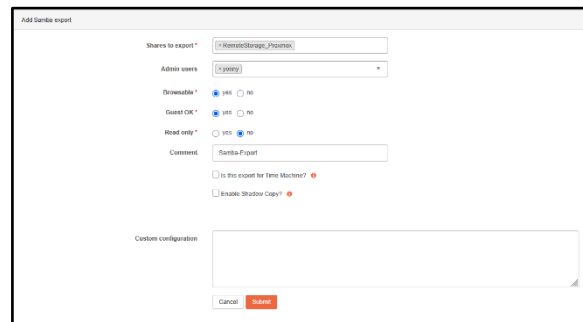
PROXMOX



Before setting up the remote storage we create a share on the NAS, still using Rockstor and then we create a SMB Export of it to use for Proxmox. For the SMB Export we use our admin user.



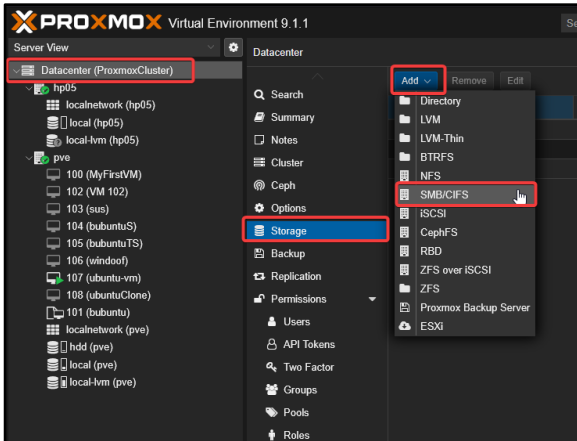
Proxmox - Screenshot 99: Creating Proxmox file share



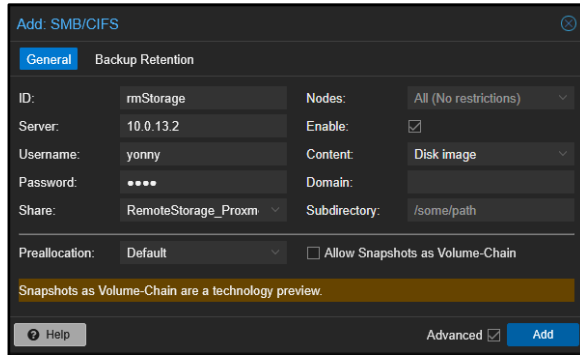
Proxmox - Screenshot 100: Creating SMB Export

For the remote storage on the Proxmox side we deliberately used a different protocol from the VMware half. The project asks for two different storage protocols across the whole setup, and since VMware already mounts its NAS over NFS, we added the Proxmox remote storage over SMB/CIFS instead, so the requirement is met with two distinct protocols.

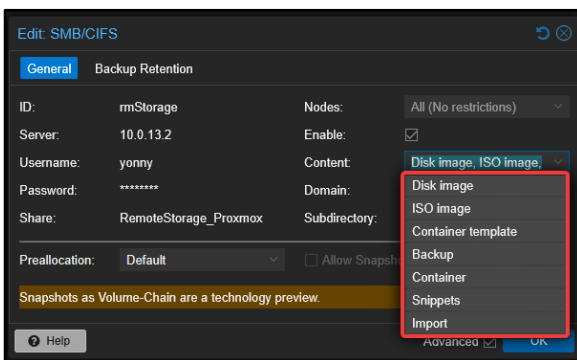
Adding it is done at the Datacenter level rather than per node: Datacenter → Storage → Add → SMB/CIFS. We point it at the NAS server, give it the share, and Proxmox mounts it under `/mnt/pve/<storage-id>` on every node in the cluster. Adding it at Datacenter level after the cluster was formed matters, because the storage configuration lives in the shared `/etc/pve`, so both HP05 and HP07 inherit the mount cleanly. We also tick the content types this storage is allowed to hold (disk images and, importantly, VZDump backup files) so it can be used as a backup target as well.



Proxmox - Screenshot 101: Adding a SMB/CIFS Storage



Proxmox - Screenshot 102: Entering Storage information



Proxmox - Screenshot 103: Editing Content settings of storage

rmStorage	SMB/CIFS	Disk image	/mnt/pve/rmStorage	Yes	Yes
-----------	----------	------------	--------------------	-----	-----

Proxmox - Screenshot 104: The storage in the list

Compared to VMware, the mechanics are similar – point the hypervisor at a NAS export and mount it – but Proxmox attaches the storage once at the cluster level and every node picks it up, rather than mounting the datastore on each ESXi host individually. The other difference is the deliberate choice of protocol (SMB/CIFS here versus NFS on VMware) to satisfy the two-protocol requirement.

LIVE MIGRATION

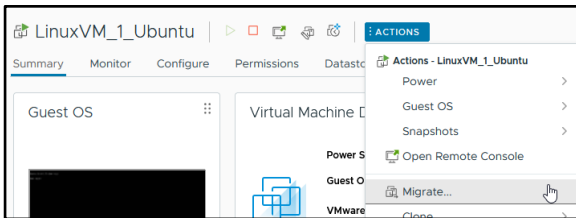
VMWARE



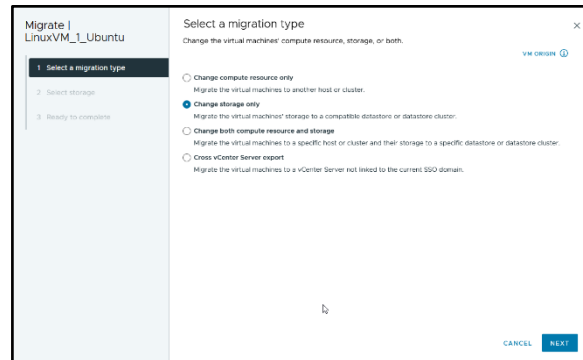
VM STORAGE MIGRATION

For our VM Storage Migration we'll be using our Remote Storage datastore we created before.

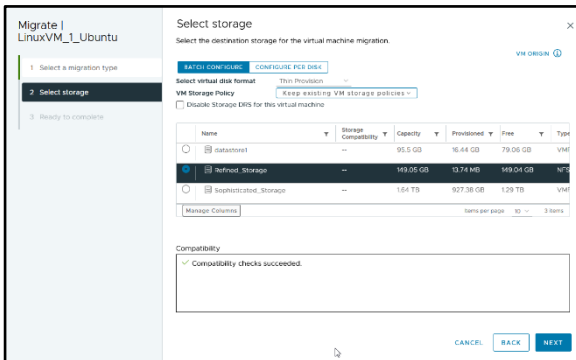
Firstly, to migrate a VM from one storage to another, use the action menu of the VM and click on "Migrate". Then in the window, select "Change Storage only" and select the Remote Storage datastore.



VMWare - Screenshot 142: Migrating VM action menu

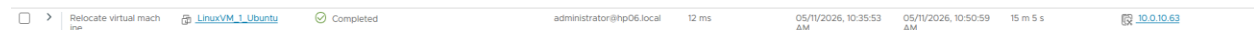


VMWare - Screenshot 143: VM Storage migration - selecting migration type

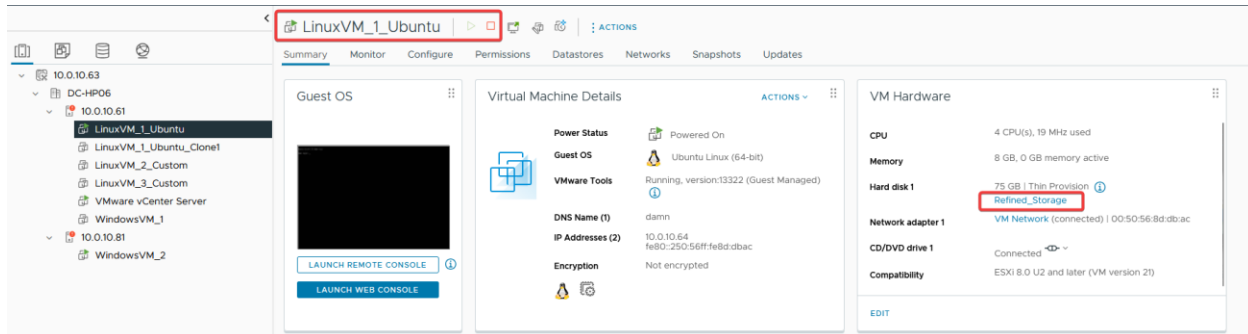


VMWare - Screenshot 144: Selecting migration storage destination

With that the VM will begin transferring over to the storage, note that we transferred over a running VM, after the transfer we see that the VM is still running.

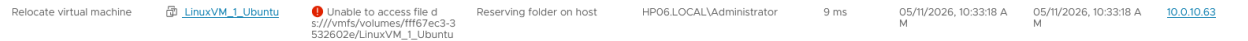


VMWare - Screenshot 145: After migrating VM Storage

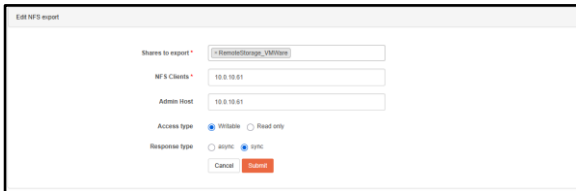


VMWare - Screenshot 146: VM after migrating storage

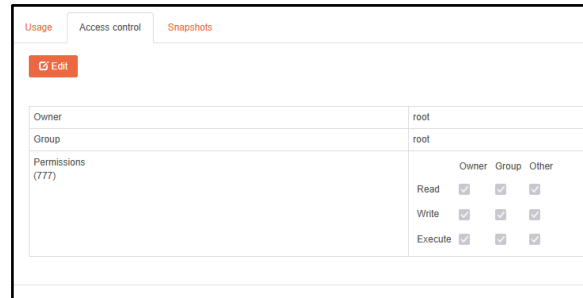
One error that we encountered was the ESXi host being unable to access the volumes of the Remote Storage, this was fixed by changing the admin host of the share and also due to being unable to get a correct fix, despite being insecure, giving the permissions for others to write to the share. Do not implement this fix in a real scenario, use a NAS Software that supports the usage of ESXi hosts.



VMWare - Screenshot 147: VM storage migration error



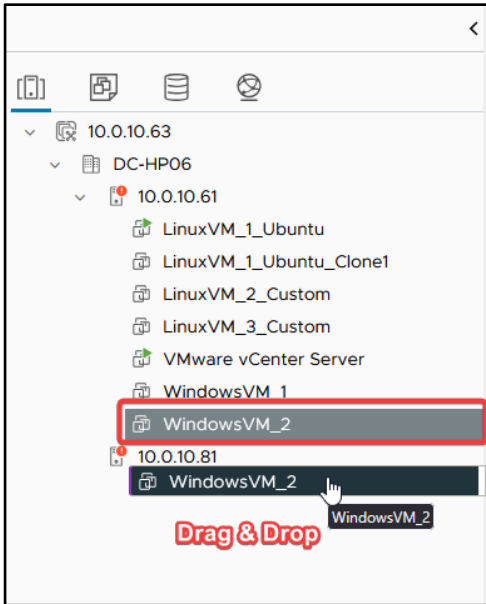
VMWare - Screenshot 148: Changing Admin host of the share



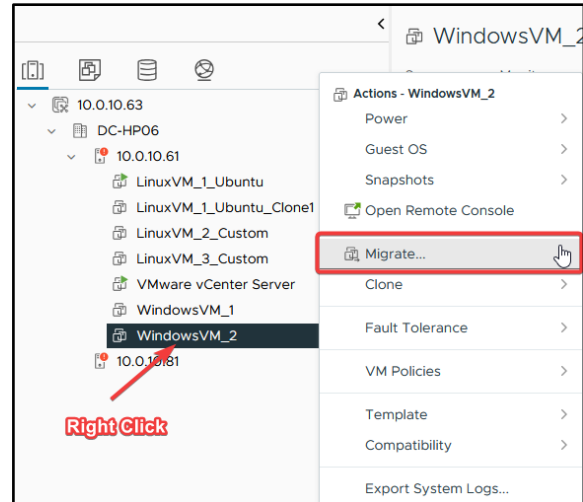
VMWare - Screenshot 149: Changing permissions of the share

VM HOST MIGRATION

Transferring a VM across ESXi hosts can be done in 2 ways, one through the action menu like we did for the Storage, or by dragging and dropping the VM onto the ESXi Host.



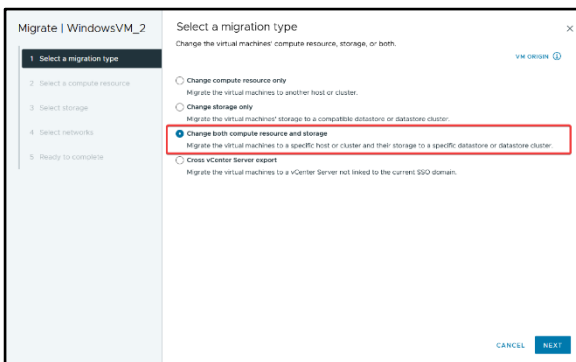
VMWare - Screenshot 150: VM Host migration via Drag & Drop



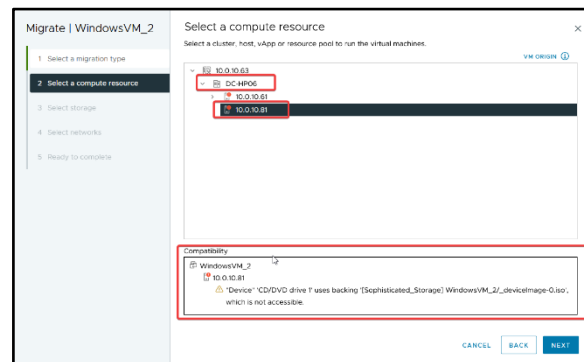
VMWare - Screenshot 151: VM host migration via action menu

Next similar to the migration of storage, we select our migration type for this, we choose “Change both compute resource and storage”. Next select our other host and it’s corresponding HDD datastore we made for it. Newly we also need to select a different Network that’s present on the ESXi host.

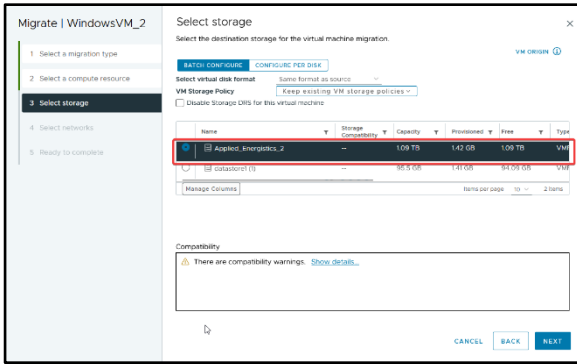
You may get a compatibility error, this is from the ISO image within the Disk drive that’s only accessible to the source host, but if you already had started and installed the VM, you won’t need to mind the issue.



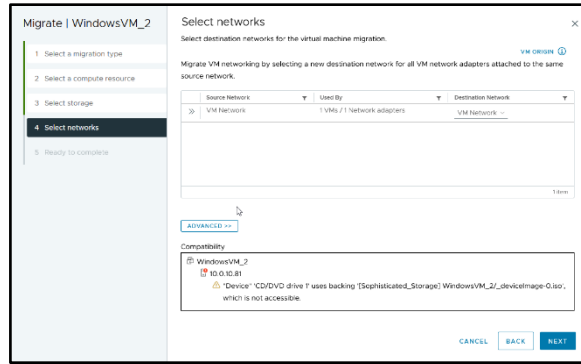
VMWare - Screenshot 152: VM Host Migration selecting migration type



VMWare - Screenshot 153: Selecting migration host destination



VMWare - Screenshot 154: Selecting Storage of new host of transferred VM

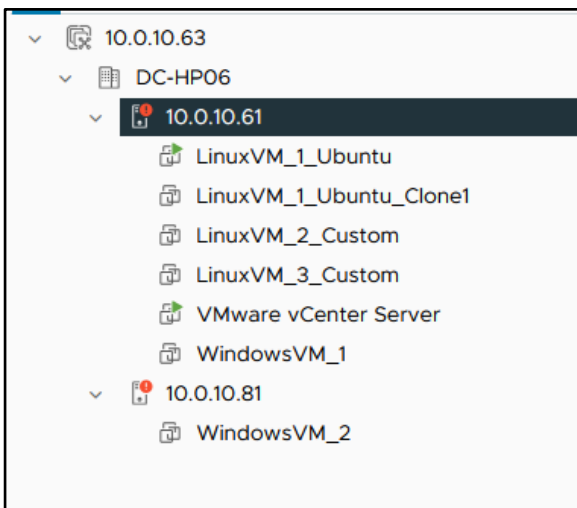


VMWare - Screenshot 155: Selecting network of migrated VM

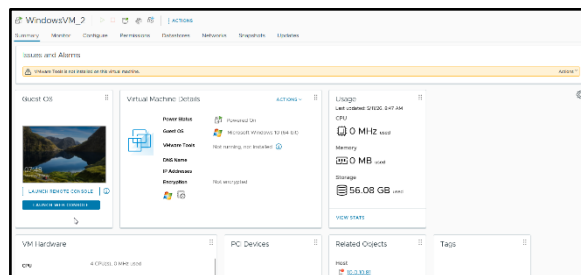
Once the information has been confirmed and commenced, you'll see it in the task bar below, afterwards you'll be able to confirm that the VM successfully migrated, by checking it's location and if it starts.

Task Name	Target	Status	Details	Initiator	Quarantined For	Start Time	Completion Time	Server
Relocate virtual machine	WindowsVM_2	20%	Copying Virtual Machine Files	HPO6LOCAL\Administrator	8 ms	05/11/2026, 8:32:12 AM		10.0.10.63

VMWare - Screenshot 156: After commencing Host migration



VMWare - Screenshot 157: VM after host migration



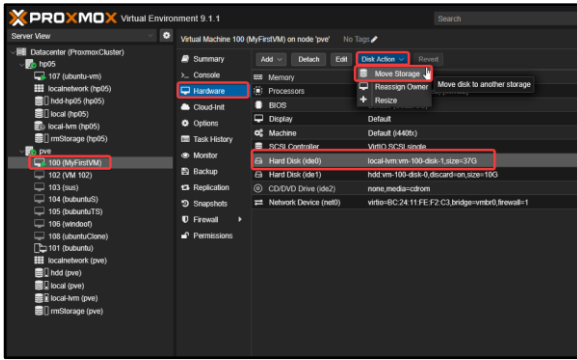
VMWare - Screenshot 158: VM starting successfully after host migration

PROXMOX

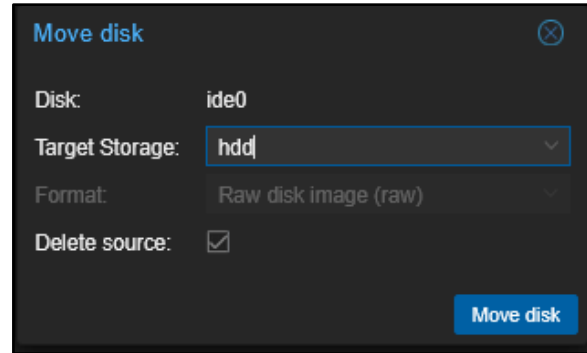


VM STORAGE MIGRATION

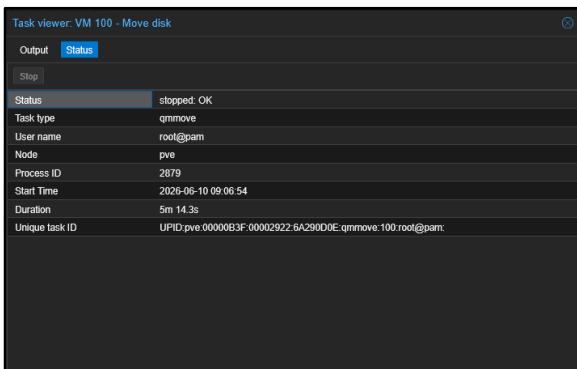
To move a VM's disk from one storage to another, head to the Hardware tab of the VM and select the hard disk. Then open the "Disk Action" dropdown and choose "Move Storage". In the dialog we pick the target storage to move the disk onto and, if we want the original copy cleaned up automatically, tick "Delete source". Proxmox copies the disk across and the task can be followed in the log. This also works while the VM is running, so the disk can be relocated with no downtime – the storage-side equivalent of the host migration below.



Proxmox - Screenshot 105: Moving VM Storage



Proxmox - Screenshot 106: Selecting target storage

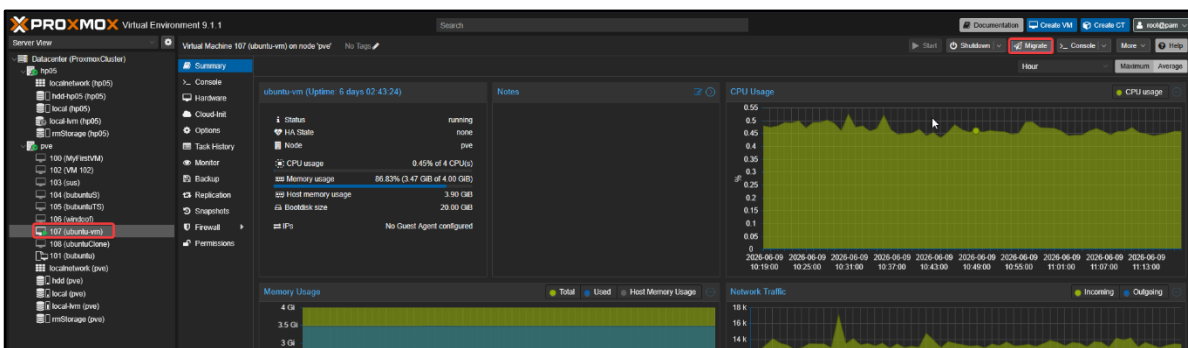


Proxmox - Screenshot 107: Storage migration status

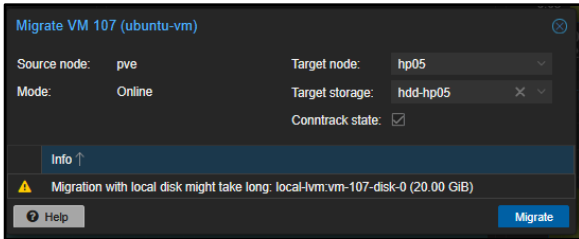
VM HOST MIGRATION

Migration in Proxmox moves a VM from one node to the other, here between HP05 and HP07. The shared NAS storage we added earlier is the prerequisite that makes this smooth: because both nodes already see the same datastore, the VM's disk doesn't have to be copied across when it lives on shared storage. We migrate from the VM's "Migrate" button, pick the target node, and start it.

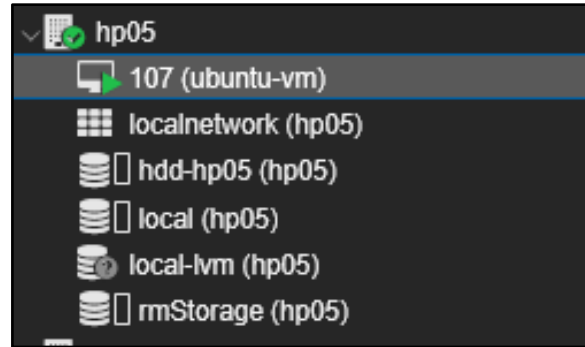
Before being able to migrate, make sure that no media is left within the CD/DVD Drive, for that just head to the hardware and Edit that drive to have no media.



Proxmox - Screenshot 108: Migrating VM



Proxmox - Screenshot 109: Selecting Target host and storage



Proxmox - Screenshot 110: VM after migrating to new host

We tested both states. With the VM powered off it's an offline migration: Proxmox simply moves the configuration (and any local disks) and the VM ends up on the other node. With the VM running it performs a live migration, streaming the memory state across so the VM keeps running throughout with no downtime, which is the more impressive case to demonstrate.

Jun 09 11:21:19	Jun 09 11:21:21	hp05	root@cam	VM 107 - Start	OK
Jun 09 11:21:17	Jun 09 11:24:24	pve	root@cam	VM 107 - Migrate	OK

Proxmox - Screenshot 111: VM Migration status of live migrating

Two things are worth flagging. First, a gotcha we have to keep in mind: if a VM's CPU type is set to "host" (which we used for performance), live migration can fail when the two nodes have different CPU generations, because the guest is pinned to features the destination may not have – in that case a more generic CPU type is the safer choice for migration. Second, the two-node cluster caveat applies here too: with only HP05 and HP07, the cluster needs both nodes up to keep quorum, so if one node is down the other can't happily make cluster changes on its own.

This lines up with VMware's vMotion and Storage vMotion: the same two ideas of moving compute between hosts and moving a VM's storage, both relying on shared storage being reachable by the destination. The main difference is that Proxmox rolls host and storage migration into the one "Migrate" dialog.

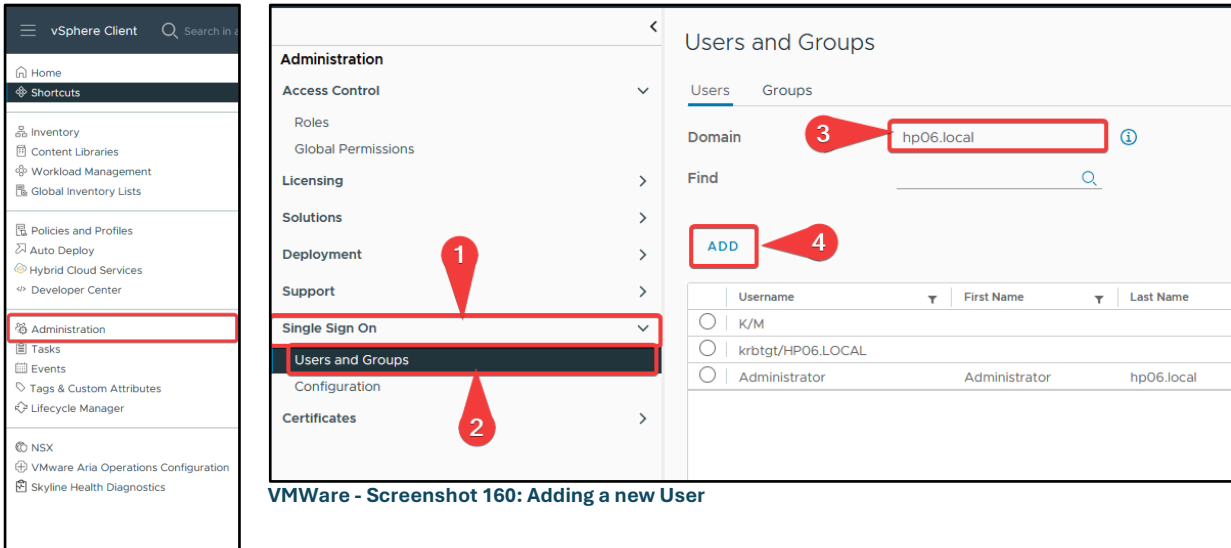
MANAGE USERS

VMWARE



To add new users and groups to vSphere, navigate to the Administration tab in either the side menu or home page of the client and find the “Single Sign On” section under that “Users and Groups”.

To add a user make sure you’re on the right domain for managing the vSphere client, and click “Add” in the Users tab. Give this user a name and password, and further information if necessary.



VMWare - Screenshot 160: Adding a new User

VMWare - Screenshot 159: Administration tab navigation

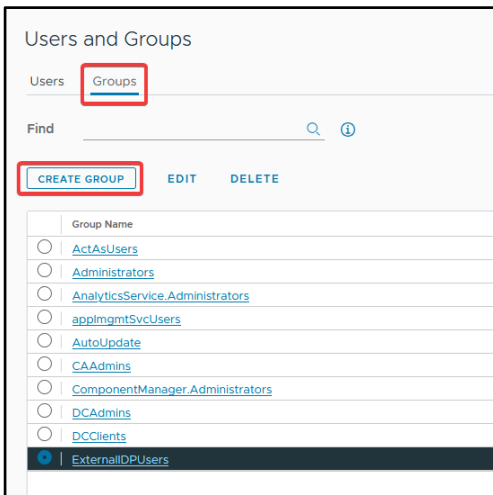
The 'Add User' dialog box is shown with the following fields:

- Username: Yonny
- Password: [masked]
- Confirm Password: [masked]
- First Name: [empty]
- Last Name: [empty]
- Email: [empty]
- Description: [empty]

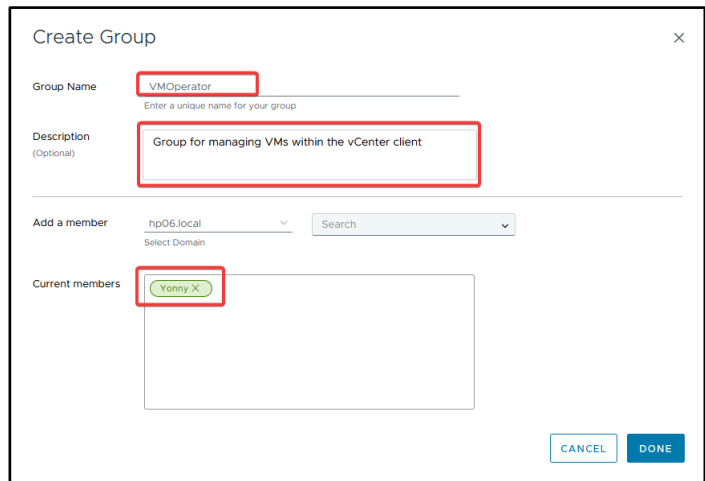
Buttons: CANCEL, ADD

VMWare - Screenshot 161: New User information

Once the user is added, create a new group to add our user into. In the Group tab of Users and Groups, click on “Create Group”. Give this Group a name and description, below search for the user we just created and add them into current members.



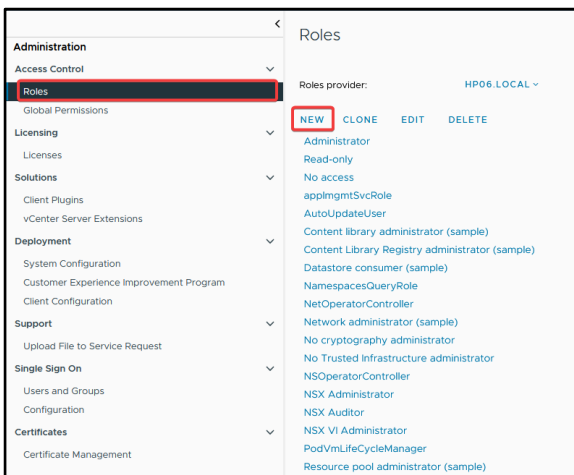
VMWare - Screenshot 162: Creating new group



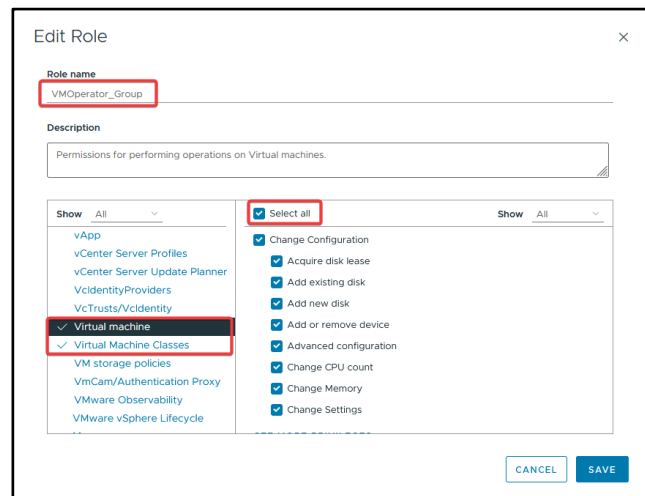
VMWare - Screenshot 163: New group information

Once that is done, it is now possible to give them roles, so a set rules that tell them what they’re allowed to do on the vSphere client.

For that navigate to the “Roles” tab of the “Access Control” section within the “Administration” screen and create a new Role. Next we want our user to manage VMs within vSphere so we give the role the necessary permissions to do so. So select all for Virtual Machine and Virtual Machine Classes permissions.



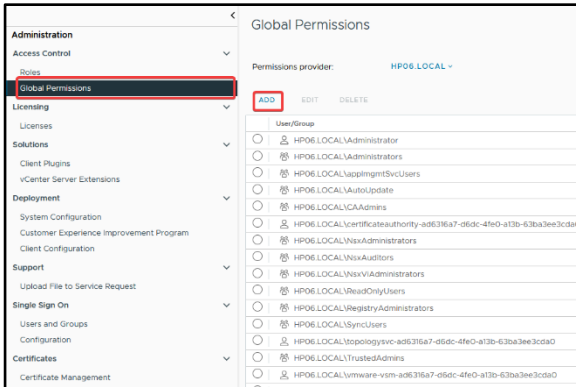
VMWare - Screenshot 164: Creating new Role



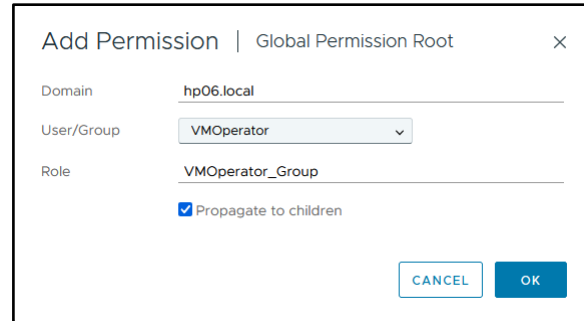
VMWare - Screenshot 165: New Role information

Once we created the role we also need to assign the permissions to our new group we created.

So head to the “Global Permission” tab under the same section and click “Add”. Here we want to specify our domain and the group and role we want to assign, make sure to tick the “Propagate to children” option so that our User also gets these permissions.

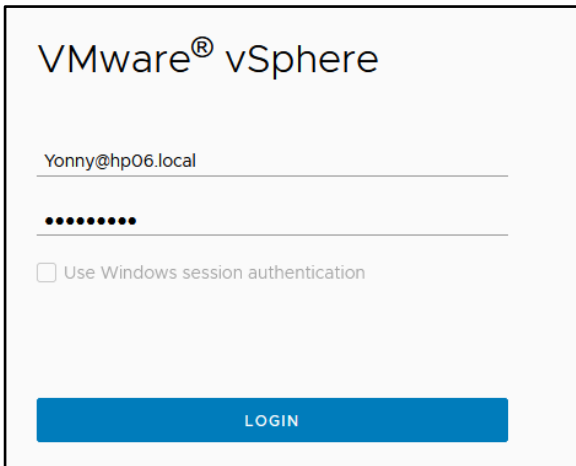


VMWare - Screenshot 166: Adding new Global permission

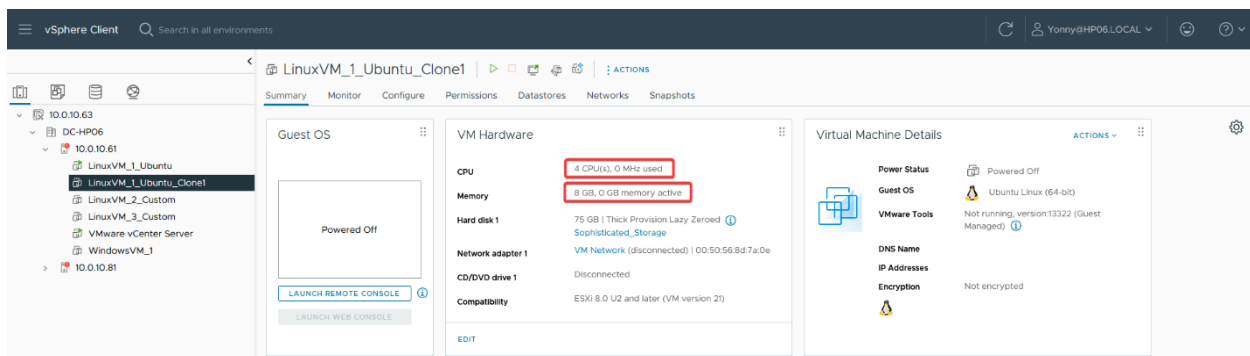


VMWare - Screenshot 167: Assigning Permission to group

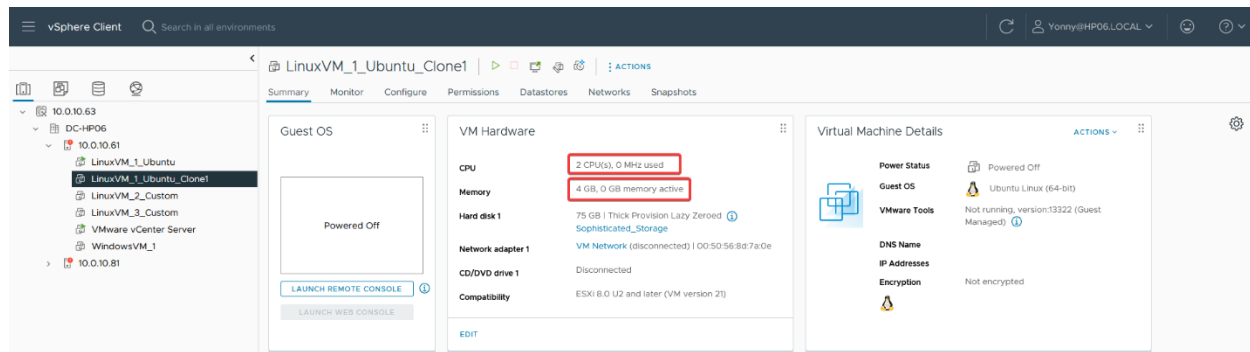
With that done we can now test out the new permissions with the user. Log out of the administrator account and into the new user account. From here select any VM and attempt to edit their settings, if everything was set correctly you'll be able to successfully edit the settings of the VM.



VMWare - Screenshot 168: Logging in with new User



VMWare - Screenshot 169: Before changing settings with new user

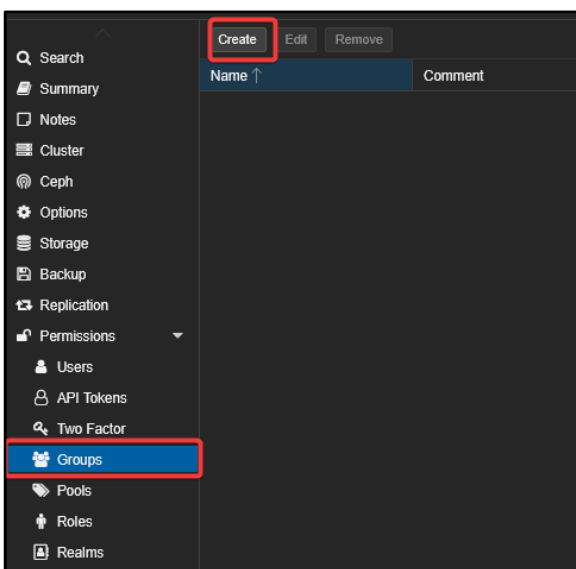


VMWare - Screenshot 170: After changing settings with new user

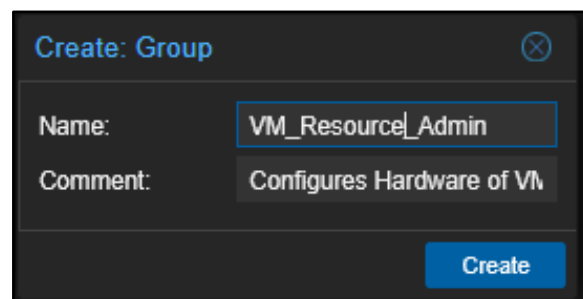
PROXMOX



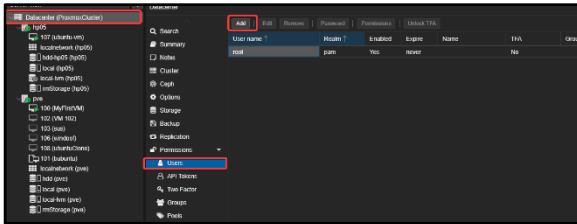
User management in Proxmox is built around realms (authentication sources). The two that matter here are PAM and PVE. A PAM user is a real Linux account on the host, so it can only be created if that system user already exists – we hit exactly this when setting a password for a PAM user “donny” failed because there was no matching Linux account. For self-contained users that live only inside Proxmox, the PVE realm (“Proxmox VE authentication server”) is the one to use, so that’s what we use to create our user under Datacenter → Permissions → Users.



Proxmox - Screenshot 112: Creating Group



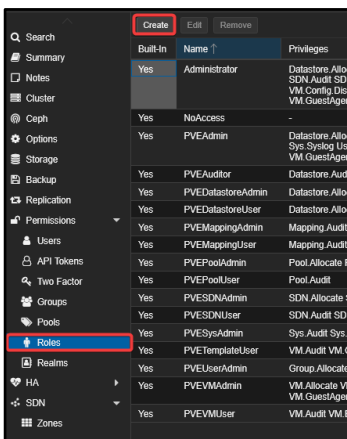
Proxmox - Screenshot 113: Entering group information



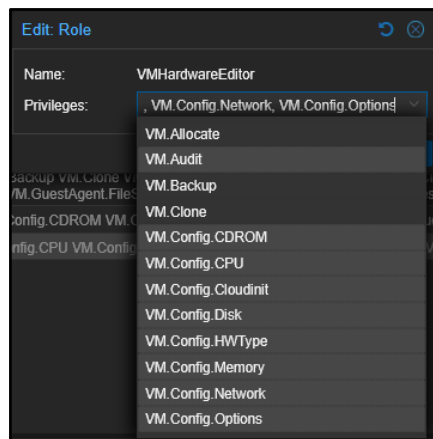
Proxmox - Screenshot 114: Adding User

Proxmox - Screenshot 115: Entering user information for PVE

Permissions are then granted by combining a role (a set of privileges) with a path. We create a custom role under Datacenter → Permissions → Roles with the privileges our user needs. An important catch here: a role needs the VM.Audit privilege for VMs to even be visible to the user – leave it out and the user logs in to an empty screen even though the VM.Config.* privileges are set. So we include VM.Audit alongside the VM.Config privileges that let them actually change settings.

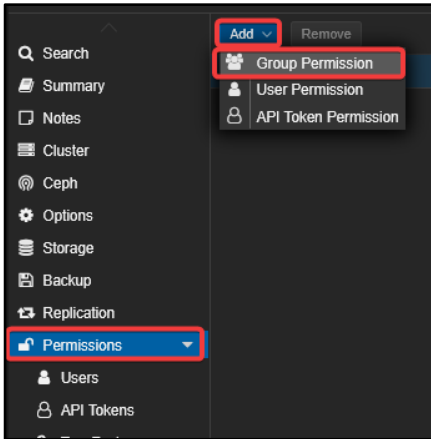


Proxmox - Screenshot 116: Creating Role

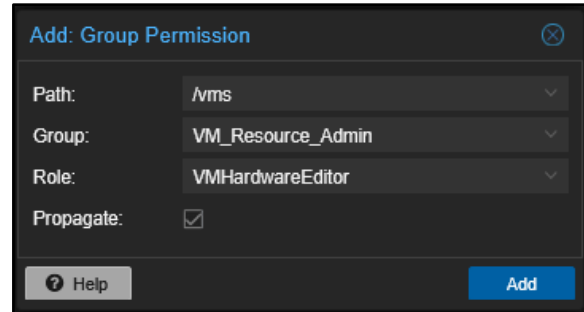


Proxmox - Screenshot 117: Editing role privileges

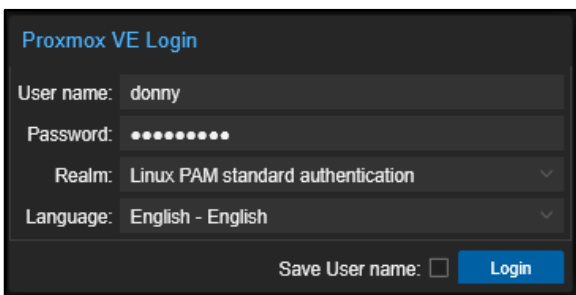
Finally we assign the permission under Datacenter → Permissions → Add. This is where Proxmox differs most from VMware: instead of attaching a role to an object in an inventory tree, you attach the role to an ACL path. Paths describe what the permission covers – “/” for everything, “/vms” for all VMs, “/vms/<vmid>” for a single VM, and similarly “/nodes”, “/storage”, “/pool/<id>” and “/access”. We tick “Propagate” so the permission also applies to everything beneath the chosen path. After assigning it we log in as the user to confirm they can see and edit only what we allowed.



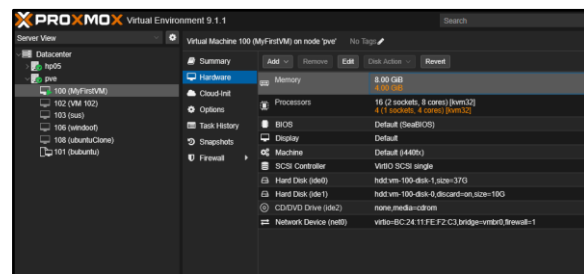
Proxmox - Screenshot 118: Adding Group Permission



Proxmox - Screenshot 119: Assigning Group permission



Proxmox - Screenshot 120: Logging in with new user



Proxmox - Screenshot 121: After editing settings with the user

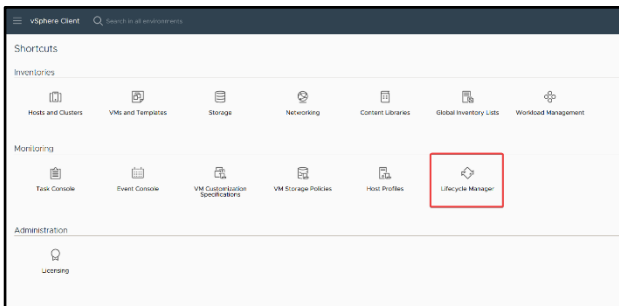
Compared to vSphere, the building blocks are the same – users, roles and propagation – but the models differ in two ways. Proxmox separates authentication into explicit realms (PAM vs PVE), and it assigns permissions to text-based ACL paths rather than to objects picked from an inventory tree. The VM.Audit requirement is also a Proxmox-specific quirk: visibility is an explicit privilege, whereas vSphere tends to show objects implicitly once any permission is granted.

UPDATE & PATCHING

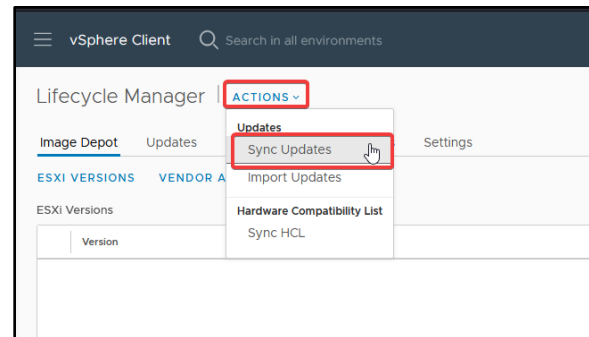
VMWARE



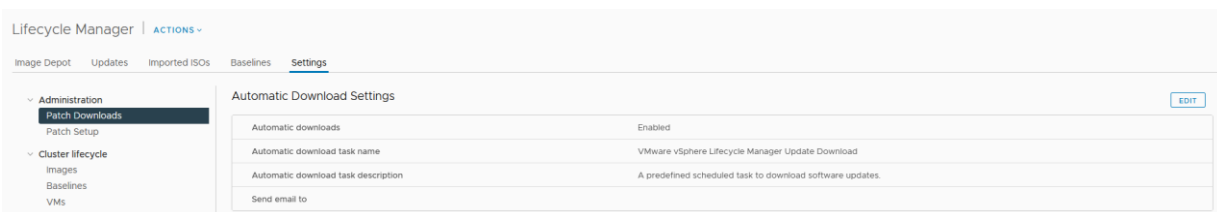
To manage the updates and patching of VMWare, head to the Lifecycle Manager of vSphere, in either the home page or side bar. From there open the action menu and click “Sync updates”, this syncs all of new patches currently available for vSphere or ESXi. To make sure your patches are up to date, enable “Automatic downloads in the Settings tab, and then under the Administration section in the “Patch downloads” tab.



VMWare - Screenshot 171: Lifecycle Manager navigation



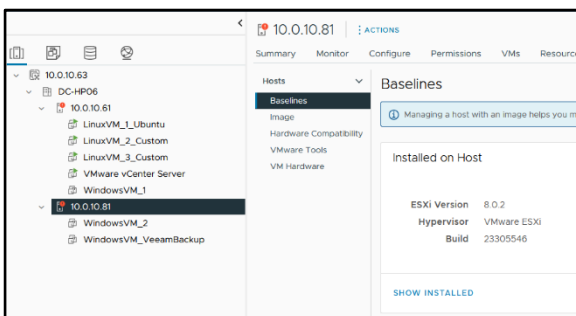
VMWare - Screenshot 172: Syncing updates



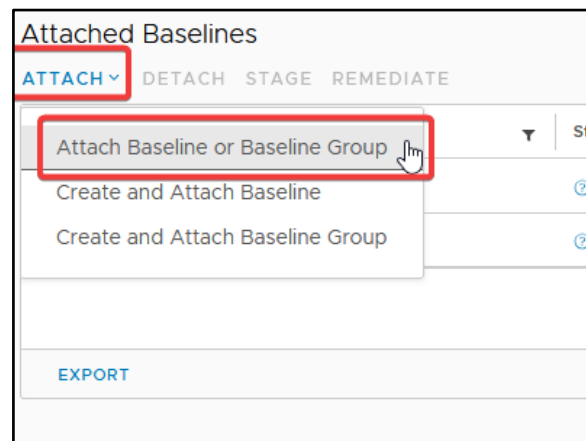
VMWare - Screenshot 173: Enabling Automatic Patch downloads

Before patching a host, you have to make sure every VM on that host is powered off.

To actually patch a host on vSphere, head back to “the Hosts and Clusters” Menu, and select a host. Then head to the “Baseline” tab and Attach a baseline.

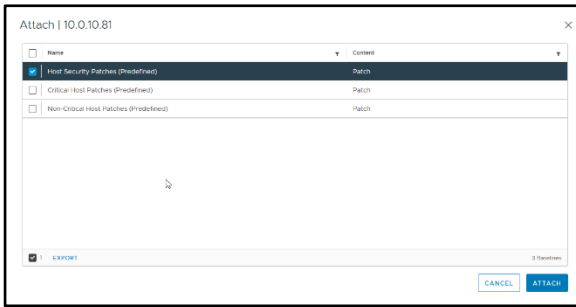


VMWare - Screenshot 174: Host Baselines tab

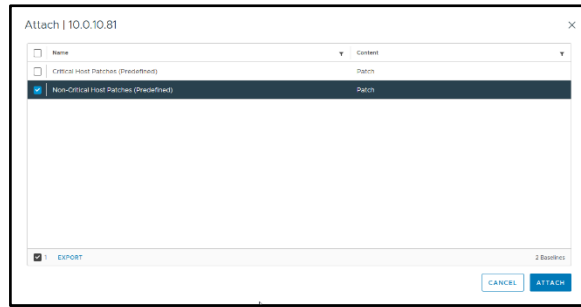


VMWare - Screenshot 175: Attaching baseline to host

For this lab attach both the “Host Security Patches” and “Non-Critical Host Patches”.

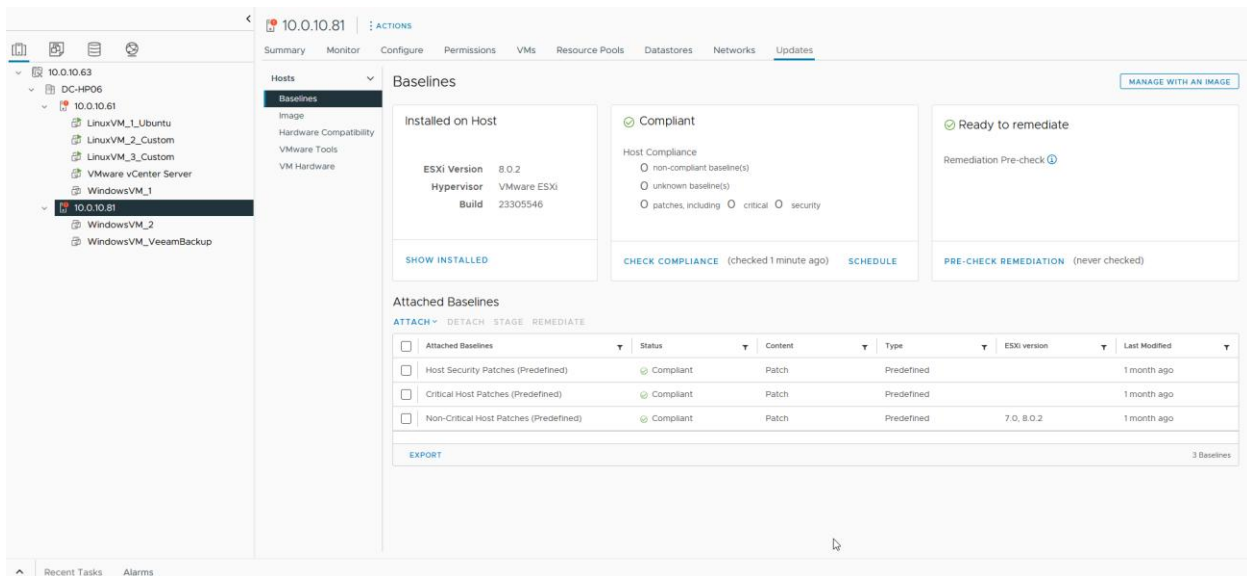


VMWare - Screenshot 176: Attaching Security Patches



VMWare - Screenshot 177: Attaching Non-Critical Host Patches

Now you'll be able to view the patches in the baselines of the ESXi host.



You can also create custom images with your own patches on them and upload them to the ESXi to create your baselines, however this lab won't cover them.

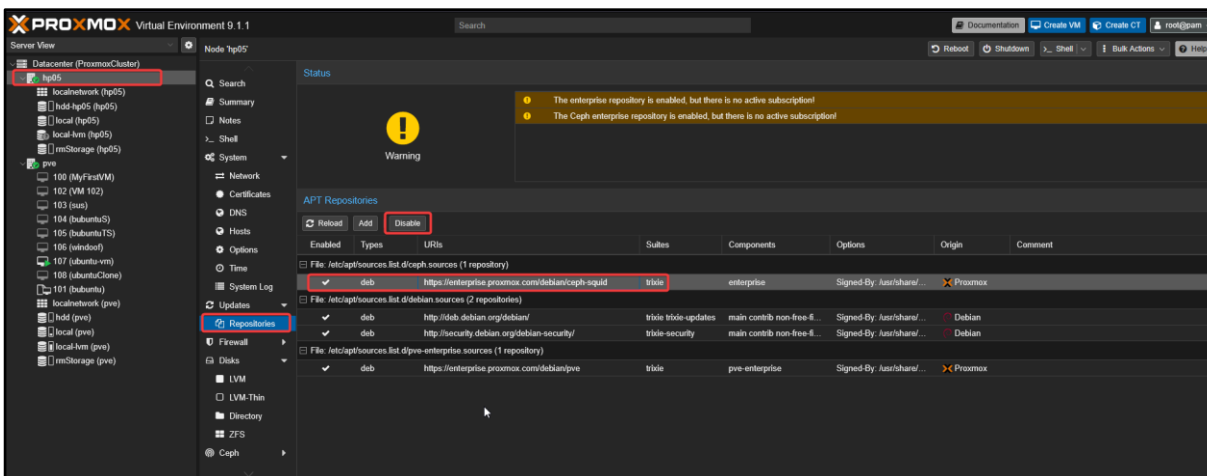
PROXMOX



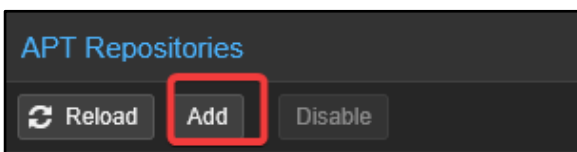
Out of the box Proxmox VE is configured to use the enterprise repository, which needs a paid subscription, so on a no-subscription lab host the updates fail until the repositories are switched over. The fix is to disable the enterprise repo and enable the no-subscription repo, which can be done straight from the GUI under Node → Updates → Repositories, or by editing the apt sources files. Once the no-subscription repo is enabled we refresh the package lists and apply the updates.

apt update && apt dist-upgrade

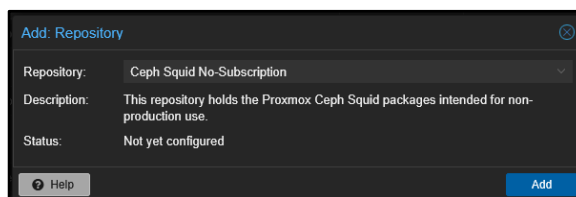
We use dist-upgrade rather than a plain upgrade because Proxmox's kernel and package updates sometimes need to add or remove dependencies, which a normal upgrade would hold back. After it finishes a reboot may be needed if the kernel was updated.



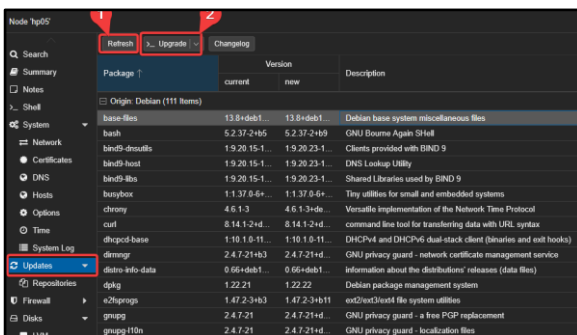
Proxmox - Screenshot 122: Disabling enterprise subscription



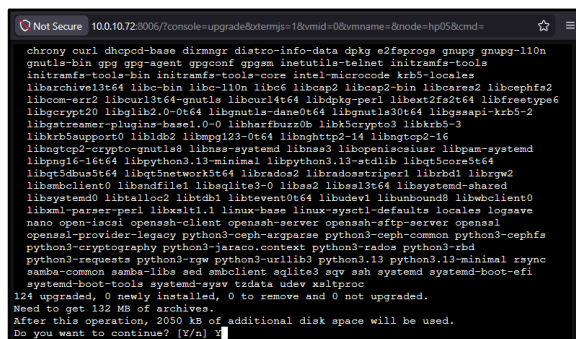
Proxmox - Screenshot 123: Adding Repository



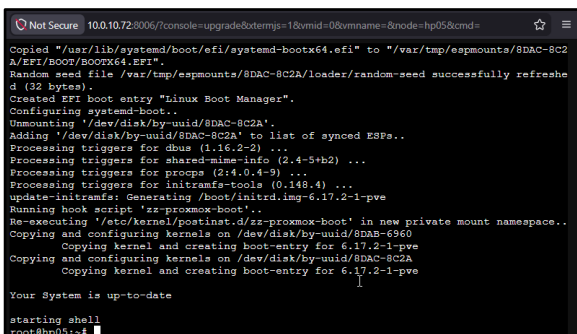
Proxmox - Screenshot 124: Selecting no subscription repository



Proxmox - Screenshot 125: Refreshing and Upgrading Updates



Proxmox - Screenshot 126: Confirming upgrade



Proxmox - Screenshot 127: After installation of new patches

The contrast with VMware is mainly in approach: vSphere patches hosts through the Lifecycle Manager with attached baselines and requires the host's VMs to be powered off first, whereas Proxmox patches the host like any other Debian machine with apt, once the repositories are pointed at the right place.